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# Do Language Barriers Shift Bargaining Power from Employers to Workers? Evidence from Migrant-Owned Small Businesses

Hong Minh Dang (Addison) Nguyen

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Department of Econometrics & Business Statistics

Supervised by: Professor Dan Zhu

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# Abstract

Labour regulation presumes employers hold bargaining power over employees. This presumption may fail in small businesses whose migrant owners have limited English and depend on employees' local and institutional knowledge. We formalise this power reversal as bargaining under compliance risk and test three implications using the Household, Income and Labour Dynamics in Australia Survey. Fixed-effects regressions exploiting industry variation in migrant ownership by English proficiency show the 11 per cent wage penalty in small independent workplaces narrows where English-proficient migrant owners are common, but not where owners have limited English. Employers out-earn employees of their own origin but bear greater income volatility, with no detectable difference by origin. Entry into ownership follows labour market detachment and low earnings, with the push from unemployment strongest for migrants with limited English. Although we find little evidence of direct power reversal, migrants with limited English often enter ownership from weak economic positions.

**Keywords:** power reversal, migrant entrepreneurship, small business

# 1 Introduction

Labour regulation in most developed economies rests on a familiar premise that employers hold bargaining power over workers through their ability to set wages, dictate terms and replace staff (Manning, 2003). Australian workplace law reflects this premise by placing strict liability for compliance on the employer. Yet the premise may fail in small businesses run by migrants with limited English. Such owners operate with thin margins and little administrative capacity, and they rely on employees, contractors and advisers for the language skills and institutional knowledge needed to run the business. If this reliance shifts de facto bargaining power towards workers, regulation designed to protect employees may, in part of the small-business sector, shift cost and risk onto owners who are themselves economically vulnerable. Whether it does matters for small-business regulation, migrant integration policy and the enforcement of workers' entitlements.

This paper asks whether migrant ownership and limited English proficiency weaken the bargaining position of small-business owners relative to their workers, and whether any such reversal leaves a detectable imprint on labour market outcomes. We formalise the mechanism in a model of owner–agent bargaining under compliance risk, which yields three testable implications, and test them with twenty-four waves of the Household, Income and Labour Dynamics in Australia (HILDA) Survey. If workers hold more bargaining power, employees of small migrant-owned workplaces should earn more, migrant owners should earn less or bear more income risk than comparable employees of their own origin, and entry into ownership should reflect weak outside options. The central difficulty is that HILDA, like most household surveys, does not identify respondents' employers. We therefore measure each worker's exposure to migrant ownership by the share of business owners in their industry and year who are migrants, separately by English proficiency, and ask how the wage gap between small independent and larger workplaces in the same industry and year changes with this share. We compare owners with employees of the same origin on the level and the risk of their income, and relate entry into ownership to each person's labour market position before entry.

We document three findings. First, employees of small single-site workplaces earn about 11 per cent less than comparable employees of larger workplaces in the same industry and year. This penalty narrows to about 2 per cent where a third of owners are non-English-speaking migrants with good English, but not where owners with limited English are common, the group for which the framework predicts the largest effect. Second, in every origin group, owners with employees earn more than employees of their own origin but face far more volatile income; among the native-born, owners' income volatility exceeds employees' by about 27 percentage points. We detect no migrant-specific difference in this trade-off. Third, entry into ownership follows unemployment and low pay, and for low-English migrants unemployment raises the odds of entry about four times as much as for the native-born, with a push that persists two years before entry. Together, the findings offer little support

for a direct reversal of power, but they show that migrants with limited English enter ownership from weak positions and then face the high income risk that ownership brings in every group.

This paper contributes to three literatures. First, it brings two literatures on migrants together at the point where they meet. Research on migrant workers shows how migrants are disadvantaged as employees, through sorting into lower-paying jobs (Han & Hermansen, 2024; Hermansen et al., 2025) and widespread underpayment (Berg & Farbenblum, 2017). Research on migrant entrepreneurship asks why migrants become self-employed, often out of necessity (Azoulay et al., 2022; Fairlie & Fossen, 2020). Neither examines the employment relationship once the migrant is the employer. The few studies that do are qualitative case studies of dependency between migrant owners and their local staff (Cooke et al., 2022; David et al., 2026; David & Terstriep, 2025) or typologies of power within firms (Audretsch & Fiedler, 2023). To our knowledge, this paper provides the first quantitative test, in nationally representative data, of whether the informational disadvantage of migrant owners shifts the distribution of surplus between owners and their workers. It also separates language barriers from migrant status, which existing work typically pools.

Second, we propose a measure of exposure to migrant ownership for settings in which employers and employees cannot be linked. The measure is a survey-weighted, leave-one-out share of business owners in each industry and year, by origin and English proficiency, with an optional decay by years in Australia whose rate we select by cross-validation. Combined with workplace types that separate independent small businesses from small sites of chains, and with a comparison against those chain sites, it allows the wage effects of migrant ownership to be studied in any household panel that records industry and workplace size.

Third, we extend the study of necessity entrepreneurship beyond the point of entry. Existing work explains why migrants become self-employed; we follow them into ownership and compare their income and income risk with those of employees of the same origin, the comparison that isolates the return to ownership from origin gaps that already exist among employees. We show that the push into ownership from unemployment is stronger and more persistent for low-English migrants, and we test it against anticipation of entry and stable personal traits. This connects the motives for entry with their consequences. The migrants with the weakest alternatives enter ownership, and once they do, they face the same trade-off between higher income and higher risk as other owners.

Our findings are consistent with existing evidence on self-employment and migrant entrepreneurship. That owners with employees earn more than employees while own-account workers earn less mirrors evidence that owners of incorporated businesses out-earn employees while owners of small unincorporated businesses earn considerably less (Levine & Rubinstein, 2017), and it helps reconcile our results with studies that pool all the self-employed and conclude that self-employment does not pay (Hamilton, 2000). The push into ownership accords with research that defines necessity entrepreneurship by prior work status (Fairlie & Fossen, 2020), and its persistence among low-English migrants suggests that their entry reflects a sustained lack of alternatives, not a single spell of unemployment. Finally, because the wage penalty narrows where English-proficient migrant owners are common but

not where owners have limited English, migrant ownership appears to shape pay through industry pay structures or ethnic networks, not through owners' language barriers.

Australia is an informative setting for this test. Migrants account for a substantial share of its small-business sector, which operates under one of the more actively enforced workplace relations frameworks among comparable economies (Wright & Clibborn, 2020). Under the Fair Work Act 2009 (Cth), penalties for underpayment fall on the employer regardless of intent or knowledge, while employees can leave the relationship without residual liability, which is precisely the asymmetry that could shift leverage away from an owner who relies on better-informed workers. The mechanism rests on strict compliance regimes and migrant business density, not on features unique to Australia, so the findings should apply, at least qualitatively, to other economies with large migrant-owned small-business sectors, such as the United Kingdom, the United States and Canada.

The paper proceeds as follows. Section 2 sets out the model of owner–agent bargaining under compliance risk. Section 3 describes the data and the owner-share measure, and Section 4 the estimating equations. Section 5 reports the results, Section 6 discusses them and their limitations, and Section 7 concludes.

# 2 Conceptual Framework

We propose viewing the small migrant-owned firm as a constrained agent facing four structural sources of asymmetry.

1. *Information asymmetry.* Employees, contractors and external advisors often possess more complete knowledge of regulatory compliance, local institutional norms and communication channels than the owner does. Where the owner faces language barriers or general unfamiliarity with local institutions, they must delegate operational and compliance decisions to agents whose local expertise they cannot independently verify.
2. *Outside options.* Differences in mobility reshape the relative bargaining strength of workers and owners. Skilled employees can generally re-enter the broader labour market with little friction, whereas older migrant owners face substantial immobility, the result of sunk capital investment, visa-related constraints and limited alternative employment prospects.
3. *Concentrated risk.* Legal and financial liabilities fall almost exclusively on the firm owner. Under frameworks such as the Fair Work Act, statutory penalties for non-compliance apply irrespective of the owner's intent or information state, while employees face negligible direct exposure to regulatory enforcement.
4. *Credibility constraints.* Language barriers and institutional unfamiliarity limit the owner's capacity to negotiate from an informed position. They restrict an owner's ability to contest advice, inde-

pendently verify regulatory claims or credibly threaten to withhold compensation or terminate a contract, eroding their negotiating position in any wage or contract dispute.

Together, these four asymmetries shift de facto bargaining power toward workers. The remainder of this section formalises this reversal and derives the predictions we test empirically in Section 5.

To formalise the conditions under which this reversal occurs, consider a small firm operated by a migrant entrepreneur (Owner, $O$) who faces language and institutional barriers, and who employs or contracts a more skilled agent (Agent, $A$) who holds private information about regulatory compliance and administrative requirements.

The firm's output is $y = \theta + a + \varepsilon$, where $a \geq 0$ denotes the agent's effort, $\theta$ is baseline profitability and $\varepsilon$ is mean-zero noise. Compliance requirements are uncertain. Let

$$c \in \{c_L, c_H\}, \quad \Pr(c = c_H) = p, \tag{2.1}$$

where $c$ denotes the minimum compliance expenditure required to avoid a violation. The realisation of $c$ is privately observed by the agent and unobserved by the owner. The owner chooses compliance spending $s \geq 0$, and a violation occurs whenever spending falls short of the true requirement,

$$\pi(s, c) = \begin{cases} 1, s < c \\ 0, s \geq c \end{cases} \tag{2.2}$$

A violation triggers a penalty $F > 0$ borne entirely by the owner, reflecting the risk asymmetry described above. The owner's expected payoff is thus

$$U_O = \mathbb{E}[y - w - s - \pi(s, c)F] \tag{2.3}$$

and the agent's payoff is

$$U_A = \mathbb{E}[w] - \psi(a) + bs, \tag{2.4}$$

where $\psi(a) = \frac{\gamma}{2}a^2$ is the disutility of effort and $b \geq 0$ captures the agent's private benefit from higher compliance spending, whether through convenience, commissions, a lighter workload or simply greater influence over firm decisions.

Compensation follows a standard incentive contract, $w = \alpha + \beta y$, where $\alpha$ is the base wage and $\beta$ is a performance incentive. Hence, the agent's effort choice is

$$a^* = \frac{\beta}{\gamma}. \tag{2.5}$$

The owner delegates the compliance decision by committing to a spending rule $s(\hat{c})$ based on the agent's report $\hat{c}$ of the true requirement. The agent's expected wage does not depend on $\hat{c}$, so the agent does not internalise the cost of compliance and therefore has no incentive to report truthfully. Instead, the agent reports whichever value maximises claimed requirements,

$$\hat{c} \in \arg\max_{\tilde{c}} s(\tilde{c}). \tag{2.6}$$

**Result 1 (Systematic Overstatement).** If the agent benefits from spending ($b > 0$) and the owner spends more when requirements are high ($s(c_H) > s(c_L)$), the agent always reports $\hat{c} = c_H$, regardless of the true requirement.

Anticipating this misreporting, the owner cannot learn the true requirement from the agent's report and must instead choose a report-invariant policy, comparing two scenarios.

1. *High-spending scenario:* $\Pi_H = \mathbb{E}[y - w] - c_H$, and
2. *Low-spending scenario:* $\Pi_L = \mathbb{E}[y - w] - c_L - pF$

**Result 2 (Over-Compliance).** The owner over-complies, choosing $s = c_H$ even when the true requirement may be $c_L$, if and only if

$$c_H - c_L \leq pF. \tag{2.7}$$

This result captures how regulatory risk amplifies the underlying information problem. Stricter enforcement (a higher $F$), greater uncertainty (a higher $p$) or a stronger reliance on delegation all increase the owner's incentive to over-comply as a precaution against the agent's informational advantage. Unable to verify the true compliance requirement, the owner internalises a regulatory risk that should properly be borne by the better-informed agent.

We next consider how surplus is divided between owner and agent once production and compliance decisions are determined. The central mechanism here is the owner's inability to exit the relationship, itself a product of migration constraints, sunk costs and limited outside employability, which asymmetrically shifts bargaining power toward labour.

Let net surplus be

$$S = \Pi - C - pF, \tag{2.8}$$

where $\Pi$ is revenue net of effort, $C$ captures compliance and administrative costs and $pF$ is the expected penalty. Payoffs to the owner and agent are then $U_O = S - w$ and $U_A = w - \psi$, respectively.

The agent's outside option reflects their labour-market alternatives, $\bar{u}_A = u_0$, and is relatively strong for a skilled worker with language proficiency and regulatory knowledge. The owner, by contrast, faces an exit loss

$$\bar{u}_O = -K, \quad K > 0, \tag{2.9}$$

that captures their immobility. Leaving Australia would forfeit sunk capital in housing and business equipment, visa requirements and accumulated labour-market networks, while re-entering formal employment is made difficult by age, language proficiency and foregone work experience.

Wages are determined through Nash bargaining over the surplus,

$$w \in \arg\max_{w} (U_A - \bar{u}_A)^{\eta} (U_O - \bar{u}_O)^{1-\eta}, \tag{2.10}$$

where $\eta \in (0, 1)$ is the agent's bargaining weight, yielding

$$w^* = \eta S + (1-\eta)(u_0 + \psi) + \eta K. \tag{2.11}$$

**Result 3 (Power Reversal).** The equilibrium wage is increasing in the owner's exit loss $K$. Although the owner remains the formal employer and residual claimant, a greater inability to exit strengthens the agent's bargaining position.

Compliance risk further compresses the owner's payoff. Substituting the surplus equation gives

$$w^* = \eta(\Pi - C - pF) + (1-\eta)(u_0 + \psi) + \eta K. \tag{2.12}$$

An increase in compliance risk $pF$ reduces total surplus, but need not reduce wages proportionally. Wages adjust with the bargaining weight $\eta$, so workers are partly insulated from compliance risk while owners absorb the remainder, an asymmetry that is amplified when the owner's exit loss is high.

**Result 4 (Risk Concentration).** Stricter regulatory enforcement (a higher $F$) or greater compliance uncertainty (a higher $p$) disproportionately harms owners with large exit costs $K$, so that regulation amplifies the bargaining asymmetry precisely for the owners least able to exit.

Taken together, these results explain why reforms meant to improve worker outcomes can end up harming small-business owners who face language and institutional barriers. When enforcement rises without extra language or compliance support, owners with high exit costs absorb the cost through lower profits, not through lower wages or stronger worker protections.

As detailed in Section 4, we test these predictions in three analyses. In Section 4.1, we examine whether employees in small migrant-owned firms earn a wage premium, as Result 3 predicts. In Section 4.2, we turn to owner income and look at whether migrant owners, especially those with limited English proficiency, earn less, or bear more income risk, than employees of the same origin who work in workplaces of the same size. In Section 4.3, we turn to entry into business ownership and ask whether "push" factors, such as labour-market detachment, low earnings and liquidity constraints, drive migrants into entrepreneurship more than migrant origin does.

# 3 Data

We use waves 1 to 24 (2001 to 2024) of the Household, Income and Labour Dynamics in Australia (HILDA) Survey (Department of Social Services & Melbourne Institute of Applied Economic and Social Research, 2025), which is a nationally representative household panel that has interviewed the same individuals annually since 2001.[1] Three features make HILDA well suited to our questions. First, its panel structure allows us to observe year-to-year changes in income and transitions into business ownership. Second, it records each respondent's country of birth, year of arrival and self-assessed

[1]This paper uses unit record data from Household, Income and Labour Dynamics in Australia Survey (HILDA) conducted by the Australian Government Department of Social Services (DSS). The findings and views reported in this paper, however, are those of the author and should not be attributed to the Australian Government, DSS, or any of DSS' contractors or partners. We use General Release 24.0 (waves 1 to 24).

English proficiency, which allows migrants to be distinguished by their linguistic distance from the host population. Third, it records the industry, workplace size and number of employer locations of every employed respondent, which allows small independent workplaces to be identified within each industry. Immigrants make up 23.8% of the unweighted sample and 25.5% of the weighted sample.[2]

The main limitation of HILDA is that it does not identify respondents' employers. Workers therefore cannot be linked to the owners of their workplaces, nor to those owners' migrant status or English proficiency. The workplace characteristics that HILDA does record are nonetheless sufficient to distinguish small independent workplaces from large chains, even within broad industries such as retail trade. Our empirical strategy therefore does not attempt to identify individual employers. For the wage analysis, we instead classify jobs by industry and employer structure and construct a measure of migrant owner density for each industry and year (Section 3.5). Migrant owners are concentrated in small independent businesses, so the higher this density, the more likely a small single-site workplace in that industry and year is to be migrant-owned, whereas a large workplace or a site of a chain is not. Comparing the wage gap between small independent and other workplaces within the same industry and year, and asking how this gap varies with migrant owner density, therefore isolates pay differences associated with migrant ownership from industry-wide shocks to wages. The owner income and entry analyses do not require this link, since they observe owners and entrants into ownership directly. Table A.7 in the appendix summarises all HILDA variables we used in the analysis.

## 3.1 Estimation Samples

Each analysis block in Section 4 has its own sample (Table A.1). The *employee sample* for the wage analysis contains wage and salary earners in waves 1 to 21, the waves in which workplace size and the number of employer locations are recorded. For the owner income analysis, we use pairs of consecutive waves ending in waves 2 to 21. The *unrestricted sample* classifies individuals by their status at $t$ (174,962 person-pairs), while the *restricted sample* keeps only those with the same status in both waves (152,060 person-pairs), so that we do not attribute income changes caused by a change of status to ownership. For the entry analysis, we draw on all 24 waves and use 289,616 person-pairs in which the individual is not a business owner at $t-1$ (the *transition sample*), which contain 5,876 entries into ownership.

## 3.2 Origin and English Proficiency

Following Guven & Islam (2015), Islam & Parasnis (2016) and To et al. (2017), we classify respondents by country of birth as native-born Australians, migrants from main English-speaking (MES) countries[3] and migrants from other, non-English-speaking (MNES) countries. An English-speaking background brings substantial labour market advantages in Australia (Cobb-Clark & Nguyen, 2012; Nguyen et al., 2025), but the MNES group is heterogeneous, because linguistic distance from English is a primary

[2] As at 30 June 2025, 8.8 million people, or 32.0% of Australia's population, were born overseas (ABS, Australia's population by country of birth).

[3] The United Kingdom, New Zealand, Canada, the United States of America, Ireland and South Africa.

determinant of language acquisition (Adserà & Ferrer, 2021; Isphording & Otten, 2014). We therefore split MNES migrants by English proficiency, which gives four mutually exclusive groups,

$$G_i \in \mathcal{G} = \left\{\text{Native, MES, MNES}_{\text{HighEng}}, \text{MNES}_{\text{LowEng}}\right\}, \tag{3.1}$$

with the native-born as the reference group throughout. This split separates the role of language barriers, which is central to our framework, from that of migrant status in general.

HILDA records self-reported speaking ability in every wave and reading ability relative to an average Australian in waves 12, 16, 20 and 24. Following Clarke & Isphording (2017), a respondent has low proficiency if they speak English less than well or rate their reading below 4 on a scale from 0 to 10; native-born and MES respondents are proficient by construction. We assign each respondent their modal classification across all answered waves, with ties classified as proficient, for two reasons. The reading measure is collected only every four years, and MNES migrants are observed for a median of 4 waves (interquartile range 1 to 14), so most answer it once or twice. And the within-person variation appears to be reporting noise, not language acquisition. In total, 58.5% of repeat respondents change their speaking score and 80.7% their reading score at least once, but these changes alternate between adjacent categories and do not trend upwards. The measure behaves as expected. Among MNES migrants, 93.8% of those who arrived before age 11 speak English very well, against 46.2% of those who arrived later ($p < 0.001$), consistent with Lee et al. (2022). Years in Australia are the survey year minus the year of first arrival, which is reported for 8,838 of 8,855 migrants.

## 3.3 Workplace Type and Size

In waves 1 to 21, the HILDA survey asks employed respondents how many people work at their workplace, whether their employer operates from more than one location in Australia and how many people the employer employs nationally. Following the Australian Bureau of Statistics definition of a small business, a workplace is small if fewer than 20 people work there. From these questions we construct three mutually exclusive workplace types,

$$\begin{aligned}
\text{SmallSingleSite}_{it} &= \mathbf{1}\{\text{WorkplaceSize}_{it} < 20, \text{MultiLocation}_{it} = 0\}, \\
\text{SmallSiteChain}_{it} &= \mathbf{1}\{\text{WorkplaceSize}_{it} < 20, \text{MultiLocation}_{it} = 1\}, \\
\text{ChainLike}_{it} &= \mathbf{1}\{\text{WorkplaceSize}_{it} \geq 20, \text{MultiLocation}_{it} = 1, \text{EmployerSize}_{it} \geq 500\}.
\end{aligned} \tag{3.2}$$

A small single-site workplace is the closest observable counterpart of an independently owned small business, such as a family restaurant or a corner shop. A small-site chain also has fewer than 20 people at the workplace, but belongs to an employer with more than one location, such as a branch of a franchise. It shares the local scale of a small business but not its ownership structure. A chain-like workplace, such as a supermarket of a national chain, has 20 or more people and belongs to an employer with at least 500 employees nationally. The reference category is all other workplaces with 20 or more people that either belong to a single-location employer or to a multi-location employer with fewer than 500 employees nationally (Table A.2).

## 3.4 Employment Status and Outcomes

We classify each employed respondent into one of three mutually exclusive groups: employees, own-account workers and employers with employees. Own-account workers are business owners who are self-employed or own a business without employees.

*Hourly wages* are usual weekly gross earnings in the main job divided by usual weekly hours in that job, with wages below \$1 per hour treated as reporting errors. Wages are nominal, since the year fixed effects absorb common price changes,

$$w_{it} = \frac{\text{WeeklyWageMainJob}_{it}}{\text{UsualWeeklyHoursMainJob}_{it}}, \qquad w_{it} \geq 1. \tag{3.3}$$

*Labour and business income* adds financial-year gross wages and salary to net business income, so that incorporated owners who pay themselves a wage are comparable with unincorporated owners,

$$y_{it} = \text{AnnualGrossWages}_{it} + \text{BusinessIncome}_{it}^{+} - \text{BusinessIncome}_{it}^{-}. \tag{3.4}$$

Business losses are retained, since they are part of the risk of ownership.

Let $s_{it} \in \{W, E, O, N\}$ denote the labour market status of individual $i$ in year $t$, where $W$ is an employee, $E$ an owner with employees, $O$ an own-account worker and $N$ not employed. *Entry into ownership* is defined for individuals who are not owners at $t-1$, that is, with $s_{i,t-1} \in \{W, N\}$,

$$\text{Entry}_{it} = \begin{cases} 1 \text{ if } \; s_{i,t-1} \notin \{E, O\} \;\text{ and }\; s_{it} \in \{E, O\} \\ 0 \text{ if } \; s_{i,t-1} \notin \{E, O\} \;\text{ and }\; s_{it} \notin \{E, O\} \end{cases} \tag{3.5}$$

Every transition is counted, so an individual who leaves and later re-enters ownership contributes more than one entry. Respondents not interviewed at $t$ have no status at $t$ and are excluded.

The entry analysis also uses labour market position at $t-1$. $\text{Unemployed}_{i,t-1}$ and $\text{NotInLabourForce}_{i,t-1}$ record non-employment, and $\text{LowWage}_{i,t-1}$ equals one for employees in the bottom quintile of that year's hourly wage distribution; the 1 per cent of employee person-years with a missing wage are excluded. Following Bray (2001), Butterworth et al. (2012), Kiely et al. (2015) and Gerards & Welters (2020), a respondent is liquidity constrained if they report, in the self-completion questionnaire, either a cash-flow problem, such as being unable to pay bills or rent on time or asking family or friends for money, or hardship, such as going without meals, being unable to heat the home, pawning possessions or asking a welfare organisation for help,

$$\text{LiquidityConstraint}_{it} = \mathbf{1}\{\text{CashflowProblem}_{it} = 1 \text{ or } \text{Hardship}_{it} = 1\}. \tag{3.6}$$

Respondents who do not return the questionnaire are kept, with the measure set to zero and an indicator for a missing measure.[4]

[4]Dropping non-returners would remove a disproportionate share of the low-English group, since a written questionnaire is harder to complete with limited English. With the missing indicator, the liquidity coefficient compares constrained with unconstrained respondents among those observed, while every respondent contributes to the other coefficients (Dardanoni et al., 2011). Missing arrival years are treated in the same way. Because this approach can bias the other coefficients when missingness is related to them (Jones, 1996), we do not interpret the control coefficients causally.

## 3.5 Migrant Owner Density

Because HILDA does not link workers to employers, we measure each worker's exposure to migrant ownership by the share of business owners in their industry and year who belong to each origin group. For employee $i$ in industry $k$ and year $t$, the leave-one-out owner share of group $g$ is

$$S_{kt}^{g(-i)} = \frac{\sum_{m \in (k,t),\, m \neq i} \omega_{mt}\, \text{Owner}_{mt}\, \vartheta_{mt}\, \mathbf{1}[G_m = g]}{\sum_{m \in (k,t),\, m \neq i} \omega_{mt}\, \text{Owner}_{mt}}, \qquad g \in \left\{\text{MES}, \text{MNES}_{\text{HighEng}}, \text{MNES}_{\text{LowEng}}\right\} \tag{3.7}$$

where $\omega_{mt}$ is the HILDA cross-sectional responding-person weight, so that the share describes the population of owners in that year, and $\vartheta_{mt}$ is the assimilation weight defined below. The shares built with this weight are our main measure throughout. As a sensitivity check, reported in the appendix only (Section B.2), we also build them with the longitudinal responding-person weight, which is positive only for people who have responded in every wave since wave 1. Employees in the wage analysis are not owners, so leaving out individual $i$ does not change their share, which is simply the composition of the other owners in their industry and year. Industries are the 19 ANZSIC divisions, with two-digit industries used for robustness; Table A.3 summarises the distribution of the shares.

**Assimilation Weighting**

The barriers in our framework should fade as migrants settle, so we also consider down-weighting each migrant owner by their years in Australia,

$$\vartheta_{mt} = \begin{cases} \exp(-\lambda_{G_m}\, \text{YearsInAus}_{mt}) & \text{if } m \text{ is foreign-born,} \\ 0 & \text{if } m \text{ is Australian-born .} \end{cases} \tag{3.8}$$

A recent arrival receives a weight close to one and a long-settled owner a weight close to zero, with a half-life of $\ln 2/\lambda$ years; with $\lambda = 0$ the share is a plain compositional share. We compare no decay (Type 0), a single rate for all migrant owners (Type 1) and separate rates for MES and MNES owners (Type 2), and choose the rates by five-fold cross-validation over individuals, minimising the out-of-fold mean squared error of log wages from the wage equation (Equation (4.1)), over $\lambda \in [0, 0.30]$ for Type 1 and $\lambda_{\text{MES}}, \lambda_{\text{MNES}} \in [0, 0.50]$ for Type 2. These ranges span the horizons in the assimilation literature, from convergence within a decade (Akee et al., 2026; Chiswick, 1978) to considerably longer (Arellano-Bover & San, 2026). Type 1 selects $\lambda = 0.02$, a half-life of 34.7 years. Type 2 selects $\lambda_{\text{MES}} = 0.04$, a half-life of 17.3 years, and $\lambda_{\text{MNES}} = 0$, which means no decay at all, consistent with slower convergence for non-English-speaking migrants (Breunig et al., 2013; Colic-Peisker, 2011). The gain from decay is negligible, however. The best Type 1 rate lowers the cross-validated error by $1.4 \times 10^{-5}$ against an error of 0.152. We therefore use the plain shares (Type 0) for all main estimates and report the decay weightings for comparison only, since their rates are chosen on the same data, are poorly identified and change the units of the shares. Appendix F reports the grid search and the supporting literature.

## 3.6 Descriptive Statistics

Ownership is strongly stratified by origin and language (Table A.4). Construction is the largest industry of owners in every group, and native-born and MES owners have similar profiles in the remaining industries apart from agriculture, forestry and fishing. High-English MNES owners are over-represented in health care and social assistance (13.3 per cent), transport and warehousing (7.3 per cent), accommodation and food services (6.7 per cent) and financial and insurance services (3.7 per cent). Low-English MNES owners are the most distinctive group. Only 1.0 per cent operate in professional, scientific and technical services, against 11.9 to 14.6 per cent of the other groups, while they are concentrated in manufacturing (21.5 per cent), retail trade (14.9 per cent) and accommodation and food services (7.2 per cent). Across worker types (Table A.5), ownership is concentrated in low-capital, small-scale industries such as construction, professional services and agriculture, while employees are concentrated in health care, education and public administration. Industries that rely on team production, such as accommodation and food services, have a higher share of employers, while industries suited to independent contracting, such as administrative and support services, have a higher share of own-account workers.

Table 3.1 reports characteristics by origin group. Business owners are predominantly male, in their mid-forties and highly educated, with 68% holding a post-school qualification, falling to 61% among low-English MNES owners. Low-English MNES owners are also far more recent arrivals.

Panel persistence differs substantially across groups (see Table A.6). Low-English migrants are interviewed in 6.5 waves on average (median 3) against 10.3 waves for native-born respondents (median 8), and 3% of the low-English MNES group is interviewed in all 24 waves, against 12% of the native-born. Because the owner income and entry samples require consecutive observations, low-English migrants are under-represented in them relative to their population share, and our estimation results therefore rest on few individuals.

**Table 3.1:** *Characteristics of business owners by origin and English proficiency group.*

| | Native-born | | MES | | MNES (high English) | | MNES (low English) | | All owners | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Mean | SD | Mean | SD | Mean | SD |
| Age | 46.59 | 13.25 | **50.66**$^{***}$ | 11.29 | **48.71**$^{***}$ | 12.15 | 46.67 | 9.96 | 47.30 | 12.97 |
| Female | 0.35 | 0.48 | 0.31 | 0.46 | 0.36 | 0.48 | 0.44 | 0.50 | 0.35 | 0.48 |
| Married or de facto | 0.80 | 0.40 | **0.84**$^{*}$ | 0.37 | **0.84**$^{**}$ | 0.36 | **0.93**$^{***}$ | 0.26 | 0.81 | 0.39 |
| Post-school qualification | 0.67 | 0.47 | **0.74**$^{***}$ | 0.44 | **0.74**$^{***}$ | 0.44 | 0.61 | 0.49 | 0.68 | 0.47 |
| Low English proficiency | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 | 0.00 | 0.01 | 0.08 |
| Years since arrival in Australia | | | 31.88 | 14.57 | 29.02 | 15.53 | 16.13 | 10.27 | 30.08 | 15.19 |
| Own-account worker | 0.61 | 0.49 | **0.65**$^{*}$ | 0.48 | 0.59 | 0.49 | 0.50 | 0.50 | 0.61 | 0.49 |
| Owner with employees | 0.39 | 0.49 | **0.35**$^{*}$ | 0.48 | 0.41 | 0.49 | 0.50 | 0.50 | 0.39 | 0.49 |
| Market income | 60,899.33 | 105,634.01 | 65,755.71 | 121,338.11 | 64,110.74 | 111,911.24 | **42,605.48**$^{**}$ | 57,804.08 | 61,717.90 | 108,075.78 |
| Cashflow problem | 0.18 | 0.39 | 0.17 | 0.37 | 0.18 | 0.38 | **0.11**$^{**}$ | 0.31 | 0.18 | 0.38 |
| Hardship | 0.07 | 0.26 | 0.07 | 0.26 | 0.07 | 0.26 | 0.09 | 0.29 | 0.07 | 0.26 |
| Liquidity constraint | 0.20 | 0.40 | 0.19 | 0.39 | 0.20 | 0.40 | 0.16 | 0.37 | 0.20 | 0.40 |
| Home owner | 0.78 | 0.41 | **0.81**$^{*}$ | 0.39 | 0.76 | 0.43 | 0.67 | 0.47 | 0.78 | 0.41 |
| SEIFA disadvantage decile | 6.10 | 2.70 | **6.44**$^{**}$ | 2.88 | 6.30 | 2.77 | 5.55 | 3.13 | 6.16 | 2.74 |
| Person-years | 26,784 | | 3,964 | | 4,071 | | 200 | | 35,019 | |
| Persons | 5,039 | | 693 | | 818 | | 57 | | 6,607 | |

*Notes:* This table reports descriptive statistics for business owners (employers and own-account workers) by origin group, pooled over owner person-years in HILDA waves 1 to 24. Mean and SD denote the mean and standard deviation of each variable within each group. Market income denotes annual labour and business income, in dollars. Bolded figures indicate group means that differ significantly from those of native-born owners. Significance is assessed by regressing each variable on a group indicator, estimated on the pooled person-years of the respective migrant group and native-born owners, with standard errors clustered at the individual level. Low English proficiency and years since arrival are not tested, as neither is defined for the native-born. Person-years denotes owner person-years and Persons the number of distinct owners. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.01$.

# 4 Methodology

The framework in Section 2 concerns relationships that are rarely observed directly, namely the terms on which migrant owners employ workers, the risks those owners bear and the circumstances in which migrants become owners. HILDA does not link workers to their employers, so we cannot compare the same worker employed by a migrant and by a native-born owner. We instead test three observable implications of the framework, each with its own sample (Section 3.1) and source of variation. No single analysis is decisive, so we read the framework as supported only where the three point in the same direction.

## 4.1 Employee Wages

If limited English weakens an owner's bargaining position, the workers of such owners should be paid more. We test this by asking whether the wage gap between small independent workplaces and other workplaces in the same industry and year varies with migrant owner density (Section 3.5). The model is specified as follows,

$$\begin{aligned}\log w_{it} = {} & \delta_t + \mu_{kt} + \nu_{rt} \\ & + \beta_1 \text{SmallSingleSite}_{it} + \beta_2 \text{SmallSiteChain}_{it} + \beta_3 \text{ChainLike}_{it} \\ & + \beta_4 \left( \text{SmallSingleSite}_{it} \times S_{kt}^{\text{MES}} \right) \\ & + \beta_5 \left( \text{SmallSingleSite}_{it} \times S_{kt}^{\text{MNES}_{\text{HighEng}}} \right) \\ & + \beta_6 \left( \text{SmallSingleSite}_{it} \times S_{kt}^{\text{MNES}_{\text{LowEng}}} \right) + \mathbf{X}_{it}' \boldsymbol{\pi} + \epsilon_{it},\end{aligned} \tag{4.1}$$

where $w_{it}$ is employee $i$'s hourly wage in year $t$, the three workplace types are defined in Section 3.3 and $S_{kt}^{g}$ is the share of business owners in employee $i$'s industry $k$ and year $t$ who belong to origin group $g$ (Equation (3.7)). The terms $\delta_t$, $\mu_{kt}$ and $\nu_{rt}$ are year, industry-by-year and state-by-year fixed effects. The vector $\mathbf{X}_{it}$ contains age, sex, the worker's origin group, education, occupation, casual or permanent employment, occupational tenure, employer tenure, union membership and the decile of area disadvantage (SEIFA). We estimate the model by OLS without survey weights, since sampling is exogenous to wages conditional on the controls (Solon et al., 2015), and cluster standard errors by individual and by industry-year.

The coefficients $\beta_1$ to $\beta_3$ measure the wage gap of each workplace type relative to other workplaces with 20 or more people in the same industry and year. The interactions $\beta_4$ to $\beta_6$ measure how the small single-site gap changes as the share of owners from group $g$ rises. The owner shares vary only by industry and year, so $\mu_{kt}$ absorbs their main effects together with any shock common to all workers in an industry and year, such as changes in demand or award wages. The interactions are therefore identified from comparisons between workplace types within the same industry and year.

To interpret the interactions, consider three explanations. First, if owners with limited English depend on their workers for local knowledge, as in the power-reversal mechanism (Result 3), the gap should narrow where these owners are common, so $\beta_6 > 0$. Second, if limited outside options and weak enforcement dominate, the gap should widen, so the interactions should be negative and most negative for $\beta_6$. Third, if shared language and networks improve the quality of job matches, $\beta_5$ and $\beta_6$ should be non-negative. Positive interactions are therefore consistent with both power reversal and better job matching, and only their pattern across groups separates the two, since power reversal predicts a narrowing concentrated where owners have limited English. Owners from main English-speaking countries face no language barrier, so $\beta_4$ should be close to zero and serves as a benchmark for $\beta_5$ and $\beta_6$.

The estimates rely on the assumption that, conditional on the fixed effects and controls, migrant owner density is uncorrelated with unobserved determinants of wages specific to small independent workplaces. The assumption would fail if migrant owners set up firms in industries where small-firm pay is already relatively high, in which case our estimates would overstate any effect of migrant ownership. We do not include individual fixed effects, which would absorb the worker's origin group and identify the interactions only from workers who move between workplace types, so the estimates are conditional associations. We assess their sensitivity by re-estimating the model with the two decay weightings of Section 3.5.a, with small workplaces defined at 10 and 50 employees, with the owner shares built at the two-digit industry level and with the three origin-specific shares replaced by the share of all migrant owners and the share of low-English owners. We also interact the owner shares with small sites of multi-site employers and with chain-like workplaces. These workplaces are rarely migrant-owned, so if the main interactions reflect migrant ownership and not industry pay structure, these additional interactions should be close to zero.

## 4.2 Owner Income and Risk

A shift in bargaining power towards workers should also leave a mark on owners, through lower income or greater exposure to income risk. A natural question is therefore whether migrant owners with employees, and those with limited English in particular, earn less or bear more risk than employees of the same origin. The returns to ownership differ sharply between owners with and without employees (Hamilton, 2000; Sorgner et al., 2017), and owners bear income risk as residual claimants (Audoly, 2025; Bhandari et al., 2026). We therefore study employers and own-account workers separately and examine the risk of income as well as its level.

Measuring risk requires income in consecutive years, so the unit of analysis is a pair of consecutive waves $t-1$ and $t$ (Ziliak et al., 2011). We examine three outcomes built from annual labour and business income $y_{it}$. The first is its level. The second is its volatility, measured by the absolute arc per cent change (Brewer et al., 2025),

$$| a_{it} | = 100 \times \frac{|y_{it} - y_{i,t-1}|}{(| y_{it} | + | y_{i,t-1} |)/2}. \tag{4.2}$$

The third is downside risk, an indicator for an income fall of at least 25 per cent among individuals with positive income at $t-1$, $\text{LargeFall}_{it} = \mathbf{1}[y_{it} \leq 0.75\, y_{i,t-1}]$ (Hacker et al., 2014; Rohde et al., 2015). The arc per cent change is scale-invariant, so it does not understate volatility among low-income groups. It is defined when income is zero in one year and, with absolute values in the denominator, stays between 0 and 200 even when business income is negative (Ziliak et al., 2011). Its mean is the summary of volatility recommended for subgroup comparisons, since it is less sensitive to extreme changes than variance-based measures (Brewer et al., 2025).

We estimate each outcome on the unrestricted and restricted samples of Section 3.1. HILDA measures income over the financial year ending on 30 June, while employment status is recorded at the interview a few months later. For people who changed status between interviews, income at $t$ therefore partly reflects their earlier status. The restricted sample, in which status is unchanged, avoids this problem, so we treat it as the cleaner estimate and use the comparison between the two samples to show how much of the gaps comes from recent changes of status. For each outcome $Y_{it} \in \{y_{it}, | a_{it} |, \text{LargeFall}_{it}\}$ and each sample, the model is specified as follows,

$$\begin{aligned} Y_{it} = {} & \delta_t + \mu_{kt} + \nu_{rt} \\ & + \theta_S \text{SmallEmployer}_{it} + \theta_L \text{LargeEmployer}_{it} + \theta_O \text{OwnAccountWorker}_{it} \\ & + \sum_{g \in \mathcal{G} \setminus \{\text{Native}\}} \gamma_g \mathbf{1}[G_i = g] \\ & + \sum_{g \in \mathcal{G} \setminus \{\text{Native}\}} \theta_{Sg} (\text{SmallEmployer}_{it} \times \mathbf{1}[G_i = g]) \\ & + \sum_{g \in \mathcal{G} \setminus \{\text{Native}\}} \theta_{Lg} (\text{LargeEmployer}_{it} \times \mathbf{1}[G_i = g]) \\ & + \sum_{g \in \mathcal{G} \setminus \{\text{Native}\}} \theta_{Og} (\text{OwnAccountWorker}_{it} \times \mathbf{1}[G_i = g]) \\ & + \mathbf{Z}_{it}' \boldsymbol{\zeta} + u_{it}, \end{aligned} \tag{4.3}$$

where the three mutually exclusive ownership indicators are

$$\begin{aligned} \text{SmallEmployer}_{it} &= \mathbb{1}[s_{it} = E, \text{Size}_{it} < 20], \\ \text{LargeEmployer}_{it} &= \mathbb{1}[s_{it} = E, \text{Size}_{it} \geq 20], \\ \text{OwnAccountWorker}_{it} &= \mathbb{1}[s_{it} = O, \text{Size}_{it} < 20], \end{aligned} \tag{4.4}$$

with $s_{it}$ the status defined in Section 3.4, $\text{Size}_{it}$ the number of people at the individual's workplace and employees as the reference category. The term $\mathbf{1}[G_i = g]$ equals one if individual $i$ belongs to migrant group $g$, with the native-born as the reference group, and $\delta_t$, $\mu_{kt}$ and $\nu_{rt}$ are year, industry-by-year and state-by-year fixed effects. The controls $\mathbf{Z}_{it}$ contain age and its square, sex, education, occupational tenure, marital status, remoteness, the SEIFA decile and the workplace types, so small

employers are compared with employees in small workplaces and large employers with employees in large workplaces. All regressors are measured at $t$, with workplace type, industry and occupational tenure taken from $t-1$ when missing at $t$. The volatility and downside-risk specifications also control for the individual's prior income decile, the within-wave decile of average income over waves $t-5$ to $t-2$, since volatility is concentrated at the bottom of the income distribution (Tin & Tran, 2023) and a low-earning group could otherwise appear more volatile solely because of where it sits. The decile is excluded from the income-level specification, where it would absorb the gap we aim to measure. We estimate the models by OLS without survey weights and cluster standard errors by individual and by industry-year.

The coefficients $\theta_S$, $\theta_L$ and $\theta_O$ compare each type of owner with comparable employees among the native-born, and $\gamma_g$ compares migrant group $g$ with the native-born among employees. For small employers from group $g$, the sum $\theta_S + \theta_{Sg}$ compares them with employees of the same origin, while the interaction $\theta_{Sg}$, which we term the ownership premium gap, measures whether this difference departs from the native-born difference. The same holds for large employers and own-account workers. The distinction matters since some migrant groups already earn less than the native-born as employees, so a disadvantage can be attributed to ownership itself only if it exceeds this existing gap.

The power-reversal hypothesis makes three predictions. First, low-English small employers should have a negative premium gap in income, $\theta_{Sg} < 0$, reflecting lower returns to ownership. Second, they should have positive premium gaps in volatility and downside risk, reflecting greater exposure to risk. Third, these gaps should be concentrated among small employers and close to zero for MES owners. We also test jointly whether all origin by owner-type interactions are zero.

Three features of the data matter for interpretation. First, low-English owners are few, so alongside each premium gap we report its minimum detectable effect, $\text{MDE} = 2.8 \times \widehat{\text{SE}}$, the smallest true difference that a two-sided 5 per cent test detects with 80 per cent power (Bloom, 1995). An insignificant gap with a large MDE is uninformative, not evidence of no difference. Owner by origin cells with fewer than five individuals are excluded, which removes the single low-English large employer. Second, owners who are not re-interviewed, and owners whose business closes between $t-1$ and $t$, drop out of the owner comparison, so neither sample measures the income consequences of leaving ownership. Third, workplace size is partly an outcome of business success, so we treat the comparison of small and large employers as heterogeneity, not as an effect of size. The self-employed also under-report income (Hurst et al., 2014), which is a further reason to examine income risk alongside its level.

## 4.3 Entry into Business Ownership

The wage and income comparisons take ownership as given, so we finally ask who becomes an owner. A high rate of ownership among migrants is not in itself evidence of entrepreneurial opportunity, since people may also enter ownership when their alternatives in paid employment are poor. Motives for entry are rarely observed, so we follow Fairlie & Fossen (2020) and classify entry by labour market

position before entry, using the transition sample of Section 3.1, in which each person-pair is one year at risk of entry (Evans & Leighton, 1989; Fairlie, 1999). The model is specified as follows,

$$\begin{aligned}\Pr(\text{Entry}_{it}=1) = \Lambda\Big(&\delta_t + \mu_{kt} + \nu_{rt} + \sum_{g\in\mathcal{G}\setminus\{\text{Native}\}} \phi_g \mathbf{1}[G_i = g] \\ &+\phi_1 \text{Unemployed}_{i,t-1} + \phi_2 \text{NotInLabourForce}_{i,t-1} \\ &+\phi_3 \text{LowWage}_{i,t-1} + \phi_4 \text{LiquidityConstraint}_{i,t-1} \\ &+\mathbf{X}'_{i,t-1}\boldsymbol{\kappa}\Big),\end{aligned} \tag{4.5}$$

where $\Lambda$ is the logistic function and the indicators are defined in Section 3.4. The omitted category is an employee who is not low-paid at $t-1$, so each labour market coefficient compares a weaker position with stable, reasonably paid employment. The terms $\delta_t$, $\mu_{kt}$ and $\nu_{rt}$ are year, industry-by-year and state-by-year fixed effects, where industry $k$ is that of the individual's job at $t-1$ or, for those not employed, their most recent job. The controls $\mathbf{X}_{i,t-1}$ contain age and its square, sex, years in Australia with an indicator for missing arrival year, marital status, home ownership, the SEIFA decile, remoteness and an indicator for a missing liquidity measure. We add the regressors in four steps (origin alone, then labour market position and liquidity alone, then both, then both with education and employer tenure at $t-1$), so that the change in the origin coefficients $\phi_g$ shows how far origin differences in entry reflect labour market position and human capital. Standard errors are clustered by individual.

The logit is the standard model of transitions into self-employment (Evans & Leighton, 1989; Fairlie, 1999) and suits an outcome that occurs in about 2 per cent of person-years. Coefficients are log-odds, which we report as odds ratios $\exp(\beta)$. Each fixed-effect cell contains many individuals, so the logit does not suffer from the incidental parameters problem (Neyman & Scott, 1948). As a robustness check, we re-estimate every specification as a linear probability model on the same person-pairs. Its coefficients are percentage-point changes in the probability of entry (Angrist & Pischke, 2009), and its interactions are measured on the probability scale, whereas a logit interaction is a ratio of odds ratios whose implication for probabilities depends on each group's baseline entry rate (Ai & Norton, 2003). Baseline rates differ between low-English migrants and the native-born, so comparing the two models shows whether a larger push is larger in relative terms, in absolute terms or both.

Necessity entry implies that $\phi_1$ and $\phi_3$ are positive, and the framework implies that this push is strongest for migrants with limited English. To test whether the push differs by origin, we add interactions of the origin groups with unemployment and low wages to the final step,

$$\begin{aligned}\Pr(\text{Entry}_{it}=1) = \Lambda\Big(&\cdots + \sum_{g\in\mathcal{G}\setminus\{\text{Native}\}} \theta_g^U \big(\text{Unemployed}_{i,t-1} \times \mathbf{1}[G_i = g]\big) \\ &+ \sum_{g\in\mathcal{G}\setminus\{\text{Native}\}} \theta_g^W \big(\text{LowWage}_{i,t-1} \times \mathbf{1}[G_i = g]\big)\Big),\end{aligned} \tag{4.6}$$

where the omitted terms are those of Equation (4.5). Unemployment multiplies the odds of entry by $\exp(\phi_1)$ for the native-born and by $\exp(\phi_1 + \theta_g^U)$ for group $g$, so $\exp(\theta_g^U)$, the ratio of the two odds

ratios, measures how many times stronger the push from unemployment is for group $g$ than for the native-born. The push from low wages is interpreted in the same way, with $\theta_g^W$ in place of $\theta_g^U$. Necessity entry predicts $\theta_g^U > 0$ and $\theta_g^W > 0$, largest for low-English migrants.

The estimates rely on a no-anticipation assumption, under which labour market status in the year before entry is not itself a response to a decision to start a business. The assumption would fail if people left paid work to prepare a business they had already decided to start. We assess this by re-estimating Equation (4.6) with labour market status measured at $t-1$, at $t-2$ and at both, all on individuals observed at $t-2$, so that differences reflect timing and not sample composition.

The push coefficients could also reflect stable traits, such as ability, appetite for risk or family wealth, that affect both how often someone is unemployed or low-paid and how likely they are to start a business. Individual fixed effects would absorb the origin groups, which do not change over time, and a fixed-effects logit (Chamberlain, 1980) would discard the 85 per cent of individuals who never enter. We therefore add each individual's mean of the time-varying regressors (unemployment, low wages, non-participation, liquidity, home ownership and the SEIFA decile), the correlated random effects approach of Mundlak (1978) and Wooldridge (2010). The coefficients on the time-varying regressors then compare an individual with themselves at other times, as fixed effects would, while the origin groups remain identified. This distinguishes a push that occurs when a person becomes unemployed or low-paid from one that reflects which kinds of people tend to be unemployed or low-paid. As a second check, we estimate the heterogeneity specification as a linear probability model with individual fixed effects, which keeps individuals who never enter and identifies the origin interactions from changes in labour market status over time. Both within-person approaches require strict exogeneity, so past entry must not affect later unemployment or wages. This is doubtful in a transition sample, since an owner who exits rejoins the sample with a labour market history shaped by their business. We therefore treat the within-person estimates as a check on the role of stable traits, not as a preferred specification.

# 5 Results

## 5.1 Employee Wages

### Main Results

**Table 5.1:** *Main regression results for employee wages.*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | **No decay (Type 0, main)** | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| *Firm structure* | | | |
| Small single-site | -0.113*** | -0.104*** | -0.114*** |
| | (0.018) | (0.016) | (0.016) |
| Chain-like | 0.060*** | 0.060*** | 0.060*** |
| | (0.005) | (0.005) | (0.005) |
| Small-site chain | -0.030*** | -0.030*** | -0.030*** |
| | (0.005) | (0.005) | (0.005) |
| *Interactions of firm structure with owner density* | | | |
| Small single-site x MES owner density | -0.330*** | -0.595*** | -0.862*** |
| | (0.092) | (0.142) | (0.199) |
| Small single-site x MNES high-English owner density | 0.277*** | 0.379*** | 0.266*** |
| | (0.058) | (0.090) | (0.058) |
| Small single-site x MNES low-English owner density | -0.182 | -0.263 | -0.171 |
| | (0.173) | (0.206) | (0.175) |
| Observations (person-years) | 156,786 | 156,786 | 156,786 |
| R-Squared | 0.490 | 0.490 | 0.490 |
| Adjusted R-Squared | 0.488 | 0.488 | 0.488 |
| Individuals | 23,067 | 23,067 | 23,067 |
| Controls | Yes | Yes | Yes |
| Year FEs | Yes | Yes | Yes |
| Industry-year FEs | Yes | Yes | Yes |
| State-year FEs | Yes | Yes | Yes |
| F-Statistic (df) | 14.602 (3, 397) | 14.923 (3, 397) | 16.674 (3, 397) |
| F-test p-value | < 0.001 | < 0.001 | < 0.001 |

*Notes:* This table reports OLS estimates of Equation (4.1) for employees in HILDA waves 1 to 21. The dependent variable is the log hourly wage in the employee's main job. Each column represents a different weighting of the leave-one-out owner shares of Equation (3.7), namely no decay (Type 0, $\lambda = 0$), which is our main specification, pooled decay (Type 1, $\lambda = 0.02$) and origin-specific decay (Type 2, $\lambda_{\text{MES}} = 0.04$ and $\lambda_{\text{MNES}} = 0$). Workplace types are compared with other workplaces of 20 or more people, and the owner shares run from 0 to 1. All columns include covariates for age, sex, origin group, casual or permanent employment, occupation, education, SEIFA decile, occupational tenure, employer tenure and union membership, and year, industry-year and state-year fixed effects. The F-statistic, with its degrees of freedom in parentheses, tests the null hypothesis that the three owner-share interactions are jointly zero, using the clustered covariance matrix; its p-value is reported in the row below. All standard errors reported in parentheses are clustered at the individual and industry-year levels. Full results are reported in Table B.5. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table 5.1 reports OLS estimates of Equation (4.1) on 156,786 person-years from 23,067 employees. The first column is our main specification, which uses the owner shares without decay (Type 0). The second and third columns apply the pooled and origin-specific decay weightings of Section 3.5.a and are reported for comparison.

Employees of small single-site workplaces earn 10.7 per cent less than employees of other large workplaces in the same industry and year ($\beta_1 = -0.113$),[5] while employees of chain-like workplaces earn about 6 per cent more ($\beta_3 = 0.060$). Small sites of multi-site employers fall in between ($\beta_2 = -0.030$).

The small single-site wage gap narrows where owners from non-English-speaking countries with high English proficiency are more common, but not where owners with limited English are common. The interaction with the high-English MNES share is $\beta_5 = 0.277$ ($p < 0.001$), so a one-standard-deviation increase in that share (0.104) narrows the gap by 2.9 percentage points. The interaction with the low-English MNES share is $\beta_6 = -0.182$ ($p = 0.295$), negative and far from significant. The three interactions are jointly significant ($p < 0.001$).

To gauge the size of these interactions, we compute the predicted small single-site wage gap at different owner shares. Where there are no non-English-speaking migrant owners, employees of small single-site workplaces earn 10.7 per cent less than employees of large workplaces in the same industry and year. At the 95th percentile of the high-English owner share (34.7 per cent of owners), the penalty narrows to 1.7 per cent. At the 95th percentile of the low-English owner share (6.9 per cent of owners), it is 11.8 per cent, close to the penalty where there are no such owners. Working in a small independent business therefore costs employees less in industries where more such businesses are run by non-English-speaking migrants who speak English well, but not where more are run by owners with limited English. These gaps compare employees of different workplace types within the same industry and year. They are not gaps between owners and employees, nor between migrant and native-born employees.

The MES share does not behave like the native-born benchmark we expected. Its interaction is $\beta_4 = -0.330$ ($p < 0.001$), so a higher share of MES owners is associated with a *wider* small single-site gap. In terms of the explanations set out in Section 4.1, the positive high-English interaction is consistent with better job matching or with power reversal, but the absence of any narrowing where owners have limited English does not show the concentration of the effect among low-English owners that power reversal predicts.

**Robustness and Sensitivity Checks**

The wage estimates rest on several measurement choices and, for some origin groups, on very few owners. We assess how sensitive they are to each choice in turn; the corresponding tables are in Appendix B.

*Construction of migrant owner density.* HILDA observes few MNES owners. Only 47 low-English owners appear in waves 1 to 21, and the low-English share is zero for 64.1 per cent of employees, so the share in any industry-year rests on a handful of owners and depends on how each of them is weighted. We therefore vary the weighting in two ways. The decay weightings in the second and third columns of Table 5.1 down-weight long-settled migrant owners, and Table B.10 builds the shares with the longitudinal instead of the cross-sectional survey weight. The high-English interaction remains positive

[5] The percentage difference is $100 \times (e^{\beta} - 1)$. For example, $\beta = -0.113$ corresponds to a 10.7 per cent lower wage relative to the reference category.

and significant under every decay weighting. The low-English interaction is far less stable. Under the longitudinal weight, which is positive for only 14 of the 47 low-English owners, it becomes positive (0.456, $p < 0.001$). The choice of weighting therefore matters for the low-English estimate, which we treat as uninformative. Sampling error in a share that rests on few owners also acts like measurement error in a regressor and biases its interaction towards zero, which reinforces this conclusion.

*Workplace size threshold.* The 20-person cut-off follows the ABS definition of a small business, but the businesses that migrants own may be smaller or larger than this. We redefine small workplaces at 10 and 50 people (Table B.2). The high-English interaction is 0.269 ($p < 0.001$) and 0.255 ($p < 0.001$) under the two thresholds, and the low-English interaction remains negative and insignificant ($-0.272$ and $-0.199$). The results do not depend on the size threshold.

*Industry classification.* The 19 ANZSIC divisions are broad, and migrant owners may concentrate in narrower industries within them. We therefore construct the owner shares and the industry-by-year fixed effects at the two-digit level (Table B.8). The pattern weakens. The high-English interaction falls to 0.048 ($p = 0.081$) and the low-English interaction is $-0.110$ ($p = 0.163$), although the three interactions remain jointly significant. Two-digit shares rest on even fewer owners per cell, so the weaker estimates are consistent with greater attenuation as well as with a weaker underlying relationship.

*Migrant ownership versus English proficiency.* The three origin-specific shares are correlated, which makes it hard to separate migrant ownership from owners' English. We therefore replace them with the share of all migrant owners and the share of low-English owners (Table B.7), so that the low-English coefficient measures any effect of limited English over and above migrant ownership. The migrant share has a coefficient of 0.156 ($p = 0.002$), while the low-English share has a coefficient of $-0.298$ ($p = 0.126$). The narrowing is therefore associated with migrant ownership in general, not with owners' limited English.

*Chain workplaces.* If the interactions reflect migrant ownership, they should appear mainly for workplaces that migrants are likely to own. Small sites of multi-site employers share the local scale of small independent workplaces but belong to larger, mostly native-run employers, so they allow us to check whether the main interactions instead reflect the pay structure of industries with many migrant owners. We add interactions of each owner share with small-site chains and with chain-like workplaces (Table B.3). The small-site chain interactions are jointly significant ($p = 0.014$), mainly because small sites of chains also pay relatively more where high-English MNES owners are common (0.141, $p = 0.002$), and the chain-like interactions are also jointly significant ($p < 0.001$). Once these terms are included, the small single-site interactions with the high-English and low-English shares are 0.280 ($p < 0.001$) and $-0.033$ ($p = 0.841$). The high-English interaction for small independent workplaces is about twice that for small sites of chains, so the narrowing is strongest where migrants are most likely to be owners, but it is not confined to those workplaces.

Figure 5.1 shows what this extended model implies for every workplace type. Where no non-English-speaking migrants own businesses, employees of small single-site workplaces earn 11.9 per cent less

than employees of other large workplaces, employees of small sites of chains 6.5 per cent less and employees of chain-like workplaces 5.6 per cent more. As the share of high-English MNES owners rises to its 95th percentile, these differences become −2.9, −1.8, +2.8 per cent. The wage distribution across workplace types therefore compresses in industries with many high-English MNES owners. The compression is largest for small independent workplaces, but part of the main pattern reflects the pay structure of these industries and not ownership alone.

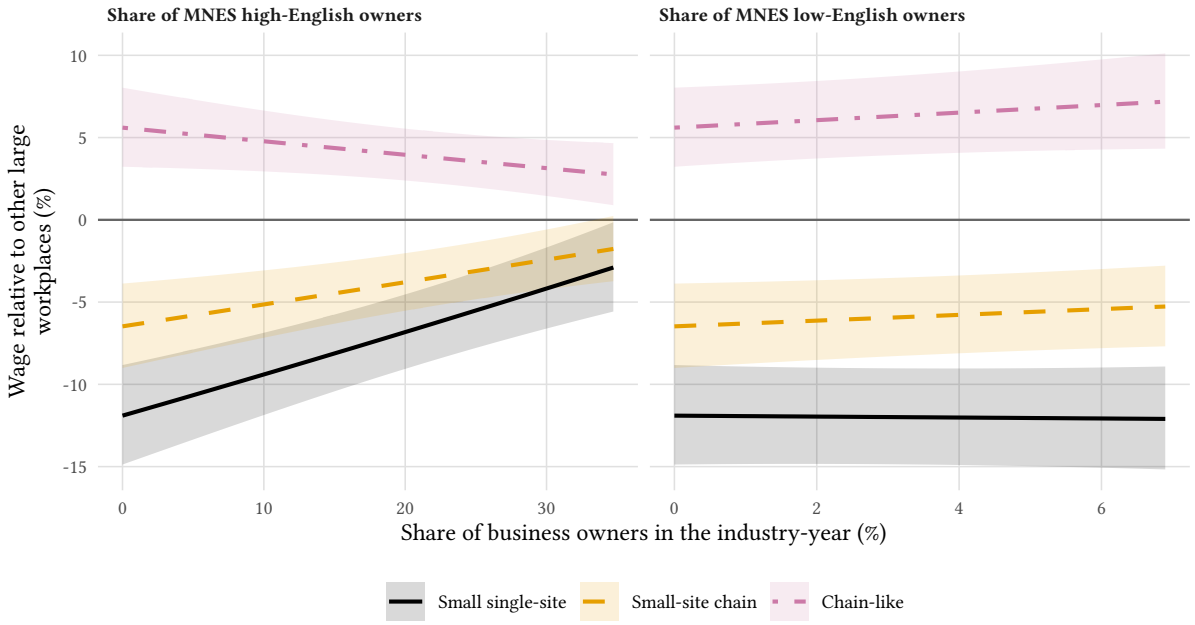



**Figure 5.1:** *Wage gaps of each workplace type relative to other large workplaces in the same industry and year, as the share of non-English-speaking migrant owners rises.*

*Notes:* This figure plots the predicted percentage difference in hourly wages between employees of each workplace type and employees of other large workplaces in the same industry and year, from the Type 0 column of Table B.3. The left panel varies the share of MNES owners with high English proficiency and the right panel the share with low English proficiency, each from zero to its 95th percentile, holding the other shares at zero. Shaded bands are 95 per cent confidence intervals based on standard errors clustered at the individual and industry-year levels.

## 5.2 Owner Income and Risk

If workers in these businesses hold more bargaining power, owners should bear part of the cost through lower income or greater income risk. We therefore compare owners with employees of the same origin, in the unrestricted sample (21,549 individuals) and the restricted sample (20,031 individuals). Low-English owners are few, with 13 low-English small employers and 13 low-English own-account workers in the restricted sample (Table C.1).

### Main Results

We report the results for small employers in two steps. Table 5.2 reports the *owner–employee gap*, the difference between small employers and employees *of the same origin*, which is $\theta_S$ for the native-born and $\theta_S + \theta_{Sg}$ for migrant group $g$ in Equation (4.3). Table 5.3 then reports whether each migrant group's gap differs from the native-born gap. This difference is the premium gap $\theta_{Sg}$ that the power-reversal

hypothesis predicts to be negative for income and positive for risk among low-English owners, and it is reported together with its minimum detectable difference.

To take one example, in the unrestricted income column of Table 5.2, native-born small employers earn $12,730 more than native-born employees, and low-English small employers earn $24,541 more than low-English employees. The difference between these two gaps, $11,810, is the first low-English entry in Panel A of Table 5.3. Its minimum detectable difference of $29,863 in Panel B means that, with 21 low-English small employers, a true difference of up to about that size in either direction could go undetected.

**Table 5.2:** *Owner–employee gaps in income and income risk for small employers.*

| | **Unrestricted sample: status at t** | | | **Restricted sample: same status at t−1 and t** | | | |
|---|---|---|---|---|---|---|---|
| Dependent variable: | Income (dollars) | Volatility (pp) | Large fall (pp) | Income (dollars) | Volatility (pp) | Large fall (pp) | Small employers (restricted sample) |
| Native-born | 12,730*** (3,136) | 27.3*** (1.2) | 15.1*** (0.7) | 14,369*** (4,168) | 32.7*** (1.4) | 17.3*** (0.8) | 1,103 |
| MES | 9,263* (5,325) | 25.4*** (2.8) | 15.4*** (1.6) | 11,665* (6,310) | 29.1*** (3.3) | 17.2*** (1.8) | 151 |
| MNES high-English | 17,217*** (4,893) | 21.8*** (2.6) | 12.3*** (1.3) | 20,298*** (5,329) | 25.8*** (3.3) | 13.8*** (1.6) | 193 |
| MNES low-English | 24,541** (10,275) | 39.1*** (12.2) | 19.5*** (7.0) | 19,169*** (7,121) | 44.3*** (14.5) | 20.7** (8.5) | 13 |
| Observations (person-pairs) | 174,962 | 174,962 | 164,973 | 152,060 | 152,060 | 148,198 | |
| Individuals | 21,549 | 21,549 | 20,618 | 20,031 | 20,031 | 19,747 | |

*Notes:* This table reports the owner–employee gap for small employers from Equation (4.3), that is, the difference between small employers (owners with employees and fewer than 20 people at their workplace) and employees of the same origin, which is $\theta_S$ for the native-born and $\theta_S + \theta_{Sg}$ for migrant group $g$. The dependent variables are annual labour and business income in dollars, income volatility (the absolute arc per cent change of Equation (4.2)) and the probability of an income fall of 25 per cent or more, the last two in percentage points. Each column represents a separate regression. The first three columns use the unrestricted sample, which classifies individuals by their status at $t$, and the last three use the restricted sample of individuals with the same status at $t-1$ and $t$. All columns include covariates for age and its square, sex, education, occupational tenure, marital status, remoteness, SEIFA decile and workplace type, and year, industry-year and state-year fixed effects; the volatility and large-fall columns also control for prior income decile. The last column is the number of distinct small employers in each group in the restricted income regression. All standard errors reported in parentheses are clustered at the individual and industry-year levels. Full results are reported in Table C.4. * p < 0.10, ** p < 0.05, *** p < 0.01.

**Table 5.3:** *Difference between each migrant group's owner–employee gap and the native-born gap, for small employers.*

| | Unrestricted sample: status at t | | | Restricted sample: same status at t–1 and t | | |
|---|---|---|---|---|---|---|
| Dependent variable: | Income (dollars) | Volatility (pp) | Large fall (pp) | Income (dollars) | Volatility (pp) | Large fall (pp) |
| *Panel A: Difference from the native-born owner–employee gap* | | | | | | |
| MES | −3,468 | −1.9 | 0.4 | −2,704 | −3.5 | −0.0 |
| | (6,054) | (3.0) | (1.8) | (7,407) | (3.6) | (2.0) |
| MNES high-English | 4,487 | −5.5* | −2.8** | 5,929 | −6.9* | −3.4** |
| | (5,773) | (2.8) | (1.4) | (6,622) | (3.6) | (1.7) |
| MNES low-English | 11,810 | 11.8 | 4.4 | 4,800 | 11.6 | 3.4 |
| | (10,666) | (12.3) | (7.1) | (8,305) | (14.5) | (8.5) |
| *Panel B: Minimum detectable difference* | | | | | | |
| MES | 16,952 | 8.5 | 4.9 | 20,740 | 10.1 | 5.5 |
| MNES high-English | 16,163 | 7.9 | 3.9 | 18,542 | 10.0 | 4.6 |
| MNES low-English | 29,863 | 34.4 | 19.8 | 23,255 | 40.7 | 23.7 |
| Joint test of all interactions (p-value) | 0.737 | 0.182 | 0.180 | 0.837 | 0.055 | 0.106 |
| Observations (person-pairs) | 174,962 | 174,962 | 164,973 | 152,060 | 152,060 | 148,198 |
| Individuals | 21,549 | 21,549 | 20,618 | 20,031 | 20,031 | 19,747 |

*Notes:* This table reports, for small employers, the difference between each migrant group's owner–employee gap in Table 5.2 and the native-born gap, which is the interaction $\theta_{Sg}$ of Equation (4.3). Each column represents a separate regression, with the dependent variables, samples, covariates and fixed effects of Table 5.2. Panel A reports the difference. Panel B reports its minimum detectable difference, $2.8 \times \text{SE}(\theta_{Sg})$, the smallest true difference that a two-sided test at the 5 per cent level would detect with 80 per cent probability. The joint test reports the p-value of an F-test of the null hypothesis that all eight origin by owner-type interactions in the regression are jointly zero, including those for large employers and own-account workers. All standard errors reported in parentheses are clustered at the individual and industry-year levels. Full results are reported in Table C.4. * p < 0.10, ** p < 0.05, *** p < 0.01.

Among the native-born, running a small business with employees involves a clear trade-off between income and risk. In the unrestricted sample, small employers earn $12,730 more per year than employees in small workplaces ($p < 0.001$), but the average year-to-year change in their income is 27.3 percentage points larger and their probability of an income fall of 25 per cent or more is 15.1 points higher.

The same trade-off holds for every migrant group. Small employers earn more than employees of their own origin, and in every group they face much higher volatility and downside risk. In the unrestricted sample the income gap is $17,217 for high-English migrants ($p < 0.001$) and $24,541 for low-English migrants ($p = 0.017$), while for MES migrants it is smaller and only marginally significant ($9,263, $p = 0.083$). Ownership with employees therefore looks like compensated risk for all groups, not the uncompensated risk that the power-reversal hypothesis predicts for owners with limited English.

None of the differences from the native-born gap for low-English small employers is significant (Table 5.3). Their risk gaps point in the predicted direction, with volatility 11.8 percentage points higher and the probability of a large fall 4.4 points higher than the native-born comparison would imply, but the income premium gap is positive ($11,810), not negative. With 21 low-English small employers in the unrestricted sample and 13 in the restricted sample, these estimates are very imprecise. The minimum detectable differences are $29,863 for income, 34.4 percentage points for volatility and 19.8

percentage points for downside risk, so the data cannot rule out large differences in either direction. High-English small employers, if anything, face less volatility than the native-born comparison would imply (−5.5 points, $p = 0.054$), and MES small employers are close to their native-born counterparts on every outcome.

**Heterogeneity by Owner Type**

Table C.2 and Table C.3 report the same comparisons for large employers and own-account workers. Large migrant employers are too few for firm conclusions. Their income estimates are dominated by a handful of very high incomes, with a standard error of $126,299 for the difference between MES and native-born large employers, so the data cannot show whether migrant-specific gaps are larger in small than in large businesses. Own-account workers earn less than employees of their own origin in every group, although not significantly so for every migrant group (−$9,379 among the native-born in the unrestricted sample), and face even more volatility than small employers. The joint test that all origin by owner-type interactions are zero is marginally significant for volatility in the restricted sample ($p = 0.055$), but not in the unrestricted sample ($p = 0.182$), nor for income or downside risk in either sample. As Table C.3 shows, the restricted-sample result comes mainly from own-account workers.

Low-English own-account workers stand out. In the restricted sample, their income appears much less volatile than the native-born comparison implies (−19.1 percentage points, $p = 0.056$). In the unrestricted sample, which also includes those who became own-account workers between the two years, the difference is smaller and not significant (−10.7 points, $p = 0.300$). The apparent protection from risk is therefore concentrated among the 13 low-English own-account workers who stay in that status in both years.

**Robustness and Sensitivity Checks**

*Changes of status between waves.* HILDA measures income over the financial year before the interview, so for people who changed status between interviews, income at $t$ partly reflects their earlier status. We therefore estimate every model on both samples. Restricting the sample to individuals whose status is unchanged gives larger native-born gaps, of $14,369 in income, 32.7 points in volatility and 17.3 points in downside risk, but the same pattern across origin groups. The comparison between owners and employees is therefore not driven by people who changed status between the two years.

*Attrition.* Owners who are not re-interviewed at $t$ drop out of both samples, which would bias the comparisons if attrition differed by origin. Attrition is similar across groups, at 5 to 8 per cent a year among small employers (Table C.1), so it is unlikely to explain the results. Neither sample, however, measures the income consequences of a business closing.

*Hours of work.* Annual income does not account for the longer hours that owners typically work. We therefore compare the median hourly income of owners with employees with the median hourly wage of employees of small independent workplaces, by industry and without controls (Section E). Owners earn less per hour than employees in several low-margin industries, such as accommodation and food

services, so the income premium in Table 5.2 does not hold in every industry once hours are taken into account. The full regression estimates, including all controls, are in Table C.4.

## 5.3 Entry into Business Ownership

Ownership with employees pays more than paid employment but brings more volatile income, which raises the question of who takes on that risk. Unconditional entry rates are 1.99 per cent a year for the native-born, 2.22 per cent for MES migrants, 2.13 per cent for high-English MNES migrants and 1.11 per cent for low-English MNES migrants (Table D.1). The low-English group contributes only 3,332 at-risk person-pairs and 37 entries, partly because its weak panel persistence (Table A.6) excludes many of its members from a sample that requires consecutive observations.

### Main Results

**Table 5.4:** *Entry into business ownership between t–1 and t (logit coefficients).*

| | Dependent variable: entry into business ownership (logit) | | | |
|---|---|---|---|---|
| | Origin only | Labour market only | Origin and labour market | Adding human capital |
| *Origin and English proficiency* | | | | |
| Migrant with English-speaking origin | 0.038<br>(0.087) | | 0.075<br>(0.087) | -0.022<br>(0.088) |
| Migrant from other origin with high English proficiency | 0.053<br>(0.084) | | 0.066<br>(0.083) | -0.059<br>(0.084) |
| Migrant from other origin with low English proficiency | -0.098<br>(0.207) | | -0.230<br>(0.208) | -0.256<br>(0.207) |
| *Labour-market position at t–1* | | | | |
| Unemployed | | 0.777***<br>(0.067) | 0.777***<br>(0.067) | 0.700***<br>(0.068) |
| Not in labour force | | 0.588***<br>(0.048) | 0.590***<br>(0.048) | 0.491***<br>(0.052) |
| Bottom wage quintile | | 0.406***<br>(0.047) | 0.408***<br>(0.047) | 0.430***<br>(0.048) |
| Liquidity constrained | | 0.151***<br>(0.038) | 0.152***<br>(0.038) | 0.149***<br>(0.038) |
| Observations (person-pairs) | 289,616 | 289,616 | 289,616 | 289,616 |
| AIC | 54352 | 53974 | 53976 | 53786 |
| BIC | 61460 | 61102 | 61136 | 61020 |
| Individuals | 28,626 | 28,626 | 28,626 | 28,626 |
| Year FEs | Yes | Yes | Yes | Yes |
| Industry-year FEs | Yes | Yes | Yes | Yes |
| State-year FEs | Yes | Yes | Yes | Yes |
| Education and employer tenure | No | No | No | Yes |
| Wald $\chi^2$ (p-value) | | 268.05 (< 0.001) | 268.83 (< 0.001) | 197.35 (< 0.001) |

*Notes:* This table reports logit estimates of Equation (4.5) for person-pairs in which the individual is not a business owner at $t-1$. The dependent variable equals one if the individual is an own-account worker or employer at $t$. Each column adds regressors in steps, starting with origin group only, then labour-market position and liquidity at $t-1$ only, then both and finally both with education and employer tenure at $t-1$. All columns include covariates for age and its square, sex, years in Australia, marital status, home ownership, SEIFA decile and remoteness, and year, industry-year and state-year fixed effects; columns with the liquidity constraint also include an indicator for a missing liquidity measure. Coefficients are log-odds, and the reference

categories are the Australian-born and employees who were not low-paid and not liquidity constrained at $t-1$. The Wald $\chi^2$ statistic, with its p-value in parentheses, tests the null hypothesis that the four labour-market and liquidity coefficients are jointly zero. All standard errors reported in parentheses are clustered at the individual level. Full results are reported in Table D.2. * p < 0.10, ** p < 0.05, *** p < 0.01.

Table 5.4 reports logit estimates of Equation (4.5), adding regressors in steps across the four columns. Weak labour market positions are the strongest predictors of entry.[6] Relative to employees who are not low-paid, unemployed individuals have 2.18 times the odds of entering ownership ($\beta = 0.777$ in the *Origin and labour market* column), those outside the labour force 1.80 times and employees in the bottom wage quintile 1.50 times. Liquidity-constrained individuals are modestly more likely to enter (odds ratio 1.16). These coefficients change little across the last three columns, and adding education and employer tenure lowers the unemployment coefficient only from 0.777 to 0.700. None of the origin coefficients is significant in any column. The low-English coefficient is the largest in magnitude (-0.256 in the last column), but its standard error of 0.21 means that the data cannot distinguish large differences from none. Entry also rises with qualifications and falls with tenure with the current employer (Table D.2), which suggests that push and pull motives operate together.

These positions can explain migrant entry only if migrants are more likely to be in them, so Table D.5 compares the origin groups among people at risk of entry. Holding age, sex and year fixed, low-English migrants are more likely than the native-born to be unemployed (by 1.2 percentage points, $p = 0.010$), outside the labour force (by 24.7 points, $p < 0.001$) and in the bottom wage quintile (by 2.2 points, $p = 0.040$), and somewhat more likely to be liquidity constrained (by 3.8 points, $p = 0.077$). The same pattern appears among those who do enter. Among low-English entrants, 13.5 per cent were unemployed in the year before entry and 24.3 per cent were in the bottom wage quintile, compared with 5.2 and 14.5 per cent of native-born entrants, although these shares rest on only 37 low-English entries.

**Heterogeneity by Origin**

Table 5.5 allows the push from unemployment and low wages to differ by origin (Equation (4.6)). For the native-born, unemployment raises the odds of entry by a factor of 1.87 and low wages by 1.44. The push is at least as strong for every migrant group. For low-English migrants, unemployment raises the odds of entry by a factor of 7.87, which is 4.21 times the native-born push ($p = 0.005$). For MES migrants, low wages raise the odds by a factor of 2.26, 1.57 times the native-born push ($p = 0.002$), and for high-English migrants the push from unemployment is 1.40 times the native-born push ($p = 0.061$). The six interactions are jointly significant ($p = 0.001$). The low-English estimate is large but rests on only 5 entries among unemployed low-English migrants, so its confidence interval is wide.

[6] The odds ratio is $\exp(\beta)$, the multiplicative change in the odds of entry for the given category relative to the reference category. For example, $\exp(0.777) = 2.18$, so the odds of entry are 118 per cent higher for the unemployed than for employees.

**Table 5.5:** *Push into business ownership from unemployment and low wages, by origin group.*

| Origin group | Dependent variable: entry into business ownership (logit) | | | | | |
|---|---|---|---|---|---|---|
| | Unemployed at t−1 | | Bottom wage quintile at t−1 | | | |
| | Odds ratio | Relative to native-born | Odds ratio | Relative to native-born | Odds ratio when stably employed | Entries among the unemployed |
| Native-born | 1.87*** | Ref. | 1.44*** | Ref. | Ref. | 240 |
| | [1.60, 2.18] | | [1.30, 1.60] | | | |
| MES | 2.23*** | 1.19 | 2.26*** | 1.57*** | 0.90 | 31 |
| | [1.49, 3.33] | [0.78, 1.82] | [1.72, 2.96] | [1.18, 2.08] | [0.75, 1.09] | |
| MNES high-English | 2.61*** | 1.40* | 1.78*** | 1.23 | 0.88 | 49 |
| | [1.89, 3.60] | [0.98, 1.98] | [1.38, 2.30] | [0.94, 1.61] | [0.74, 1.06] | |
| MNES low-English | 7.87*** | 4.21*** | 2.89** | 2.00 | 0.59** | 5 |
| | [2.93, 21.11] | [1.56, 11.38] | [1.18, 7.04] | [0.82, 4.90] | [0.36, 0.98] | |
| Observations (person-pairs) | 289,616 | | | | | |
| Individuals | 28,626 | | | | | |
| Joint test of interactions (p-value) | 0.001 | | | | | |

*Notes:* This table reports odds ratios from the logit estimation of Equation (4.6), which interacts origin group with unemployment and the bottom wage quintile at $t-1$. The dependent variable equals one if an individual who is not a business owner at $t-1$ is an own-account worker or employer at $t$. *Odds ratio* is the push into ownership for each origin group relative to employees who are not low-paid, $\exp(\phi_1 + \theta_g^U)$ for unemployment and $\exp(\phi_3 + \theta_g^W)$ for low wages. *Relative to native-born* is $\exp(\theta_g^U)$ or $\exp(\theta_g^W)$, the ratio of the group's push to the native-born push. *Odds ratio when stably employed* is $\exp(\phi_g)$, the odds of entry relative to the native-born among employees who are neither unemployed nor low-paid at $t-1$. *Entries among the unemployed* is the number of entries on which each unemployment estimate rests. The model includes the covariates and fixed effects of the *Adding human capital* column of Table 5.4. The joint test reports the p-value of a Wald test of the null hypothesis that the six origin by labour-market interactions are jointly zero. All 95 per cent confidence intervals reported in brackets are based on standard errors clustered at the individual level. Stars test the null that the odds ratio equals one. * p < 0.10, ** p < 0.05, *** p < 0.01.

Among individuals who were employed and not low-paid at $t-1$, low-English migrants are less likely than the native-born to enter (odds ratio 0.59, $p = 0.043$). The low unconditional entry rate of this group therefore reflects low entry among those in stable employment, alongside a strong push among the unemployed.

### Robustness and Sensitivity Checks

*Timing of labour market status.* A push from unemployment could run in reverse if people left work to prepare a business they had already decided to start, in which case status at $t-1$ would partly reflect the entry decision. Table D.3 therefore re-estimates the heterogeneity model with labour market status measured at $t-1$, at $t-2$ and at both, on the 256,524 person-pairs for which status two years before entry is observed. For the native-born, the push from unemployment is confined to the year before entry. Measured at $t-2$, unemployment does not predict entry (coefficient −0.027), and when status at both $t-1$ and $t-2$ is included, the $t-1$ coefficient is 0.834. Native-born entry therefore follows recent labour market weakness, which is consistent either with recent job loss pushing people into ownership or with anticipation of entry, and the data cannot separate the two. For low-English migrants, unemployment two years before entry also predicts entry strongly (interaction 1.697, $p < 0.001$). Their push reflects a persistent condition, not a spell immediately before entry, which is harder to explain by anticipation.

*Stable personal traits.* Without individual fixed effects, the push could reflect stable traits that affect both labour market position and entry. The last column of Table D.3 adds each individual's mean of the time-varying regressors, so that the push coefficients compare an individual with themselves at other times (Mundlak, 1978). The push from unemployment survives within the same person (coefficient 0.370, $p < 0.001$), as does the extra push for low-English migrants (interaction 1.377, $p = 0.006$). The push from low wages does not survive (0.087, $p = 0.119$), which suggests that low wages predict entry because persistently low-paid people enter more, not because a fall in pay pushes a given person into ownership.

*Probability scale.* Odds ratios can make an effect look large for a group whose baseline entry rate is low, so we re-estimate the main and heterogeneity specifications as linear probability models (Table D.4). The main results are unchanged on the probability scale. In the *Origin and labour market* column, unemployment at $t-1$ raises the annual probability of entry by 1.65 percentage points, against a mean entry rate of about 2 per cent, non-participation by 0.94 points and low wages by 0.73 points. The heterogeneity estimates agree in sign with the logit but not always in significance. The extra push from low wages for MES migrants (1.74 points, $p = 0.002$) remains, and the six interactions are jointly significant ($p = 0.002$), but the extra push from unemployment for low-English migrants, although large in absolute terms (3.34 points), is no longer significant ($p = 0.129$). This is the difference between the two scales described by Ai & Norton (2003). Low-English migrants rarely enter from stable employment, so a large proportional rise in their odds of entry corresponds to a smaller rise in probability, which is estimated imprecisely given the few entries involved.

With individual fixed effects (last column of Table D.4), unemployment and low wages still raise the probability of entry within the same person, by 0.87 and 0.67 points. The extra push from unemployment for low-English migrants, which survives the person-means logit, disappears with fixed effects (-1.16 points), and the six interactions are only marginally significant jointly ($p = 0.091$). The push from unemployment for the population as a whole is therefore robust to the estimator, while whether the larger push for low-English migrants operates within a person depends on the estimator, and given the few entries involved we do not draw a firm conclusion.

# 6 Discussion

## 6.1 Interpretation of Results

Our first finding concerns the wages of employees. We find that employees of small single-site workplaces earn about 11 per cent less than similar employees of large workplaces in the same industry and year. This wage penalty becomes much smaller, falling to under 2 per cent, in industries where a third of business owners are non-English-speaking migrants who speak English well. However, the penalty does not become smaller in industries with more owners who have limited English. When we replace the origin-specific owner densities with the share of all migrant owners and the share of

owners with limited English, only the share of all migrant owners matters. That is, the smaller wage penalty is linked to migrant ownership in general and not to owners' language barriers. This is not what Result 3 of our framework predicts.

The idea behind the framework is still reasonable. English proficiency is valuable for earnings in Australia (Budría & Martínez-de-Ibarreta, 2021), migrants from non-English-speaking countries take a long time to catch up with the wages of the native-born (Breunig et al., 2013), and migrant owners can come to depend on their local employees in practice (Cooke et al., 2022). However, there are very few migrant owners with limited English in HILDA, only 47 in waves 1 to 21. As a result, the owner density for this group is zero for 64 per cent of the employees in our sample. Our data therefore cannot confirm or rule out a moderate effect of these owners on wages.

The pay structure of industries offers another explanation for the wage results. Wages differ widely across firm sizes and industries (Brown & Medoff, 1989). In Australia, the share of employees paid exactly the award wage is highest in accommodation and food services (Australian Bureau of Statistics, 2019), which is also the lowest-paying industry (Australian Bureau of Statistics, 2026) and an industry in which migrants make up around a third of owners. When most employees are paid the award wage, small and large firms pay similar wages regardless of who owns them. In such industries, we would expect the wage penalty for small workplaces to be small even without any effect of migrant ownership. Our industry-by-year fixed effects remove differences in average wages between industries, but they do not remove differences in the size of the small-workplace penalty. Our robustness check with chain workplaces shows that this matters. Migrants rarely own small sites of chains, yet these workplaces also pay relatively more in industries with many English-proficient MNES owners, by about half as much as small independent workplaces. Ethnic networks offer a further explanation, since working with people from the same background can raise earnings (Edin et al., 2003), although Xie & Gough (2011) do not find this in every setting.

Our wage results also add to the literature on the pay of migrant workers. Previous studies show that the wage gap between immigrants and the native-born arises mainly because immigrants work in lower-paying jobs and firms, and not because they are paid less for the same job (Han & Hermansen, 2024; Hermansen et al., 2025). We find that the industries in which migrants own businesses also have smaller pay differences between workplace types. This pattern could reflect how wages are set in these industries, ethnic networks or bargaining, but it does not point specifically to owners with language barriers.

The bargaining power of employees also depends on their own options outside the job. Berg & Farbenblum (2017) show that employers in Australia often underpay migrant workers who depend on their visas, and that these employers are often from the same community as the workers. This suggests that bargaining power can only shift towards employees who have good alternatives, such as the skilled, English-proficient employee in our framework. Our wage measure includes all employees of small single-site workplaces. Higher pay for employees with good alternatives could therefore exist alongside the underpayment of more vulnerable employees.

One wage result was unexpected. We find that the wage penalty for small workplaces becomes larger in industries with more MES owners, even though these owners face no language barrier. This result most likely reflects the kinds of industries and businesses that MES owners run. It also means that MES owners are a weaker point of comparison than we expected.

Our second finding is that owning a business with employees brings higher income but also higher risk, and that this is true in every origin group. Native-born small employers earn about $12,700 a year more than native-born employees of small workplaces. However, their income changes by 27 percentage points more from year to year, and they are 15 percentage points more likely to experience a fall in income of a quarter or more. Small employers from every migrant group show the same pattern when compared with employees of their own origin. That is, owners are compensated for the extra risk they bear. Own-account workers, in contrast, earn less than employees. These results are consistent with evidence from the United States that owners of incorporated businesses earn more than employees, while owners of small unincorporated businesses earn much less (Levine & Rubinstein, 2017). They also help explain why studies that group all the self-employed together find that self-employment does not pay (Hamilton, 2000). Our results also support the view that owners bear income risk because they receive what is left after all costs are paid (Audoly, 2025; Bhandari et al., 2026). Our comparison of hourly income shows that much of the extra annual income of owners comes from working longer hours. In low-margin industries such as accommodation and food services, owners with employees earn less per hour than employees of small independent workplaces.

Our test of power reversal on the owner side does not give a clear answer. If power reversal holds, small employers with limited English should earn less and face more risk than native-born small employers, when each is compared with employees of their own origin. We find that the risk results point in this direction. The income of these owners is 12 percentage points more volatile, and their chance of a large fall is 4 percentage points higher, than the native-born comparison suggests. However, their income result points in the opposite direction, and none of these differences is statistically significant. Our unrestricted sample contains only 21 small employers with limited English. With so few owners, our tests could only reliably detect differences of about $30,000 a year in income and 34 percentage points in volatility. We therefore cannot conclude that these owners do as well as native-born owners. We can only conclude that our data are not precise enough to tell.

Our results for own-account workers are consistent with earlier Australian evidence. Le (1999) found that immigrants in Australia earned less in self-employment than the native-born. In our data, own-account workers with limited English also earn less, but much of this gap already exists among employees with limited English. This is consistent with evidence that immigrants who become self-employed were already underpaid as employees (Villarreal & Tamborini, 2026), and that English proficiency affects the earnings of self-employed immigrants (Dávila & Mora, 2004).

We also find that the income of own-account workers with limited English appears less volatile than the native-born comparison suggests. This result appears only in the restricted sample, which includes only people who stay in the same status for two years in a row. When we also include people who have

just become own-account workers, the difference becomes smaller and is no longer significant. We think this reflects which businesses survive. Businesses run by immigrants have a high risk of failure (Brzozowski & Lasek, 2019), so the own-account workers with limited English who stay in that status for two years are likely to be those with stable businesses. The way people report income in surveys may also play a role, since the self-employed tend to under-report their income (Hurst et al., 2014), and surveys measure immigrant self-employment differently for different origin groups (Tamborini & Villarreal, 2025).

Our third finding is that people enter business ownership from weak positions in the labour market. We find that being unemployed, being out of the labour force, earning a low wage and facing financial difficulty all make entry into ownership more likely. This is consistent with research that identifies necessity entrepreneurship by people's work status before they start a business (Fairlie & Fossen, 2020). For the native-born, being unemployed raises the odds of entering ownership by a factor of 1.9. For migrants with limited English, it raises the odds by a factor of 7.9, which is more than four times the effect for the native-born. However, this result is based on only five migrants with limited English who entered ownership after being unemployed. The difference is also not significant when we measure it as a change in probability instead of a change in odds. This is because so few migrants with limited English enter ownership from stable employment, so a large change in their odds is only a small change in their probability of entry (Ai & Norton, 2003).

The timing of entry supports this interpretation. For the native-born, only unemployment in the year just before entry predicts entry. This could mean that losing a job pushes people into ownership, but it could also mean that people leave their jobs to prepare for a business they had already planned. We cannot tell these two apart for the native-born. For migrants with limited English, unemployment two years before entry predicts entry just as strongly, and the result remains when we compare each person with themselves at other times. That is, these migrants enter ownership after a long period without good alternatives, not after a single short spell of unemployment. This pattern is difficult to explain by planning ahead, and it matches the type of necessity entry that our framework assumes.

Two further results add to this picture. First, among people in stable and reasonably paid jobs, migrants with limited English are less likely to enter ownership than the native-born, with 0.59 times the odds. The low income and limited savings that push people towards ownership may also make it harder for them to start a business when they are not pushed. Second, people with more education are more likely to enter ownership. This suggests that people are both pushed into ownership by poor alternatives and pulled into it by good opportunities (Fairlie & Fossen, 2020).

Some explanations for migrant self-employment are not included in our models. Discrimination in paid employment can push minorities into self-employment, while opportunities within their own communities can pull them in (Clark & Drinkwater, 1998). Immigrant self-employment also rises with time spent in the host country (Borjas, 1986), and in Australia it is more common in communities that share a language (Le, 2000). Studies also point to the role of overseas qualifications that are not

recognised and of visa conditions. HILDA does not record this information for most respondents, so we cannot examine these explanations.

Overall, our three analyses give a mixed picture of the framework. The wage results show that migrant ownership is linked to the pay of employees in small workplaces, but not through owners' language barriers. The income results cannot confirm or rule out the prediction that owners with limited English earn less or bear more risk. The entry results support the framework's assumption that these owners have few alternatives and find it costly to leave their business. In the framework, it is this cost of leaving, $K$, that gives workers more bargaining power. Our data therefore support the first step of the mechanism, but we cannot observe the second step. The clearest disadvantage of migrants with limited English appears at the point when they enter business ownership.

## 6.2 Limitations and Future Research

Our analysis has several limitations. We discuss each one in turn, together with the research that could address it.

First, we do not observe bargaining, compliance decisions or the delegation of tasks within firms, and HILDA does not link workers to their employers. We therefore test the predictions of power reversal instead of observing the mechanism directly. We also measure each worker's exposure to migrant ownership by the mix of owners in their industry and year. This means that every small single-site workplace in the same industry and year receives the same exposure, whether or not a migrant owns it. Our wage results therefore show the link between migrant ownership and pay at the industry level, and not the effect of working for a migrant owner. Our robustness check with chain workplaces shows that part of this link reflects the pay structure of industries. Future research could use linked employer–employee data, such as administrative data that link Australian tax records to businesses, to compare the wages of workers in migrant-owned and native-owned firms in the same industry. Future studies could also allow the small-workplace penalty to differ by industry, or measure the density of migrant workers instead of migrant owners, to separate the effect of ownership from the pay structure of industries and ethnic networks. Surveys that ask owners how they manage compliance, and how much they rely on employees and advisers for regulatory knowledge, would test the information gap in our framework directly.

Second, our estimates show associations, not causal effects. We use a rich set of controls and fixed effects instead of individual fixed effects. Individual fixed effects would remove origin and English proficiency, which do not change over time, and would estimate the interactions only from the few people who change status. Our wage results could therefore reflect migrant owners choosing to start businesses in industries where small workplaces already pay relatively well. If so, our results would overstate any effect of migrant ownership. Future research could use changes in the number of migrant business owners that are unrelated to local wages, such as changes to business and skilled visa programs, to estimate causal effects.

Third, our sample of migrants with limited English is small. HILDA records only 47 owners with limited English in waves 1 to 21, and our owner income analysis includes only 21 small employers with limited English. HILDA also under-represents recent immigrants, who are the most likely to have limited English (Watson, 2021; Watson & Ey, 2026). These small numbers affect all three analyses. The smallest differences that we can detect for owners with limited English are large. The owner density for this group changes sign depending on the survey weight we use to construct it. The larger push into ownership for migrants with limited English is based on only five entries. When an owner density rests on few owners, it also contains measurement error, which pulls its estimated effect on wages towards zero. We therefore treat many of the results for this group as uninformative, and not as evidence of no effect. Administrative data that cover all migrant business owners, or surveys that include more recent migrants with limited English, would allow future research to test our predictions for the group that the framework focuses on.

Fourth, our owner densities vary across only 19 industries and change slowly over time within each industry. Our wage results therefore rely on a small number of independent comparisons. When we define industries more narrowly, the results become weaker, partly because each narrower industry contains even fewer owners. Future studies could measure owner density at a more detailed level, such as by industry and region, although this would require larger samples of owners than HILDA provides.

Fifth, income in HILDA is reported by respondents and refers to the financial year before the interview. For people who changed status between interviews, income therefore partly reflects their previous status. We address this by comparing the restricted and unrestricted samples. The self-employed also tend to under-report their income in household surveys (Hurst et al., 2014). In addition, both of our samples classify people by their status in the later year. Neither sample therefore shows what happens to the income of owners whose business closes, and owners who leave the survey drop out of both samples. Tax records that follow owners through the start and closure of their businesses would allow future research to measure the full range of returns to ownership, including the losses from failed businesses.

Sixth, for the native-born, our entry analysis cannot separate entry that is pushed by job loss from entry that people plan in advance, since their push comes only from the year just before entry. Our two methods that compare each person with themselves at other times also disagree. They give different answers on whether low wages, and the larger push from unemployment for migrants with limited English, work within the same person. Both methods assume that starting a business does not affect a person's later employment status. This assumption is doubtful, since owners who close their business return to the sample with an employment history shaped by their business. Future research could use methods that allow for this feedback, together with a longer panel of migrants with limited English.

Future research could also extend our analysis of entry in two ways. First, our measure of entry groups together very different types of business, incorporated and unincorporated, with and without employees. People who start these businesses differ, and so do their reasons for starting them. In the United States, the incorporated self-employed earn more than similar salaried workers and have

stronger cognitive skills, while the unincorporated self-employed earn less (Levine & Rubinstein, 2017). In transition economies, own-account workers resemble the hidden unemployed, while employers resemble entrepreneurs (Earle & Sakova, 2000). People who start businesses out of necessity also differ from those who start businesses to take up an opportunity, both in their characteristics and in their later earnings (Block & Wagner, 2010; Fairlie & Fossen, 2020). A multinomial logit model of entry into each type of business would show whether the push we find, and its larger size for migrants with limited English, applies only to own-account work. Second, our sample treats a former owner who starts another business in the same way as a person starting their first business. Former owners have business experience, networks and a history of past success or failure, so their reasons for starting again may differ from those of first-time owners (Stam et al., 2008). Separating first entries from re-entries, and modelling the full sequence of entry, exit and re-entry, would show whether the push applies mainly to new owners and how long necessity entrants remain owners.

Finally, our study is based on Australia. Australian workplace law makes employers fully responsible for compliance, and it is enforced more actively than in many similar countries (Wright & Clibborn, 2020). The mechanism we study depends on strict compliance rules and on the presence of many migrant-owned businesses, and not on features unique to Australia. We therefore expect our findings to apply, at least in general terms, to other countries with many migrant-owned small businesses. Repeating our analysis with data from the United Kingdom, the United States or Canada, where enforcement differs, would show how much the strength of enforcement affects the position of migrant owners.

# 7 Conclusion

Labour regulation assumes that employers hold bargaining power over their workers. This paper asked whether this assumption fails in small businesses owned by migrants with limited English, who depend on their employees for the language skills and local knowledge needed to run the business. We developed a model of bargaining under compliance risk and tested three of its predictions with 24 waves of the HILDA Survey, using industry-level measures of migrant business ownership by English proficiency.

We find little evidence that bargaining power shifts towards workers in these businesses. The wage penalty of small independent workplaces falls from about 11 per cent to under 2 per cent in industries with many English-proficient migrant owners, but it does not fall where owners have limited English. Owners with employees in every origin group earn more than employees of their own origin in exchange for much more volatile income, and we detect no difference for migrant owners. The clearest disadvantage of migrants with limited English appears when they enter ownership. For this group, unemployment raises the odds of starting a business by a factor of 7.9, compared with 1.9 for the native-born, and this push persists over time.

This paper makes three contributions. It provides the first test of power reversal using nationally representative data. It develops a measure of exposure to migrant ownership that researchers can use when workers cannot be linked to their employers. It also follows people who start businesses out of necessity beyond the point of entry. Our results suggest that policy towards small migrant-owned businesses should focus less on the balance of power within these firms and more on the weak positions from which migrants with limited English enter ownership.

Our conclusions are limited by the small number of owners with limited English in HILDA and by the lack of links between workers and their employers. Linked administrative data covering all migrant business owners would allow future research to compare wages, compliance costs and owner incomes within migrant-owned firms directly.

## Data and Code Availability

This paper uses unit record data from General Release 24.0 of the Household, Income and Labour Dynamics in Australia (HILDA) Survey, conducted by the Australian Government Department of Social Services (DSS) (Department of Social Services & Melbourne Institute of Applied Economic and Social Research, 2025). The data are not publicly available. Researchers can apply for access to HILDA through the DSS and the National Centre for Longitudinal Data.

All data processing, estimation, tables and figures in this paper are produced by our own code, which is available at https://github.com/addisonniuen/disadvantagedmigrants-replication. The repository contains the full replication package, including the scripts that clean and construct the analysis samples from the HILDA release files, estimate every model and reproduce each table and figure, together with instructions for running them. We carried out the analysis in R version 4.5.3 (R Core Team, 2026), with the fixest package for estimation (Bergé et al., 2026) and the tidyverse for data processing (Wickham et al., 2019). We produced the tables with the modelsummary and tinytable packages (Arel-Bundock, 2022; 2026) and wrote and rendered the paper with Quarto (Posit Software, PBC, 2026). The paper uses the Monash honours thesis template provided by Rob J Hyndman (https://github.com/quarto-monash/honours-thesis), and we modified its underlying scripts where necessary.

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# Appendix

## A Data

This section reports the estimation samples and the descriptive tables referred to in Section 3.

**Table A.1:** *Estimation samples.*

| Analysis | Sample | Unit | Waves | Observations | Individuals | Low-English individuals |
|---|---|---|---|---|---|---|
| Employee wages | Employees | Person-year | 1 to 21 | 156,786 | 23,067 | 125 |
| Owner income and risk | Restricted pairs (same status) | Person-pair | 1 to 21 | 152,060 | 20,031 | 97 |
| Owner income and risk | Unrestricted pairs | Person-pair | 1 to 21 | 174,962 | 21,549 | 116 |
| Entry into ownership | Non-owners at t−1 | Person-pair | 1 to 24 | 289,616 | 28,626 | 394 |

*Notes:* This table reports the estimation sample of each analysis. Observations and individuals are those of the main specification, namely the Type 0 column of Table 5.1 for employee wages, the income regressions of Table 5.2 for owner income and risk, and the *Adding human capital* column of Table 5.4 for entry into ownership. A person-pair is an individual observed in two consecutive waves, $t-1$ and $t$. Unrestricted pairs classify everyone who is an employee, owner with employees or own-account worker at $t$ by their status at $t$; restricted pairs also require the same status at $t-1$. Low-English individuals are MNES migrants with low English proficiency.

**Table A.2:** *Distribution of firm structures observed in the employee sample.*

| Category | Person-years | Share (%) |
|---|---|---|
| Small single-site | 30,457 | 17.7 |
| Chain-like | 63,425 | 36.9 |
| Small-site chain | 30,399 | 17.7 |
| None of the three | 46,275 | 26.9 |
| Firm structure missing | 1,336 | 0.8 |

*Notes:* This table reports the distribution of workplace types (Section 3.3) across employee person-years in HILDA waves 1 to 21. The categories are mutually exclusive. Small single-site and small-site chain workplaces have fewer than 20 people at the workplace and an employer with one or several locations respectively; chain-like workplaces have 20 or more people and an employer with at least 500 employees nationally. *None of the three* is the reference category of other large workplaces, and *Firm structure missing* means that workplace size or the number of locations is unanswered. The base is every employee person-year before observations with missing wages or controls are dropped, so the total exceeds the wage sample in Table A.1.

**Table A.3:** *Distribution of the leave-one-out ownership exposure shares across industry-year (waves 1-21).*

| Share | Owners | Industry-year (N) | Zero (%) | Mean | SD | Median | Max |
|---|---|---|---|---|---|---|---|
| MES | 646 | 398 | 7.789 | 0.122 | 0.093 | 0.110 | 0.701 |
| MNES (high English proficiency) | 670 | 398 | 12.060 | 0.164 | 0.121 | 0.148 | 0.768 |
| MNES (low English proficiency) | 47 | 398 | 72.613 | 0.010 | 0.025 | 0.000 | 0.203 |

*Notes:* This table reports the distribution of the leave-one-out owner shares of Equation (3.7), without decay (Type 0), across industry-year cells in HILDA waves 1 to 21. *Owners* is the number of distinct owners in each origin group. *Industry-year (N)* counts the cells with at least one owner; 27 of the 399 possible cells have no owners and are excluded because the share is undefined. *Zero (%)* is the percentage of cells with no owner from the group.

**Table A.4:** *Industry composition of business owners (%) by origin and English proficiency group.*

| | | | MNES | |
|---|---|---|---|---|
| Industry | Native-born | MES | High English | Low English |
| Construction | 18.37 | 23.81 | 17.66 | 19.49 |

**Table A.4:** *Industry composition of business owners (%) by origin and English proficiency group. (continued)*

| | | | MNES | |
|---|---|---|---|---|
| Industry | Native-born | MES | High English | Low English |
| Professional, scientific and technical services | 14.16 | 14.58 | 11.90 | 1.03 |
| Agriculture, forestry and fishing | 13.50 | 3.40 | 3.29 | 4.62 |
| Other services | 7.86 | 7.58 | 5.02 | 5.64 |
| Health care and social assistance | 6.99 | 7.33 | 13.26 | 2.56 |
| Manufacturing | 6.97 | 6.85 | 5.27 | 21.54 |
| Administrative and support services | 5.58 | 7.00 | 5.79 | 9.23 |
| Retail trade | 5.46 | 5.15 | 6.83 | 14.87 |
| Transport, postal and warehousing | 3.92 | 4.23 | 7.35 | 3.59 |
| Education and training | 3.27 | 3.68 | 4.13 | 3.59 |
| Accommodation and food services | 3.24 | 3.12 | 6.73 | 7.18 |
| Arts and recreation services | 2.29 | 3.04 | 1.04 | 0.00 |
| Wholesale trade | 2.22 | 3.55 | 4.20 | 3.08 |
| Financial and insurance services | 2.16 | 1.75 | 3.71 | 0.51 |
| Rental, hiring and real estate services | 1.69 | 1.37 | 1.46 | 0.51 |
| Information media and telecommunications | 1.38 | 2.26 | 1.56 | 2.56 |
| Public administration and safety | 0.42 | 0.41 | 0.37 | 0.00 |
| Mining | 0.33 | 0.63 | 0.12 | 0.00 |
| Electricity, gas, water and waste services | 0.20 | 0.28 | 0.32 | 0.00 |
| Person-years | 26,627 | 3,944 | 4,043 | 195 |
| Persons | 5,024 | 692 | 813 | 56 |

*Notes:* This table reports the percentage of business owners (employers and own-account workers) in each origin group who work in each industry, pooled over HILDA waves 1 to 24. Industries are the 19 ANZSIC divisions, and each column sums to 100. The bottom rows give owner person-years and distinct owners; these are slightly below the counts in Table 3.1 because owners without a reported industry are excluded. *High English* and *Low English* are MNES migrants with high and low English proficiency.

**Table A.5:** *Industry composition by worker type.*

| **Industry** | **Employees** | **Self-employed worker with employees** | **Own-account worker** |
|---|---|---|---|
| Construction | 6.4 | 17.7 | 19.7 |
| Professional, scientific and technical services | 6.6 | 13.8 | 13.9 |
| Agriculture, forestry and fishing | 1.6 | 11.0 | 11.2 |
| Manufacturing | 8.4 | 8.9 | 5.5 |
| Retail trade | 10.3 | 7.9 | 4.2 |
| Health care and social assistance | 15.8 | 7.1 | 8.1 |
| Accommodation and food services | 6.9 | 6.6 | 1.8 |
| Other services | 3.2 | 6.3 | 8.3 |
| Administrative and support services | 2.5 | 4.3 | 6.8 |
| Transport, postal and warehousing | 4.3 | 3.7 | 4.8 |
| Wholesale trade | 3.1 | 3.2 | 2.2 |
| Financial and insurance services | 3.7 | 2.6 | 2.1 |
| Rental, hiring and real estate services | 1.3 | 2.0 | 1.4 |
| Education and training | 11.2 | 2.0 | 4.3 |
| Information media and telecommunications | 1.9 | 1.1 | 1.8 |

**Table A.5:** *Industry composition by worker type. (continued)*

| Industry | Employees | Self-employed worker with employees | Own-account worker |
|---|---|---|---|
| Arts and recreation services | 1.8 | 1.1 | 3.0 |
| Public administration and safety | 7.6 | 0.4 | 0.5 |
| Mining | 2.0 | 0.2 | 0.4 |
| Electricity, gas, water and waste services | 1.2 | 0.2 | 0.2 |
| Person-years | 198,204 | 13,636 | 21,191 |
| Persons | 26,611 | 3,163 | 5,357 |

*Notes:* This table reports the percentage of employees, employers (self-employed with employees) and own-account workers (self-employed without employees) who work in each industry, pooled over HILDA waves 1 to 24. Industries are the 19 ANZSIC divisions, ordered by the share of employers, and each column sums to 100. The bottom rows give person-years and distinct individuals in each group.

**Table A.6:** *Panel persistence of all respondents by origin group.*

| Group | Persons | Mean waves | Median waves | 1 wave (%) | 10+ waves (%) | Adjacent (%) | All waves (%) |
|---|---|---|---|---|---|---|---|
| Native-born | 28,323 | 10.3 | 8 | 16.9 | 46.7 | 83.3 | 12.4 |
| MES | 3,329 | 10.32 | 8 | 19.6 | 47.5 | 85.2 | 13.7 |
| MNES (high English) | 4,762 | 8.43 | 5 | 31.4 | 38.1 | 84.9 | 9.6 |
| MNES (low English) | 676 | 6.51 | 3 | 38.9 | 28.6 | 82.4 | 3.3 |

*Notes:* This table reports the panel persistence of every respondent, owners and non-owners, by origin group, counting the waves (out of 24) in which each person is interviewed. *1 wave (%)* and *10+ waves (%)* are the percentages interviewed in only one wave and in ten or more waves, *Adjacent (%)* the percentage whose interviews form one unbroken run of consecutive waves and *All waves (%)* the percentage interviewed in all 24 waves. Samples that need consecutive waves, such as the owner income and entry samples, lose more people from groups with less persistent participation.

**Table A.7:** *HILDA variables used in the analysis.*

| Variable | Description | Used for |
|---|---|---|
| *Survey weights* | | |
| HHWTRP | Responding-person cross-sectional weight | Owner shares (main measure) |
| LNWTRP | Responding-person longitudinal weight | Owner shares (sensitivity check) |
| *Origin and English proficiency* | | |
| ANBCOB, ANCOB | Country of birth (brief; SACC code) | Origin groups |
| HGEAB, ANEAB | How well speaks English (wave 1; later waves) | English proficiency |
| SKTAREA | Reading ability relative to an average Australian (waves 12, 16, 20, 24) | English proficiency |
| ANYOA | Year first came to Australia to live | Years in Australia; decay weights |
| *Employment and workplace* | | |
| ESEMPDT | Employment status in main job | Employee, employer, own-account worker |
| JBMMPL, JBMMPLR, JBMMPLY | Description of employer or business | Employment status |
| ESDTL | Labour force status | Unemployed, not in the labour force |
| JBMWPSZ, JBMWPS | Number of people at the workplace (waves 1 to 4; 5 to 21) | Workplace types; small and large employers |
| JBMMWP | Employer operates from more than one location | Workplace types |
| JBMEMSZ | Number of people employed by the employer in Australia | Chain-like workplaces |
| JBMI61, JBMI62 | Industry of main job (19 ANZSIC divisions; two-digit) | Owner shares; industry-year fixed effects |
| JBMO61 | Occupation of main job (one-digit ANZSCO) | Wage controls |

**Table A.7:** *HILDA variables used in the analysis. (continued)*

| Variable | Description | Used for |
|---|---|---|
| JBCASAB | Casual or permanent employment | Wage controls |
| JBOCCT, JBEMPT | Tenure in occupation; tenure with employer | Controls |
| JBMTUEA | Trade union or employee association member | Wage controls |
| *Income* | | |
| WSCMEI, JBMHRUC | Weekly gross wages and usual weekly hours in main job | Hourly wage |
| WSFEI | Financial-year gross wages and salary | Labour and business income |
| BIFIP, BIFIN | Financial-year business income (positive; negative) | Labour and business income |
| *Financial stress (self-completion questionnaire)* | | |
| FIPRBEG, FIPRBMR, FIPRBFH | Could not pay bills, or mortgage or rent, on time; asked family or friends for help | Cash-flow problem |
| FIPRBWM, FIPRBUH, FIPRBPS, FIPRBWO | Went without meals; unable to heat home; pawned or sold something; asked a welfare organisation for help | Hardship |
| *Demographics and location* | | |
| HGAGE, HGSEX | Age; sex | Controls |
| EDHIGH1 | Highest education level achieved | Controls |
| MRCURR | Current marital status | Controls |
| HSTENR | Housing tenure | Home ownership |
| HHSTATE | State of residence | State-year fixed effects |
| HHS3RA, HHS3ADD | Remoteness area; SEIFA decile of disadvantage | Controls |

*Notes:* This table lists the HILDA variables used in the analysis, by group. Variable names omit the one-letter wave prefix (for example, HHWTRP is AHHWTRP in wave 1). Workplace size, the number of employer locations and the employer's national size are asked only in waves 1 to 21. Variable definitions are from the HILDA Online Data Dictionary.

# B Employee Wages

This section reports the distribution of the owner shares, the full estimates behind Table 5.1 and the robustness and sensitivity checks discussed in Section 5.1. It also reports two further robustness checks, inference that allows for correlation within industries (Section B.1), and owner shares built with the longitudinal survey weight (Section B.2).

**Table B.1:** *Distribution of the leave-one-out owner shares across employees in the wage sample.*

| Density | Mean | Std. Dev. | Q1 | Median | Q3 | Max | % Zero |
|---|---|---|---|---|---|---|---|
| *Type 0 (no weighting)* | | | | | | | |
| MES owner density | 0.116 | 0.072 | 0.078 | 0.107 | 0.143 | 0.701 | 5.1 |
| MNES high English owner density | 0.190 | 0.104 | 0.119 | 0.189 | 0.258 | 0.768 | 6.2 |
| MNES low English owner density | 0.014 | 0.028 | 0.000 | 0.000 | 0.017 | 0.203 | 63.9 |
| *Type 1 (pooled - same $\lambda$ applied to all migrants)* | | | | | | | |
| MES owner density | 0.065 | 0.043 | 0.041 | 0.059 | 0.079 | 0.410 | 5.1 |
| MNES high English owner density | 0.117 | 0.070 | 0.066 | 0.115 | 0.164 | 0.709 | 6.2 |
| MNES low English owner density | 0.011 | 0.022 | 0.000 | 0.000 | 0.012 | 0.180 | 63.9 |
| *Type 2 ($\lambda$ is split by migrant origin group)* | | | | | | | |
| MES owner density | 0.040 | 0.028 | 0.022 | 0.035 | 0.049 | 0.324 | 5.1 |
| MNES high English owner density | 0.190 | 0.104 | 0.119 | 0.189 | 0.258 | 0.768 | 6.2 |
| MNES low English owner density | 0.014 | 0.028 | 0.000 | 0.000 | 0.017 | 0.203 | 63.9 |

*Notes:* This table reports the distribution of the leave-one-out owner shares of Equation (3.7) across employee person-years in the wage sample of Table 5.1, so each employee contributes the share of their own industry and year. The shares run from 0 to 1. Q1 and Q3 are the 25th and 75th percentiles, and *% Zero* is the percentage of person-years whose share is exactly zero. Type 0 applies no decay, Type 1 a pooled decay rate ($\lambda = 0.02$) and Type 2 origin-specific rates ($\lambda_{\text{MES}} = 0.04$ and $\lambda_{\text{MNES}} = 0$).

**Table B.2:** *Log hourly wages under alternative small-firm size thresholds.*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | 10 employees | 20 employees (baseline) | 50 employees |
| *Firm structure* | | | |
| Small single-site | -0.115*** | -0.113*** | -0.132*** |
| | (0.020) | (0.018) | (0.016) |
| Chain-like | 0.073*** | 0.060*** | 0.044*** |
| | (0.005) | (0.005) | (0.005) |
| Small-site chain | -0.025*** | -0.030*** | -0.055*** |
| | (0.006) | (0.005) | (0.004) |
| *Interactions of firm structure with owner density* | | | |
| Small single-site x MES owner density | -0.331*** | -0.330*** | -0.259*** |
| | (0.105) | (0.092) | (0.083) |
| Small single-site x MNES high-English owner density | 0.269*** | 0.277*** | 0.255*** |
| | (0.063) | (0.058) | (0.055) |
| Small single-site x MNES low-English owner density | -0.272 | -0.182 | -0.199 |
| | (0.197) | (0.173) | (0.177) |
| *Demographics* | | | |
| Age | 0.006*** | 0.006*** | 0.006*** |
| | (0.000) | (0.000) | (0.000) |
| Female | -0.081*** | -0.080*** | -0.080*** |
| | (0.005) | (0.005) | (0.005) |
| MES worker | 0.017** | 0.017** | 0.015* |
| | (0.008) | (0.008) | (0.008) |
| MNES high-English worker | -0.025*** | -0.026*** | -0.028*** |
| | (0.008) | (0.008) | (0.008) |
| MNES low-English worker | -0.157*** | -0.163*** | -0.162*** |
| | (0.029) | (0.030) | (0.029) |
| SEIFA decile | 0.013*** | 0.013*** | 0.013*** |
| | (0.001) | (0.001) | (0.001) |
| *Occupation* | | | |
| Professionals | -0.061*** | -0.060*** | -0.061*** |
| | (0.010) | (0.010) | (0.010) |
| Technicians and trades | -0.235*** | -0.234*** | -0.234*** |
| | (0.011) | (0.011) | (0.011) |
| Community and personal service | -0.238*** | -0.236*** | -0.234*** |
| | (0.012) | (0.012) | (0.012) |
| Clerical and administrative | -0.226*** | -0.226*** | -0.226*** |
| | (0.010) | (0.010) | (0.010) |
| Sales | -0.284*** | -0.283*** | -0.282*** |
| | (0.012) | (0.012) | (0.012) |
| Machinery operators and drivers | -0.278*** | -0.278*** | -0.277*** |
| | (0.013) | (0.013) | (0.013) |
| Labourers | -0.322*** | -0.323*** | -0.323*** |
| | (0.012) | (0.012) | (0.012) |
| *Education* | | | |
| Postgraduate | 0.346*** | 0.344*** | 0.340*** |
| | (0.013) | (0.013) | (0.013) |
| Graduate diploma or certificate | 0.287*** | 0.286*** | 0.283*** |
| | (0.012) | (0.012) | (0.012) |
| Bachelor or honours | 0.260*** | 0.259*** | 0.258*** |
| | (0.009) | (0.009) | (0.009) |
| Advanced diploma or diploma | 0.171*** | 0.171*** | 0.170*** |
| | (0.009) | (0.009) | (0.009) |

**Table B.2:** *Log hourly wages under alternative small-firm size thresholds. (continued)*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | 10 employees | 20 employees (baseline) | 50 employees |
| Certificate III or IV | 0.133*** | 0.133*** | 0.133*** |
| | (0.007) | (0.007) | (0.007) |
| Year 12 | 0.110*** | 0.110*** | 0.109*** |
| | (0.008) | (0.008) | (0.008) |
| *Job characteristics* | | | |
| Permanent (vs casual) | -0.021*** | -0.020*** | -0.019*** |
| | (0.006) | (0.006) | (0.006) |
| Occupational tenure (years) | 0.004*** | 0.004*** | 0.004*** |
| | (0.000) | (0.000) | (0.000) |
| Employer tenure (years) | 0.001*** | 0.001*** | 0.001*** |
| | (0.000) | (0.000) | (0.000) |
| Union member | 0.047*** | 0.046*** | 0.045*** |
| | (0.006) | (0.006) | (0.006) |
| Observations (person-years) | 156,668 | 156,786 | 156,301 |
| R-Squared | 0.490 | 0.490 | 0.492 |
| Adjusted R-Squared | 0.488 | 0.488 | 0.490 |
| Individuals | 23,057 | 23,067 | 23,018 |
| Year FEs | Yes | Yes | Yes |
| Industry-year FEs | Yes | Yes | Yes |
| State-year FEs | Yes | Yes | Yes |
| F-Statistic (df) | 11.959 (3, 397) | 14.602 (3, 397) | 12.783 (3, 397) |
| F-test p-value | < 0.001 | < 0.001 | < 0.001 |

*Notes:* This table reports OLS estimates of Equation (4.1) with small workplaces defined as having fewer than 10, 20 (the baseline of Table 5.1) or 50 people. The dependent variable is the log hourly wage in the employee's main job. Each column represents a different size threshold, which applies to both small single-site and small-site chain workplaces, and all columns use the owner shares without decay (Type 0). All columns include covariates for age, sex, origin group, casual or permanent employment, occupation, education, SEIFA decile, occupational tenure, employer tenure and union membership, and year, industry-year and state-year fixed effects. Reference categories are managers and administrators (occupation), Year 11 or below (education), native-born (origin), casual employment and no union membership. The F-statistic, with its degrees of freedom in parentheses, tests the null hypothesis that the three owner-share interactions are jointly zero, using the clustered covariance matrix; its p-value is reported in the row below. All standard errors reported in parentheses are clustered at the individual and industry-year levels. * p < 0.10, ** p < 0.05, *** p < 0.01.

**Table B.3:** *Owner density interacted with chain workplaces.*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| *Firm structure* | | | |
| Small single-site | -0.127*** | -0.118*** | -0.127*** |
| | (0.018) | (0.016) | (0.016) |
| Small-site chain | -0.067*** | -0.064*** | -0.064*** |
| | (0.014) | (0.013) | (0.013) |
| Chain-like | 0.055*** | 0.049*** | 0.052*** |
| | (0.012) | (0.011) | (0.011) |
| *Main interactions* | | | |
| Small single-site x MES owner density | -0.239** | -0.431*** | -0.608*** |
| | (0.093) | (0.141) | (0.192) |
| Small single-site x MNES high-English owner density | 0.280*** | 0.386*** | 0.267*** |
| | (0.056) | (0.087) | (0.056) |
| Small single-site x MNES low-English owner density | -0.033 | -0.055 | -0.028 |
| | (0.163) | (0.194) | (0.164) |
| *Small sites of multi-site employers* | | | |
| Small-site chain x MES owner density | 0.058 | 0.094 | 0.118 |
| | (0.060) | (0.100) | (0.146) |

**Table B.3:** *Owner density interacted with chain workplaces. (continued)*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| Small-site chain x MNES high-English owner density | 0.141*** (0.044) | 0.210*** (0.064) | 0.139*** (0.044) |
| Small-site chain x MNES low-English owner density | 0.186 (0.129) | 0.215 (0.155) | 0.178 (0.129) |
| *Chain-like workplaces* | | | |
| Chain-like x MES owner density | 0.155*** (0.044) | 0.321*** (0.078) | 0.549*** (0.131) |
| Chain-like x MNES high-English owner density | -0.079* (0.040) | -0.109* (0.063) | -0.081** (0.040) |
| Chain-like x MNES low-English owner density | 0.215 (0.152) | 0.334* (0.180) | 0.210 (0.153) |
| *Demographics* | | | |
| Age | 0.006*** (0.000) | 0.006*** (0.000) | 0.006*** (0.000) |
| Female | -0.080*** (0.005) | -0.080*** (0.005) | -0.080*** (0.005) |
| MES worker | 0.017** (0.008) | 0.017** (0.008) | 0.017** (0.008) |
| MNES high-English worker | -0.026*** (0.008) | -0.026*** (0.008) | -0.026*** (0.008) |
| MNES low-English worker | -0.163*** (0.029) | -0.163*** (0.030) | -0.162*** (0.030) |
| SEIFA decile | 0.013*** (0.001) | 0.013*** (0.001) | 0.013*** (0.001) |
| *Occupation* | | | |
| Professionals | -0.060*** (0.010) | -0.060*** (0.010) | -0.060*** (0.010) |
| Technicians and trades | -0.233*** (0.011) | -0.233*** (0.011) | -0.233*** (0.011) |
| Community and personal service | -0.236*** (0.012) | -0.236*** (0.012) | -0.236*** (0.012) |
| Clerical and administrative | -0.226*** (0.010) | -0.225*** (0.010) | -0.226*** (0.010) |
| Sales | -0.283*** (0.012) | -0.282*** (0.012) | -0.282*** (0.012) |
| Machinery operators and drivers | -0.277*** (0.013) | -0.277*** (0.013) | -0.277*** (0.013) |
| Labourers | -0.322*** (0.012) | -0.322*** (0.013) | -0.322*** (0.013) |
| *Education* | | | |
| Postgraduate | 0.345*** (0.013) | 0.345*** (0.013) | 0.344*** (0.013) |
| Graduate diploma or certificate | 0.286*** (0.012) | 0.286*** (0.012) | 0.286*** (0.012) |
| Bachelor or honours | 0.259*** (0.009) | 0.259*** (0.009) | 0.258*** (0.009) |
| Advanced diploma or diploma | 0.170*** (0.009) | 0.170*** (0.009) | 0.170*** (0.009) |
| Certificate III or IV | 0.133*** (0.007) | 0.133*** (0.007) | 0.133*** (0.007) |
| Year 12 | 0.110*** (0.008) | 0.110*** (0.008) | 0.110*** (0.008) |

**Table B.3:** *Owner density interacted with chain workplaces. (continued)*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| *Job characteristics* | | | |
| Permanent (vs casual) | -0.020*** | -0.020*** | -0.020*** |
| | (0.006) | (0.006) | (0.006) |
| Occupational tenure (years) | 0.004*** | 0.004*** | 0.004*** |
| | (0.000) | (0.000) | (0.000) |
| Employer tenure (years) | 0.001*** | 0.001*** | 0.001*** |
| | (0.000) | (0.000) | (0.000) |
| Union member | 0.047*** | 0.047*** | 0.047*** |
| | (0.006) | (0.006) | (0.006) |
| Observations (person-years) | 156,786 | 156,786 | 156,786 |
| Individuals | 23,067 | 23,067 | 23,067 |
| Joint test, small-site chain (p-value) | 0.014 | 0.010 | 0.015 |
| Joint test, chain-like (p-value) | < 0.001 | < 0.001 | < 0.001 |

*Notes:* This table reports OLS estimates of Equation (4.1) extended with interactions of the three owner shares with small-site chain and chain-like workplaces, as a robustness check. The dependent variable is the log hourly wage in the employee's main job. Each column represents a different weighting of the owner shares, namely no decay (Type 0, as in our main specification), pooled decay (Type 1) and origin-specific decay (Type 2). All columns include covariates for age, sex, origin group, casual or permanent employment, occupation, education, SEIFA decile, occupational tenure, employer tenure and union membership, and year, industry-year and state-year fixed effects. Reference categories are managers and administrators (occupation), Year 11 or below (education), native-born (origin), casual employment and no union membership. The joint tests report the p-value of a Wald test of the null hypothesis that the three owner-share interactions for that workplace type are jointly zero. All standard errors reported in parentheses are clustered at the individual and industry-year levels. * p < 0.10, ** p < 0.05, *** p < 0.01.

**Table B.4:** *Log hourly wages under 10-employee small-firm size threshold (full estimation results).*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| *Firm structure* | | | |
| Small single-site | -0.115*** | -0.105*** | -0.115*** |
| | (0.020) | (0.018) | (0.018) |
| Chain-like | 0.073*** | 0.073*** | 0.073*** |
| | (0.005) | (0.005) | (0.005) |
| Small-site chain | -0.025*** | -0.025*** | -0.025*** |
| | (0.006) | (0.006) | (0.006) |
| *Interactions of firm structure with owner density* | | | |
| Small single-site x MES owner density | -0.331*** | -0.596*** | -0.871*** |
| | (0.105) | (0.158) | (0.215) |
| Small single-site x MNES high-English owner density | 0.269*** | 0.362*** | 0.253*** |
| | (0.063) | (0.097) | (0.062) |
| Small single-site x MNES low-English owner density | -0.272 | -0.383* | -0.261 |
| | (0.197) | (0.232) | (0.198) |
| *Demographics* | | | |
| Age | 0.006*** | 0.006*** | 0.006*** |
| | (0.000) | (0.000) | (0.000) |
| Female | -0.081*** | -0.081*** | -0.081*** |
| | (0.005) | (0.005) | (0.005) |
| MES worker | 0.017** | 0.017** | 0.017** |
| | (0.008) | (0.008) | (0.008) |
| MNES high-English worker | -0.025*** | -0.025*** | -0.025*** |
| | (0.008) | (0.008) | (0.008) |
| MNES low-English worker | -0.157*** | -0.158*** | -0.158*** |
| | (0.029) | (0.029) | (0.029) |
| SEIFA decile | 0.013*** | 0.013*** | 0.013*** |
| | (0.001) | (0.001) | (0.001) |

**Table B.4:** *Log hourly wages under 10-employee small-firm size threshold (full estimation results). (continued)*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| *Occupation* | | | |
| Professionals | -0.061*** | -0.061*** | -0.061*** |
| | (0.010) | (0.010) | (0.010) |
| Technicians and trades | -0.235*** | -0.235*** | -0.235*** |
| | (0.011) | (0.011) | (0.011) |
| Community and personal service | -0.238*** | -0.238*** | -0.238*** |
| | (0.012) | (0.012) | (0.012) |
| Clerical and administrative | -0.226*** | -0.226*** | -0.226*** |
| | (0.010) | (0.010) | (0.010) |
| Sales | -0.284*** | -0.284*** | -0.284*** |
| | (0.012) | (0.012) | (0.012) |
| Machinery operators and drivers | -0.278*** | -0.279*** | -0.279*** |
| | (0.013) | (0.013) | (0.013) |
| Labourers | -0.322*** | -0.322*** | -0.322*** |
| | (0.012) | (0.012) | (0.012) |
| *Education* | | | |
| Postgraduate | 0.346*** | 0.346*** | 0.346*** |
| | (0.013) | (0.013) | (0.013) |
| Graduate diploma or certificate | 0.287*** | 0.287*** | 0.287*** |
| | (0.012) | (0.012) | (0.012) |
| Bachelor or honours | 0.260*** | 0.260*** | 0.260*** |
| | (0.009) | (0.009) | (0.009) |
| Advanced diploma or diploma | 0.171*** | 0.171*** | 0.171*** |
| | (0.009) | (0.009) | (0.009) |
| Certificate III or IV | 0.133*** | 0.133*** | 0.133*** |
| | (0.007) | (0.007) | (0.007) |
| Year 12 | 0.110*** | 0.110*** | 0.110*** |
| | (0.008) | (0.008) | (0.008) |
| *Job characteristics* | | | |
| Permanent (vs casual) | -0.021*** | -0.021*** | -0.021*** |
| | (0.006) | (0.006) | (0.006) |
| Occupational tenure (years) | 0.004*** | 0.004*** | 0.004*** |
| | (0.000) | (0.000) | (0.000) |
| Employer tenure (years) | 0.001*** | 0.001*** | 0.001*** |
| | (0.000) | (0.000) | (0.000) |
| Union member | 0.047*** | 0.047*** | 0.047*** |
| | (0.006) | (0.006) | (0.006) |
| Observations (person-years) | 156,668 | 156,668 | 156,668 |
| R-Squared | 0.490 | 0.490 | 0.490 |
| Adjusted R-Squared | 0.488 | 0.488 | 0.488 |
| Individuals | 23,057 | 23,057 | 23,057 |
| Year FEs | Yes | Yes | Yes |
| Industry-year FEs | Yes | Yes | Yes |
| State-year FEs | Yes | Yes | Yes |
| F-Statistic (df) | 11.959 (3, 397) | 12.533 (3, 397) | 14.286 (3, 397) |
| F-test p-value | < 0.001 | < 0.001 | < 0.001 |

*Notes:* This table reports OLS estimates of Equation (4.1) with small workplaces defined as having fewer than 10 people. The dependent variable is the log hourly wage in the employee's main job. Each column represents a different weighting of the owner shares, namely no decay (Type 0, $\lambda = 0$), pooled decay (Type 1, $\lambda = 0.02$) and origin-specific decay (Type 2, $\lambda_{\text{MES}} = 0.04$ and $\lambda_{\text{MNES}} = 0$). All columns include covariates for age, sex, origin group, casual or permanent employment, occupation, education, SEIFA decile, occupational tenure, employer tenure and union membership, and year, industry-year and state-year fixed effects. Reference categories are managers and administrators (occupation), Year 11 or below (education), native-born (origin), casual

employment and no union membership. The F-statistic, with its degrees of freedom in parentheses, tests the null hypothesis that the three owner-share interactions are jointly zero, using the clustered covariance matrix; its p-value is reported in the row below. All standard errors reported in parentheses are clustered at the individual and industry-year levels. * p < 0.10, ** p < 0.05, *** p < 0.01.

**Table B.5:** *Employee wage results under the three weightings of the owner shares, with all controls.*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| *Firm structure* | | | |
| Small single-site | -0.113*** | -0.104*** | -0.114*** |
| | (0.018) | (0.016) | (0.016) |
| Chain-like | 0.060*** | 0.060*** | 0.060*** |
| | (0.005) | (0.005) | (0.005) |
| Small-site chain | -0.030*** | -0.030*** | -0.030*** |
| | (0.005) | (0.005) | (0.005) |
| *Interactions of firm structure with owner density* | | | |
| Small single-site x MES owner density | -0.330*** | -0.595*** | -0.862*** |
| | (0.092) | (0.142) | (0.199) |
| Small single-site x MNES high-English owner density | 0.277*** | 0.379*** | 0.266*** |
| | (0.058) | (0.090) | (0.058) |
| Small single-site x MNES low-English owner density | -0.182 | -0.263 | -0.171 |
| | (0.173) | (0.206) | (0.175) |
| *Demographics* | | | |
| Age | 0.006*** | 0.006*** | 0.006*** |
| | (0.000) | (0.000) | (0.000) |
| Female | -0.080*** | -0.080*** | -0.080*** |
| | (0.005) | (0.005) | (0.005) |
| MES worker | 0.017** | 0.017** | 0.017** |
| | (0.008) | (0.008) | (0.008) |
| MNES high-English worker | -0.026*** | -0.026*** | -0.026*** |
| | (0.008) | (0.008) | (0.008) |
| MNES low-English worker | -0.163*** | -0.163*** | -0.163*** |
| | (0.030) | (0.030) | (0.030) |
| SEIFA decile | 0.013*** | 0.013*** | 0.013*** |
| | (0.001) | (0.001) | (0.001) |
| *Occupation* | | | |
| Professionals | -0.060*** | -0.060*** | -0.060*** |
| | (0.010) | (0.010) | (0.010) |
| Technicians and trades | -0.234*** | -0.233*** | -0.233*** |
| | (0.011) | (0.011) | (0.011) |
| Community and personal service | -0.236*** | -0.236*** | -0.236*** |
| | (0.012) | (0.012) | (0.012) |
| Clerical and administrative | -0.226*** | -0.226*** | -0.226*** |
| | (0.010) | (0.010) | (0.010) |
| Sales | -0.283*** | -0.283*** | -0.283*** |
| | (0.012) | (0.012) | (0.012) |
| Machinery operators and drivers | -0.278*** | -0.278*** | -0.278*** |
| | (0.013) | (0.013) | (0.013) |
| Labourers | -0.323*** | -0.323*** | -0.323*** |
| | (0.012) | (0.012) | (0.012) |
| *Education* | | | |
| Postgraduate | 0.344*** | 0.344*** | 0.344*** |
| | (0.013) | (0.013) | (0.013) |

**Table B.5:** *Employee wage results under the three weightings of the owner shares, with all controls. (continued)*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| Graduate diploma or certificate | 0.286*** | 0.286*** | 0.286*** |
| | (0.012) | (0.012) | (0.012) |
| Bachelor or honours | 0.259*** | 0.259*** | 0.259*** |
| | (0.009) | (0.009) | (0.009) |
| Advanced diploma or diploma | 0.171*** | 0.171*** | 0.171*** |
| | (0.009) | (0.009) | (0.009) |
| Certificate III or IV | 0.133*** | 0.133*** | 0.133*** |
| | (0.007) | (0.007) | (0.007) |
| Year 12 | 0.110*** | 0.110*** | 0.110*** |
| | (0.008) | (0.008) | (0.008) |
| *Job characteristics* | | | |
| Permanent (vs casual) | -0.020*** | -0.020*** | -0.020*** |
| | (0.006) | (0.006) | (0.006) |
| Occupational tenure (years) | 0.004*** | 0.004*** | 0.004*** |
| | (0.000) | (0.000) | (0.000) |
| Employer tenure (years) | 0.001*** | 0.001*** | 0.001*** |
| | (0.000) | (0.000) | (0.000) |
| Union member | 0.046*** | 0.046*** | 0.046*** |
| | (0.006) | (0.006) | (0.006) |
| Observations (person-years) | 156,786 | 156,786 | 156,786 |
| R-Squared | 0.490 | 0.490 | 0.490 |
| Adjusted R-Squared | 0.488 | 0.488 | 0.488 |
| Individuals | 23,067 | 23,067 | 23,067 |
| Year FEs | Yes | Yes | Yes |
| Industry-year FEs | Yes | Yes | Yes |
| State-year FEs | Yes | Yes | Yes |
| F-Statistic (df) | 14.602 (3, 397) | 14.923 (3, 397) | 16.674 (3, 397) |
| F-test p-value | < 0.001 | < 0.001 | < 0.001 |

*Notes:* This table reports OLS estimates of Equation (4.1) with small workplaces defined as having fewer than 20 people. These are the full results of Table 5.1. The dependent variable is the log hourly wage in the employee's main job. Each column represents a different weighting of the owner shares, namely no decay (Type 0, $\lambda = 0$), pooled decay (Type 1, $\lambda = 0.02$) and origin-specific decay (Type 2, $\lambda_{\text{MES}} = 0.04$ and $\lambda_{\text{MNES}} = 0$). All columns include covariates for age, sex, origin group, casual or permanent employment, occupation, education, SEIFA decile, occupational tenure, employer tenure and union membership, and year, industry-year and state-year fixed effects. Reference categories are managers and administrators (occupation), Year 11 or below (education), native-born (origin), casual employment and no union membership. The F-statistic, with its degrees of freedom in parentheses, tests the null hypothesis that the three owner-share interactions are jointly zero, using the clustered covariance matrix; its p-value is reported in the row below. All standard errors reported in parentheses are clustered at the individual and industry-year levels. * p < 0.10, ** p < 0.05, *** p < 0.01.

**Table B.6:** *Log hourly wages under 50-employee small-firm size threshold (full estimation results).*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| *Firm structure* | | | |
| Small single-site | -0.132*** | -0.122*** | -0.128*** |
| | (0.016) | (0.015) | (0.015) |
| Chain-like | 0.044*** | 0.044*** | 0.044*** |
| | (0.005) | (0.005) | (0.005) |
| Small-site chain | -0.055*** | -0.055*** | -0.055*** |
| | (0.004) | (0.004) | (0.004) |
| *Interactions of firm structure with owner density* | | | |
| Small single-site x MES owner density | -0.259*** | -0.511*** | -0.787*** |
| | (0.083) | (0.130) | (0.180) |

**Table B.6:** *Log hourly wages under 50-employee small-firm size threshold (full estimation results). (continued)*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| Small single-site x MNES high-English owner density | 0.255*** | 0.359*** | 0.245*** |
| | (0.055) | (0.084) | (0.055) |
| Small single-site x MNES low-English owner density | -0.199 | -0.301 | -0.194 |
| | (0.177) | (0.212) | (0.180) |
| *Demographics* | | | |
| Age | 0.006*** | 0.006*** | 0.006*** |
| | (0.000) | (0.000) | (0.000) |
| Female | -0.080*** | -0.080*** | -0.080*** |
| | (0.005) | (0.005) | (0.005) |
| MES worker | 0.015* | 0.015* | 0.015* |
| | (0.008) | (0.008) | (0.008) |
| MNES high-English worker | -0.028*** | -0.028*** | -0.028*** |
| | (0.008) | (0.008) | (0.008) |
| MNES low-English worker | -0.162*** | -0.161*** | -0.161*** |
| | (0.029) | (0.029) | (0.029) |
| SEIFA decile | 0.013*** | 0.013*** | 0.013*** |
| | (0.001) | (0.001) | (0.001) |
| *Occupation* | | | |
| Professionals | -0.061*** | -0.061*** | -0.061*** |
| | (0.010) | (0.010) | (0.010) |
| Technicians and trades | -0.234*** | -0.234*** | -0.234*** |
| | (0.011) | (0.011) | (0.011) |
| Community and personal service | -0.234*** | -0.234*** | -0.234*** |
| | (0.012) | (0.012) | (0.012) |
| Clerical and administrative | -0.226*** | -0.226*** | -0.226*** |
| | (0.010) | (0.010) | (0.010) |
| Sales | -0.282*** | -0.281*** | -0.281*** |
| | (0.012) | (0.012) | (0.012) |
| Machinery operators and drivers | -0.277*** | -0.277*** | -0.277*** |
| | (0.013) | (0.013) | (0.013) |
| Labourers | -0.323*** | -0.323*** | -0.323*** |
| | (0.012) | (0.012) | (0.012) |
| *Education* | | | |
| Postgraduate | 0.340*** | 0.340*** | 0.340*** |
| | (0.013) | (0.013) | (0.013) |
| Graduate diploma or certificate | 0.283*** | 0.283*** | 0.283*** |
| | (0.012) | (0.012) | (0.012) |
| Bachelor or honours | 0.258*** | 0.258*** | 0.257*** |
| | (0.009) | (0.009) | (0.009) |
| Advanced diploma or diploma | 0.170*** | 0.170*** | 0.170*** |
| | (0.009) | (0.009) | (0.009) |
| Certificate III or IV | 0.133*** | 0.133*** | 0.133*** |
| | (0.007) | (0.007) | (0.007) |
| Year 12 | 0.109*** | 0.109*** | 0.109*** |
| | (0.008) | (0.008) | (0.008) |
| *Job characteristics* | | | |
| Permanent (vs casual) | -0.019*** | -0.019*** | -0.019*** |
| | (0.006) | (0.006) | (0.006) |
| Occupational tenure (years) | 0.004*** | 0.004*** | 0.004*** |
| | (0.000) | (0.000) | (0.000) |
| Employer tenure (years) | 0.001*** | 0.001*** | 0.001*** |
| | (0.000) | (0.000) | (0.000) |

**Table B.6:** *Log hourly wages under 50-employee small-firm size threshold (full estimation results). (continued)*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| Union member | 0.045*** | 0.045*** | 0.045*** |
| | (0.006) | (0.006) | (0.006) |
| Observations (person-years) | 156,301 | 156,301 | 156,301 |
| R-Squared | 0.492 | 0.492 | 0.492 |
| Adjusted R-Squared | 0.490 | 0.490 | 0.490 |
| Individuals | 23,018 | 23,018 | 23,018 |
| Year FEs | Yes | Yes | Yes |
| Industry-year FEs | Yes | Yes | Yes |
| State-year FEs | Yes | Yes | Yes |
| F-Statistic (df) | 12.783 (3, 397) | 14.246 (3, 397) | 16.334 (3, 397) |
| F-test p-value | < 0.001 | < 0.001 | < 0.001 |

*Notes:* This table reports OLS estimates of Equation (4.1) with small workplaces defined as having fewer than 50 people. The dependent variable is the log hourly wage in the employee's main job. Each column represents a different weighting of the owner shares, namely no decay (Type 0, $\lambda = 0$), pooled decay (Type 1, $\lambda = 0.02$) and origin-specific decay (Type 2, $\lambda_{\text{MES}} = 0.04$ and $\lambda_{\text{MNES}} = 0$). All columns include covariates for age, sex, origin group, casual or permanent employment, occupation, education, SEIFA decile, occupational tenure, employer tenure and union membership, and year, industry-year and state-year fixed effects. Reference categories are managers and administrators (occupation), Year 11 or below (education), native-born (origin), casual employment and no union membership. The F-statistic, with its degrees of freedom in parentheses, tests the null hypothesis that the three owner-share interactions are jointly zero, using the clustered covariance matrix; its p-value is reported in the row below. All standard errors reported in parentheses are clustered at the individual and industry-year levels. * p < 0.10, ** p < 0.05, *** p < 0.01.

**Table B.7:** *Log hourly wages under 20-employee small-firm size threshold, migrant and English-proficiency owner share specification, by industry classification granularity.*

| | Dependent variable: log hourly wage in main job | |
|---|---|---|
| | One-digit industry | Two-digit industry |
| *Firm structure* | | |
| Small single-site | -0.148*** | -0.101*** |
| | (0.017) | (0.010) |
| Chain-like | 0.059*** | 0.047*** |
| | (0.005) | (0.006) |
| Small-site chain | -0.031*** | -0.033*** |
| | (0.005) | (0.006) |
| *Interactions of firm structure with owner density* | | |
| Small single-site x Migrant owner density | 0.156*** | 0.012 |
| | (0.051) | (0.024) |
| Small single-site x Low-English owner density | -0.298 | -0.089 |
| | (0.194) | (0.080) |
| *Demographics* | | |
| Age | 0.006*** | 0.006*** |
| | (0.000) | (0.000) |
| Female | -0.080*** | -0.068*** |
| | (0.005) | (0.006) |
| MES worker | 0.017** | 0.013* |
| | (0.008) | (0.008) |
| MNES high-English worker | -0.026*** | -0.031*** |
| | (0.008) | (0.008) |
| MNES low-English worker | -0.162*** | -0.154*** |
| | (0.029) | (0.029) |
| SEIFA decile | 0.013*** | 0.013*** |
| | (0.001) | (0.001) |
| *Occupation* | | |

**Table B.7:** *Log hourly wages under 20-employee small-firm size threshold, migrant and English-proficiency owner share specification, by industry classification granularity. (continued)*

| | **Dependent variable: log hourly wage in main job** | |
|---|---|---|
| | One-digit industry | Two-digit industry |
| Professionals | -0.060*** | -0.066*** |
| | (0.010) | (0.010) |
| Technicians and trades | -0.235*** | -0.241*** |
| | (0.011) | (0.011) |
| Community and personal service | -0.235*** | -0.205*** |
| | (0.012) | (0.012) |
| Clerical and administrative | -0.226*** | -0.229*** |
| | (0.010) | (0.010) |
| Sales | -0.284*** | -0.277*** |
| | (0.012) | (0.012) |
| Machinery operators and drivers | -0.277*** | -0.268*** |
| | (0.013) | (0.012) |
| Labourers | -0.324*** | -0.307*** |
| | (0.012) | (0.013) |
| *Education* | | |
| Postgraduate | 0.345*** | 0.335*** |
| | (0.013) | (0.013) |
| Graduate diploma or certificate | 0.286*** | 0.289*** |
| | (0.012) | (0.012) |
| Bachelor or honours | 0.259*** | 0.259*** |
| | (0.009) | (0.009) |
| Advanced diploma or diploma | 0.171*** | 0.174*** |
| | (0.009) | (0.009) |
| Certificate III or IV | 0.133*** | 0.135*** |
| | (0.007) | (0.007) |
| Year 12 | 0.110*** | 0.110*** |
| | (0.008) | (0.008) |
| *Job characteristics* | | |
| Permanent (vs casual) | -0.020*** | -0.024*** |
| | (0.006) | (0.006) |
| Occupational tenure (years) | 0.004*** | 0.004*** |
| | (0.000) | (0.000) |
| Employer tenure (years) | 0.001*** | 0.001** |
| | (0.000) | (0.000) |
| Union member | 0.046*** | 0.041*** |
| | (0.006) | (0.006) |
| Observations (person-years) | 156,786 | 144,607 |
| R-Squared | 0.490 | 0.502 |
| Adjusted R-Squared | 0.488 | 0.496 |
| Individuals | 23,067 | 22,756 |
| Industry classification | 1-digit (19 categories) | 2-digit (85 categories) |
| Year FEs | Yes | Yes |
| Industry-year FEs | Yes | Yes |
| State-year FEs | Yes | Yes |
| F-Statistic (df) | 5.302 (2, 397) | 0.649 (2, 397) |
| F-test p-value | 0.005 | 0.523 |

*Notes:* This table reports OLS estimates of an alternative to Equation (4.1) that replaces the three origin-specific owner shares with the share of all migrant owners, $MO_{kt}^{(-i)}$, and the share of low-English migrant owners, $LEMO_{kt}^{(-i)}$,

$$
\begin{aligned}
\log w_{it} = {} & \delta_t + \mu_{kt} + \nu_{rt} \\
& + \beta_1 \text{SmallSingleSite}_{it} + \beta_2 \text{SmallSiteChain}_{it} + \beta_3 \text{ChainLike}_{it} \\
& + \beta_4 \left( \text{SmallSingleSite}_{it} \times \text{MO}_{kt}^{(-i)} \right) + \beta_5 \left( \text{SmallSingleSite}_{it} \times \text{LEMO}_{kt}^{(-i)} \right) \\
& + \mathbf{X}_{it}' \boldsymbol{\pi} + \epsilon_{it},
\end{aligned}
\tag{B.1}
$$

where $\text{MO}_{kt}^{(-i)} = \frac{\sum_{m \in (k,t), m \neq i} \omega_{mt} \text{Owner}_{mt} \text{Migrant}_m}{\sum_{m \in (k,t), m \neq i} \omega_{mt} \text{Owner}_{mt}}$ and $\text{LEMO}_{kt}^{(-i)} = \frac{\sum_{m \in (k,t), m \neq i} \omega_{mt} \text{Owner}_{mt} \text{Migrant}_m \text{LowEnglish}_m}{\sum_{m \in (k,t), m \neq i} \omega_{mt} \text{Owner}_{mt}}$ are leave-one-out shares in industry $k$ and year $t$, without decay. Low-English owners are also migrants, so $\beta_5$ measures how the small single-site gap changes with the share of low-English owners over and above the share of migrant owners. The dependent variable is the log hourly wage in the employee's main job. Each column represents a different level at which the industry cells, and so the owner shares and industry-year fixed effects, are defined, namely the 19 one-digit ANZSIC divisions used throughout, or two-digit ANZSIC industries. All columns include covariates for age, sex, origin group, casual or permanent employment, occupation, education, SEIFA decile, occupational tenure, employer tenure and union membership, and year, industry-year and state-year fixed effects. Reference categories are managers and administrators (occupation), Year 11 or below (education), native-born (origin), casual employment and no union membership. The F-statistic, with its degrees of freedom in parentheses, tests the null hypothesis that the two owner-share interactions are jointly zero, using the clustered covariance matrix; its p-value is reported in the row below. All standard errors reported in parentheses are clustered at the individual and one-digit industry-year levels. $^{*}$ p < 0.10, $^{**}$ p < 0.05, $^{***}$ p < 0.01.

**Table B.8:** *Baseline regression results for employee wages, two-digit ANZSIC industry classification (full estimation results).*

| | **Dependent variable: log hourly wage in main job** | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| *Firm structure* | | | |
| Small single-site | -0.094*** | -0.094*** | -0.094*** |
| | (0.010) | (0.010) | (0.010) |
| Chain-like | 0.047*** | 0.047*** | 0.047*** |
| | (0.006) | (0.006) | (0.006) |
| Small-site chain | -0.033*** | -0.033*** | -0.033*** |
| | (0.006) | (0.006) | (0.006) |
| *Interactions of firm structure with owner density* | | | |
| Small single-site x MES owner density | -0.100** | -0.184** | -0.287*** |
| | (0.047) | (0.076) | (0.109) |
| Small single-site x MNES high-English owner density | 0.048* | 0.070 | 0.046* |
| | (0.027) | (0.043) | (0.027) |
| Small single-site x MNES low-English owner density | -0.110 | -0.096 | -0.108 |
| | (0.078) | (0.098) | (0.078) |
| *Demographics* | | | |
| Age | 0.006*** | 0.006*** | 0.006*** |
| | (0.000) | (0.000) | (0.000) |
| Female | -0.068*** | -0.068*** | -0.068*** |
| | (0.006) | (0.006) | (0.006) |
| MES worker | 0.013* | 0.013* | 0.013* |
| | (0.008) | (0.008) | (0.008) |
| MNES high-English worker | -0.031*** | -0.031*** | -0.031*** |
| | (0.008) | (0.008) | (0.008) |
| MNES low-English worker | -0.156*** | -0.156*** | -0.156*** |
| | (0.029) | (0.029) | (0.029) |
| SEIFA decile | 0.013*** | 0.013*** | 0.013*** |
| | (0.001) | (0.001) | (0.001) |
| *Occupation* | | | |
| Professionals | -0.066*** | -0.066*** | -0.066*** |
| | (0.010) | (0.010) | (0.010) |
| Technicians and trades | -0.241*** | -0.241*** | -0.241*** |
| | (0.011) | (0.011) | (0.011) |
| Community and personal service | -0.205*** | -0.205*** | -0.205*** |
| | (0.012) | (0.012) | (0.012) |

**Table B.8:** *Baseline regression results for employee wages, two-digit ANZSIC industry classification (full estimation results). (continued)*

| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| Clerical and administrative | -0.229*** | -0.229*** | -0.229*** |
| | (0.010) | (0.010) | (0.010) |
| Sales | -0.277*** | -0.277*** | -0.277*** |
| | (0.012) | (0.012) | (0.012) |
| Machinery operators and drivers | -0.268*** | -0.268*** | -0.268*** |
| | (0.012) | (0.012) | (0.012) |
| Labourers | -0.307*** | -0.306*** | -0.307*** |
| | (0.013) | (0.013) | (0.013) |
| *Education* | | | |
| Postgraduate | 0.335*** | 0.335*** | 0.335*** |
| | (0.013) | (0.013) | (0.013) |
| Graduate diploma or certificate | 0.289*** | 0.289*** | 0.289*** |
| | (0.012) | (0.012) | (0.012) |
| Bachelor or honours | 0.259*** | 0.259*** | 0.259*** |
| | (0.009) | (0.009) | (0.009) |
| Advanced diploma or diploma | 0.174*** | 0.174*** | 0.174*** |
| | (0.009) | (0.009) | (0.009) |
| Certificate III or IV | 0.135*** | 0.135*** | 0.135*** |
| | (0.007) | (0.007) | (0.007) |
| Year 12 | 0.110*** | 0.110*** | 0.110*** |
| | (0.008) | (0.008) | (0.008) |
| *Job characteristics* | | | |
| Permanent (vs casual) | -0.024*** | -0.024*** | -0.024*** |
| | (0.006) | (0.006) | (0.006) |
| Occupational tenure (years) | 0.004*** | 0.004*** | 0.004*** |
| | (0.000) | (0.000) | (0.000) |
| Employer tenure (years) | 0.001** | 0.001** | 0.001** |
| | (0.000) | (0.000) | (0.000) |
| Union member | 0.041*** | 0.041*** | 0.041*** |
| | (0.006) | (0.006) | (0.006) |
| Observations (person-years) | 144,607 | 144,607 | 144,607 |
| R-Squared | 0.502 | 0.502 | 0.502 |
| Adjusted R-Squared | 0.496 | 0.496 | 0.496 |
| Individuals | 22,756 | 22,756 | 22,756 |
| Year FEs | Yes | Yes | Yes |
| Industry-year FEs (2-digit) | Yes | Yes | Yes |
| State-year FEs | Yes | Yes | Yes |
| F-Statistic (df) | 3.461 (3, 397) | 3.369 (3, 397) | 4.344 (3, 397) |
| F-test p-value | 0.016 | 0.019 | 0.005 |

*Notes:* This table reports OLS estimates of Equation (4.1) with the owner shares and the industry-year fixed effects defined over two-digit ANZSIC industries instead of the 19 one-digit divisions used in Table 5.1. The dependent variable is the log hourly wage in the employee's main job. Each column represents a different weighting of the owner shares, namely no decay (Type 0, $\lambda = 0$), pooled decay (Type 1, $\lambda = 0.02$) and origin-specific decay (Type 2, $\lambda_{\text{MES}} = 0.04$ and $\lambda_{\text{MNES}} = 0$). All columns include covariates for age, sex, origin group, casual or permanent employment, occupation, education, SEIFA decile, occupational tenure, employer tenure and union membership, and year, industry-year and state-year fixed effects. Reference categories are managers and administrators (occupation), Year 11 or below (education), native-born (origin), casual employment and no union membership. The F-statistic, with its degrees of freedom in parentheses, tests the null hypothesis that the three owner-share interactions are jointly zero, using the clustered covariance matrix; its p-value is reported in the row below. All standard errors reported in parentheses are clustered at the individual and one-digit industry-year levels, as in Table 5.1. * p < 0.10, ** p < 0.05, *** p < 0.01.

**Inference with Few Industry Clusters**

The owner shares in Equation (4.1) vary only across the 19 one-digit industries and the years within them, and they change slowly within an industry. The standard errors in Table 5.1 are clustered by individual and by industry-year. This allows the errors of employees in the same industry and year to be correlated, but treats different years of the same industry as independent. If the errors have a persistent industry component, for example because small firms in some industries pay well for reasons that the controls do not capture, then both the owner shares and the errors are correlated across the years of an industry, and these standard errors understate the uncertainty (Moulton, 1990).

Clustering by industry allows for this correlation, but it creates a second problem. After the fixed effects are partialled out, the cluster-robust variance estimator is

$$\hat{V} = (X'X)^{-1}\left(\sum_{g=1}^{G} X_g'\hat{u}_g\hat{u}_g'X_g\right)(X'X)^{-1}, \tag{B.2}$$

where $X_g$ and $\hat{u}_g$ are the regressors and residuals of industry $g$. With $G = 19$, the middle term is built from only 19 outer products. The estimator is consistent only as the number of clusters grows. With few clusters it is biased downward, because least squares fits each cluster's residuals partly to its own data, and the resulting t-statistics have thicker tails than the normal distribution. Conventional tests therefore reject a true null hypothesis too often (Cameron et al., 2008).

To address this problem, we use the wild cluster bootstrap with the null hypothesis imposed (Cameron et al., 2008). To test $H_0 : \beta_j = 0$, we first estimate Equation (4.1) with $\beta_j$ set to zero, which gives the restricted fitted values $X_g\tilde{\beta}$ and residuals $\tilde{u}_g$. We then construct $B = 9,999$ bootstrap samples,

$$y_g^{*b} = X_g\tilde{\beta} + v_g^b\,\tilde{u}_g, \qquad g = 1, \ldots, G, \quad b = 1, \ldots, B, \tag{B.3}$$

where $v_g^b$ is drawn independently for each industry from the six-point distribution of Webb (2023), which takes the values $\pm\sqrt{1/2}$, $\pm 1$ and $\pm\sqrt{3/2}$ with equal probability. In each sample we re-estimate the unrestricted model and compute the t-statistic $t_j^{*b}$ with the industry-clustered variance of Equation (B.2). The bootstrap p-value is the share of these t-statistics that are at least as large in absolute value as the t-statistic $\hat{t}_j$ from the data,

$$p_j^* = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}\left[\,\left|t_j^{*b}\right| \geq \left|\hat{t}_j\right|\,\right]. \tag{B.4}$$

The joint test is constructed in the same way, with all three interactions set to zero and the Wald statistic in place of the t-statistic.

The procedure addresses both problems. First, every observation of an industry is multiplied by the same $v_g^b$, which has mean zero and variance one. Each bootstrap sample therefore keeps the correlation of the errors within an industry, across workers and across years, while industries remain independent. Second, $t_j^{*b}$ is computed with the same estimator and the same 19 clusters as $\hat{t}_j$, so the bootstrap distribution inherits the downward bias and the thick tails of the clustered t-statistic. We compare $\hat{t}_j$

with this distribution instead of the normal distribution, so we do not need the clustered standard error itself to be accurate. Imposing the null hypothesis ensures that the bootstrap samples are generated under the hypothesis being tested.

Table B.9 reports the results. Clustering by industry roughly doubles the standard errors of the MES and high-English interactions, from 0.092 to 0.180 and from 0.058 to 0.102, which indicates that the errors are correlated across the years of an industry. Under the wild cluster bootstrap, the high-English interaction remains significant at the 5 per cent level ($p = 0.044$), while the MES interaction ($p = 0.136$) and the joint test of all three interactions ($p = 0.315$) are not significant. The low-English interaction is insignificant under both approaches ($p = 0.692$). The narrowing of the small single-site wage gap where high-English MNES owners are common is therefore robust to correlation within industries, but the MES interaction is not. The bootstrap is least reliable for the low-English interaction, because that share is zero in most industry-years and its variation comes from a few industries (MacKinnon & Webb, 2018).

**Table B.9:** *Owner-share interactions with p-values that cluster by industry (wild cluster bootstrap).*

| Owner share | Coefficient | p-value, individual and industry-year clusters | Standard error, industry clusters | Wild bootstrap p-value, industry clusters |
|---|---|---|---|---|
| *No decay (Type 0)* | | | | |
| Small single-site x MES owner share | −0.330 | < 0.001 | 0.180 | 0.136 |
| Small single-site x MNES high-English owner share | 0.277 | < 0.001 | 0.102 | 0.044 |
| Small single-site x MNES low-English owner share | −0.182 | 0.295 | 0.188 | 0.692 |
| All three interactions jointly | | < 0.001 | | 0.315 |
| *Pooled decay (Type 1)* | | | | |
| Small single-site x MES owner share | −0.595 | < 0.001 | 0.269 | 0.110 |
| Small single-site x MNES high-English owner share | 0.379 | < 0.001 | 0.159 | 0.074 |
| Small single-site x MNES low-English owner share | −0.263 | 0.204 | 0.209 | 0.581 |
| All three interactions jointly | | < 0.001 | | 0.214 |
| *Origin-specific decay (Type 2)* | | | | |
| Small single-site x MES owner share | −0.862 | < 0.001 | 0.353 | 0.120 |
| Small single-site x MNES high-English owner share | 0.266 | < 0.001 | 0.099 | 0.053 |
| Small single-site x MNES low-English owner share | −0.171 | 0.329 | 0.195 | 0.691 |
| All three interactions jointly | | < 0.001 | | 0.223 |

*Notes:* This table reports the owner-share interactions of Table 5.1 with p-values from two clustering schemes, with one panel for each weighting of the owner shares. *p-value, individual and industry-year clusters* is from the standard errors of Table 5.1, and for the joint row from its F-test. The last two columns cluster at the industry level instead, which allows the errors of all years of an industry to be correlated. *Standard error, industry clusters* is the conventional clustered standard error with 19 industry clusters. *Wild bootstrap p-value, industry clusters* is from a wild cluster bootstrap that imposes the null hypothesis (Cameron et al., 2008), with the six-point weights of Webb (2023) and 9,999 replications. For the joint row, the null is that all three interactions are zero.

### Owner Shares Built with the Longitudinal Weight

Our main owner shares weight each owner by the HILDA cross-sectional responding-person weight, because each share describes the owners of an industry in a single wave. As a sensitivity check,

Table B.10 re-estimates the main specification and the two decay weightings with the shares weighted instead by the longitudinal responding-person weight. The decay rates are calibrated on these shares by an earlier version of the cross-validation in Section 3.5, which used a narrower employee sample and did not control for the worker's origin, giving $\lambda = 0.02$ for Type 1 and $\lambda_{\text{MES}} = 0.44$ and $\lambda_{\text{MNES}} = 0.01$ for Type 2. This weight is zero in wave 1 and positive only for people who have responded in every wave since wave 1, so it gives weight to a shrinking subset of owners. Over waves 2 to 21 it is positive for 57 per cent of owner person-years, but for only 24 per cent of the person-years of low-English owners, and for 14 of the 47 low-English owners observed in waves 1 to 21.

Without decay, the MES and high-English interactions have the same signs and similar sizes under both weights. The low-English interaction does not. Without decay it is 0.456 ($p < 0.001$) with the longitudinal weight but $-0.182$ ($p = 0.295$) with the cross-sectional weight (Table 5.1), and the wild cluster bootstrap finds it insignificant under both weights (Table B.9 and Table B.10). The estimate for low-English owners therefore depends on which few owners receive weight, and we do not treat it as evidence in either direction.

**Table B.10:** *Employee wage results with owner shares weighted by the longitudinal survey weight (sensitivity check).*

| | **Dependent variable: log hourly wage in main job** | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| *Firm structure* | | | |
| Small single-site | -0.112*** | -0.117*** | -0.150*** |
| | (0.015) | (0.014) | (0.011) |
| Chain-like | 0.059*** | 0.059*** | 0.058*** |
| | (0.005) | (0.005) | (0.005) |
| Small-site chain | -0.031*** | -0.031*** | -0.031*** |
| | (0.005) | (0.005) | (0.005) |
| *Interactions of firm structure with owner density* | | | |
| Small single-site x MES owner density | -0.254*** | -0.502*** | 4.242 |
| | (0.074) | (0.139) | (5.962) |
| Small single-site x MNES high-English owner density | 0.217*** | 0.451*** | 0.367*** |
| | (0.058) | (0.089) | (0.072) |
| Small single-site x MNES low-English owner density | 0.456*** | 0.666*** | 0.676*** |
| | (0.135) | (0.215) | (0.171) |
| *Demographics* | | | |
| Age | 0.006*** | 0.006*** | 0.006*** |
| | (0.000) | (0.000) | (0.000) |
| Female | -0.080*** | -0.080*** | -0.080*** |
| | (0.005) | (0.005) | (0.005) |
| MES worker | 0.016* | 0.016* | 0.016* |
| | (0.008) | (0.008) | (0.008) |
| MNES high-English worker | -0.028*** | -0.028*** | -0.028*** |
| | (0.008) | (0.008) | (0.008) |
| MNES low-English worker | -0.162*** | -0.162*** | -0.162*** |
| | (0.032) | (0.032) | (0.032) |
| SEIFA decile | 0.013*** | 0.013*** | 0.013*** |
| | (0.001) | (0.001) | (0.001) |
| *Occupation* | | | |
| Professionals | -0.060*** | -0.060*** | -0.060*** |
| | (0.010) | (0.010) | (0.010) |

**Table B.10:** *Employee wage results with owner shares weighted by the longitudinal survey weight (sensitivity check). (continued)*

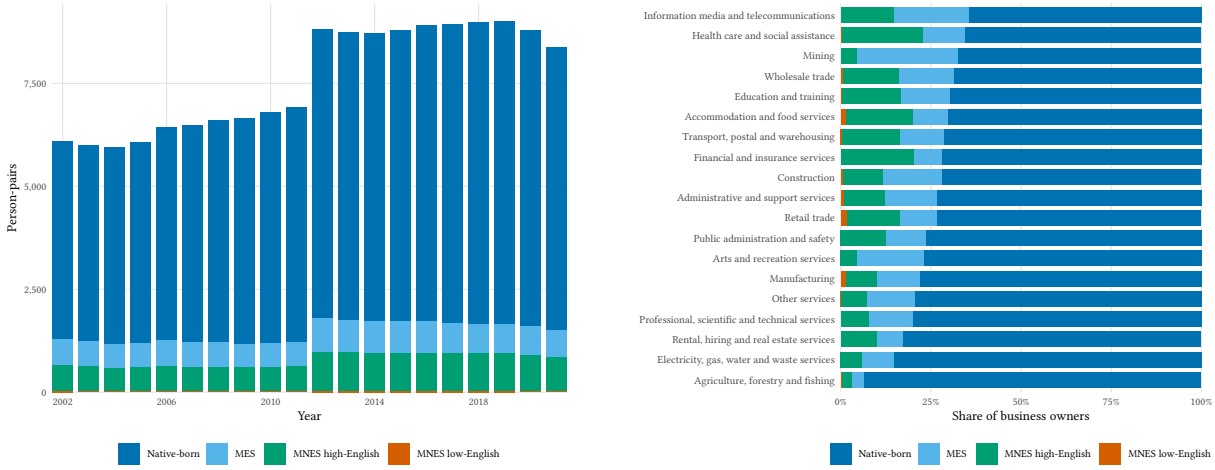
| | Dependent variable: log hourly wage in main job | | |
|---|---|---|---|
| | No decay (Type 0) | Pooled decay (Type 1) | Origin-specific decay (Type 2) |
| Technicians and trades | -0.237*** | -0.237*** | -0.237*** |
| | (0.011) | (0.011) | (0.011) |
| Community and personal service | -0.238*** | -0.238*** | -0.238*** |
| | (0.012) | (0.012) | (0.012) |
| Clerical and administrative | -0.228*** | -0.228*** | -0.228*** |
| | (0.010) | (0.010) | (0.010) |
| Sales | -0.287*** | -0.287*** | -0.287*** |
| | (0.013) | (0.013) | (0.013) |
| Machinery operators and drivers | -0.278*** | -0.278*** | -0.278*** |
| | (0.013) | (0.013) | (0.013) |
| Labourers | -0.324*** | -0.324*** | -0.324*** |
| | (0.013) | (0.013) | (0.013) |
| *Education* | | | |
| Postgraduate | 0.349*** | 0.349*** | 0.349*** |
| | (0.013) | (0.013) | (0.013) |
| Graduate diploma or certificate | 0.288*** | 0.288*** | 0.288*** |
| | (0.012) | (0.012) | (0.012) |
| Bachelor or honours | 0.261*** | 0.261*** | 0.262*** |
| | (0.009) | (0.009) | (0.009) |
| Advanced diploma or diploma | 0.172*** | 0.172*** | 0.173*** |
| | (0.009) | (0.009) | (0.009) |
| Certificate III or IV | 0.136*** | 0.136*** | 0.137*** |
| | (0.007) | (0.007) | (0.007) |
| Year 12 | 0.112*** | 0.111*** | 0.112*** |
| | (0.008) | (0.008) | (0.008) |
| *Job characteristics* | | | |
| Permanent (vs casual) | -0.021*** | -0.021*** | -0.021*** |
| | (0.006) | (0.006) | (0.006) |
| Occupational tenure (years) | 0.004*** | 0.004*** | 0.004*** |
| | (0.000) | (0.000) | (0.000) |
| Employer tenure (years) | 0.001*** | 0.001*** | 0.001*** |
| | (0.000) | (0.000) | (0.000) |
| Union member | 0.045*** | 0.045*** | 0.045*** |
| | (0.006) | (0.006) | (0.006) |
| Observations (person-years) | 149,673 | 149,673 | 149,673 |
| R-Squared | 0.481 | 0.482 | 0.481 |
| Adjusted R-Squared | 0.479 | 0.480 | 0.479 |
| Individuals | 22,335 | 22,335 | 22,335 |
| Year FEs | Yes | Yes | Yes |
| Industry-year FEs | Yes | Yes | Yes |
| State-year FEs | Yes | Yes | Yes |
| F-Statistic (df) | 17.109 (3, 371) | 23.551 (3, 371) | 15.013 (3, 371) |
| F-test p-value | < 0.001 | < 0.001 | < 0.001 |
| Wild bootstrap p-value, MES | 0.077 | 0.042 | 0.578 |
| Wild bootstrap p-value, MNES high-English | 0.087 | 0.013 | 0.031 |
| Wild bootstrap p-value, MNES low-English | 0.151 | 0.173 | 0.216 |
| Wild bootstrap p-value, joint | 0.022 | 0.021 | 0.033 |

*Notes:* This table reports OLS estimates of Equation (4.1) with each owner in the leave-one-out shares of Equation (3.7) weighted by the HILDA longitudinal responding-person weight instead of the cross-sectional weight. This weight is zero in wave 1 and positive only for people who have responded in every wave since, so there are fewer observations than in Table 5.1 where

no other owner in the industry-year has a positive weight. The dependent variable is the log hourly wage in the employee's main job. Each column represents a different weighting of the owner shares, with decay rates calibrated on the longitudinally weighted shares, namely no decay (Type 0), pooled decay (Type 1, $\lambda = 0.02$) and origin-specific decay (Type 2, $\lambda_{\text{MES}} = 0.44$ and $\lambda_{\text{MNES}} = 0.01$). All columns include covariates for age, sex, origin group, casual or permanent employment, occupation, education, SEIFA decile, occupational tenure, employer tenure and union membership, and year, industry-year and state-year fixed effects. Reference categories are managers and administrators (occupation), Year 11 or below (education), native-born (origin), casual employment and no union membership. The F-statistic, with its degrees of freedom in parentheses, tests the null hypothesis that the three owner-share interactions are jointly zero, using the clustered covariance matrix; its p-value is reported in the row below. The wild bootstrap p-values test each interaction, and all three jointly, with errors clustered at the industry level, as in Table B.9. All standard errors reported in parentheses are clustered at the individual and industry-year levels. $^{*}$ p < 0.10, $^{**}$ p < 0.05, $^{***}$ p < 0.01.

# C Owner Income and Risk

This section reports the composition of the owner income samples, the estimates for large employers and own-account workers, and the full estimates behind Table 5.2.




(a) Person-pairs by year. (b) Business owners by industry.

**Figure C.1:** *Composition of the estimation sample by origin group.*

*Notes:* This figure shows the composition of the restricted owner income sample of Table 5.2 by origin group, in HILDA waves 1 to 21. Panel (a) shows the number of person-pairs by year. Panel (b) shows the share of owners (with and without employees) in each industry who belong to each origin group, with industries ordered by the total migrant share. Both panels are unweighted. The rise in 2011 reflects the general top-up of the HILDA sample in that year.

**Table C.1:** *Sample cells, raw outcomes and attrition by ownership status, business size and origin group.*

| Group | Individuals | Person-pairs | Mean income | Median income | Volatility | Large fall (%) | Not re-interviewed (%) |
|---|---|---|---|---|---|---|---|
| Employees, small workplace | | | | | | | |
| Native-born | 9,494 | 37,609 | 46,136 | 40,000 | 36.3 | 10.3 | |
| MES | 875 | 3,347 | 52,646 | 46,000 | 29.4 | 9.8 | |
| MNES high-English | 1,035 | 3,522 | 48,690 | 43,000 | 35.8 | 10.1 | |
| MNES low-English | 54 | 152 | 33,869 | 30,000 | 42.3 | 11.9 | |
| Employees, large workplace | | | | | | | |
| Native-born | 11,737 | 71,161 | 66,538 | 57,000 | 27.4 | 6.9 | |
| MES | 1,246 | 7,805 | 78,539 | 65,000 | 23.3 | 6.5 | |
| MNES high-English | 1,525 | 9,430 | 68,675 | 60,000 | 26.0 | 7.0 | |

**Table C.1:** *Sample cells, raw outcomes and attrition by ownership status, business size and origin group. (continued)*

| Group | Individuals | Person-pairs | Mean income | Median income | Volatility | Large fall (%) | Not re-interviewed (%) |
|---|---|---|---|---|---|---|---|
| MNES low-English | 52 | 224 | 43,448 | 40,000 | 23.4 | 5.1 | |
| Small employers | | | | | | | |
| Native-born | 1,103 | 5,372 | 76,585 | 52,000 | 58.4 | 27.9 | 6.9 |
| MES | 151 | 757 | 79,691 | 55,205 | 53.5 | 27.2 | 7.8 |
| MNES high-English | 193 | 758 | 75,860 | 52,000 | 52.8 | 24.9 | 10.1 |
| MNES low-English | 13 | 49 | 58,724 | 52,000 | 75.5 | 31.1 | 9.5 |
| Large employers | | | | | | | |
| Native-born | 158 | 615 | 144,267 | 94,000 | 47.1 | 18.6 | 5.7 |
| MES | 20 | 79 | 290,924 | 108,000 | 43.7 | 26.0 | 3.1 |
| MNES high-English | 28 | 143 | 157,739 | 110,000 | 57.6 | 27.4 | 6.7 |
| Own-account workers | | | | | | | |
| Native-born | 2,029 | 8,373 | 43,936 | 30,000 | 72.3 | 33.7 | 7.0 |
| MES | 317 | 1,436 | 48,056 | 34,000 | 64.5 | 33.1 | 6.0 |
| MNES high-English | 286 | 1,198 | 46,586 | 30,000 | 69.7 | 31.1 | 7.7 |
| MNES low-English | 13 | 30 | 28,366 | 20,500 | 58.4 | 21.4 | 16.4 |

*Notes:* This table reports the number of individuals and person-pairs, raw outcomes and attrition by employment status, business size and origin group, in the restricted sample of individuals with the same status at $t-1$ and $t$. Income is annual labour and business income in dollars, volatility is the mean absolute arc per cent change of Equation (4.2) and a large fall is a fall in income of 25 per cent or more. *Not re-interviewed (%)* is the percentage of individuals at $t-1$ who are not interviewed at $t$, before any other sample restriction; these individuals drop out of both samples. Large employers with low English proficiency are omitted, since only one is observed. All figures are unweighted.

**Table C.2:** *Income and risk of large employers relative to employees of the same origin.*

| | Unrestricted sample: status at t | | | Restricted sample: same status at t–1 and t | | | |
|---|---|---|---|---|---|---|---|
| Dependent variable: | Income (dollars) | Volatility (pp) | Large fall (pp) | Income (dollars) | Volatility (pp) | Large fall (pp) | Large employers (restricted sample) |
| *Panel A: Owner–employee gap* | | | | | | | |
| Native-born | 55,750*** (18,871) | 22.3*** (3.4) | 10.2*** (1.9) | 57,824*** (21,708) | 25.7*** (3.7) | 11.7*** (2.1) | 158 |
| MES | 172,247 (125,637) | 24.5*** (7.1) | 15.8*** (4.6) | 205,027 (145,935) | 24.6*** (7.6) | 19.0*** (4.9) | 20 |
| MNES high-English | 62,529*** (18,505) | 33.3*** (7.7) | 18.0*** (4.1) | 69,505*** (20,371) | 37.0*** (8.7) | 21.0*** (4.7) | 28 |
| MNES low-English | – | – | – | – | – | – | – |
| *Panel B: Difference from the native-born owner–employee gap* | | | | | | | |
| MES | 116,497 (126,299) | 2.2 (7.7) | 5.6 (5.0) | 147,203 (146,738) | −1.1 (8.5) | 7.3 (5.5) | |
| MNES high-English | 6,779 (26,389) | 10.9 (8.3) | 7.8* (4.6) | 11,681 (29,620) | 11.3 (9.4) | 9.3* (5.1) | |
| MNES low-English | – | – | – | – | – | – | |
| *Panel C: Minimum detectable difference* | | | | | | | |
| MES | 353,637 | 21.7 | 14.1 | 410,865 | 23.8 | 15.4 | |

**Table C.2:** *Income and risk of large employers relative to employees of the same origin. (continued)*

| Dependent variable: | Unrestricted sample: status at t | | | Restricted sample: same status at t–1 and t | | | Large employers (restricted sample) |
|---|---|---|---|---|---|---|---|
| | Income (dollars) | Volatility (pp) | Large fall (pp) | Income (dollars) | Volatility (pp) | Large fall (pp) | |
| MNES high-English | 73,890 | 23.1 | 12.9 | 82,936 | 26.2 | 14.4 | |
| MNES low-English | – | – | – | – | – | – | |
| Observations (person-pairs) | 174,962 | 174,962 | 164,973 | 152,060 | 152,060 | 148,198 | |
| Individuals | 21,549 | 21,549 | 20,618 | 20,031 | 20,031 | 19,747 | |

*Notes:* This table reports the owner–employee gap for large employers (owners with employees and 20 or more people at their workplace), relative to employees of large workplaces of the same origin, from the regressions of Equation (4.3) in Table 5.2. Panel A reports the gap, $\theta_L$ for the native-born and $\theta_L + \theta_{Lg}$ for migrant group $g$. Panel B reports the difference from the native-born gap, $\theta_{Lg}$, and Panel C its minimum detectable difference, 2.8 × SE ($\theta_{Lg}$). The dependent variables are income in dollars, and volatility and large falls in percentage points. Each column represents a separate regression on the unrestricted or restricted sample, with the covariates and fixed effects of Table 5.2. The last column is the number of distinct large employers in the restricted income regression. A dash marks the single low-English large employer, who is excluded from estimation. All standard errors reported in parentheses are clustered at the individual and industry-year levels. * p < 0.10, ** p < 0.05, *** p < 0.01.

**Table C.3:** *Income and risk of own-account workers relative to employees of the same origin.*

| Dependent variable: | Unrestricted sample: status at t | | | Restricted sample: same status at t–1 and t | | | Own-account workers (restricted sample) |
|---|---|---|---|---|---|---|---|
| | Income (dollars) | Volatility (pp) | Large fall (pp) | Income (dollars) | Volatility (pp) | Large fall (pp) | |
| *Panel A: Owner–employee gap* | | | | | | | |
| Native-born | −9,379*** (1,169) | 33.7*** (1.1) | 19.8*** (0.5) | −13,463*** (1,536) | 41.0*** (1.3) | 22.3*** (0.6) | 2,029 |
| MES | −12,288*** (2,690) | 30.1*** (2.0) | 20.0*** (1.2) | −14,412*** (2,972) | 35.7*** (2.3) | 22.4*** (1.3) | 317 |
| MNES high-English | −4,294 (6,084) | 30.9*** (2.3) | 17.5*** (1.3) | −6,615 (7,473) | 37.6*** (2.9) | 19.6*** (1.6) | 286 |
| MNES low-English | −4,844 (10,842) | 23.0** (10.3) | 21.3*** (6.6) | −18,907*** (7,062) | 21.9** (9.9) | 11.1 (7.4) | 13 |
| *Panel B: Difference from the native-born owner–employee gap* | | | | | | | |
| MES | −2,908 (2,767) | −3.6* (2.1) | 0.2 (1.2) | −949 (3,112) | −5.4** (2.5) | 0.1 (1.4) | |
| MNES high-English | 5,086 (6,258) | −2.8 (2.5) | −2.3 (1.4) | 6,848 (7,664) | −3.4 (3.1) | −2.7 (1.7) | |
| MNES low-English | 4,535 (10,929) | −10.7 (10.3) | 1.5 (6.7) | −5,444 (7,252) | −19.1* (10.0) | −11.2 (7.4) | |
| *Panel C: Minimum detectable difference* | | | | | | | |
| MES | 7,747 | 5.9 | 3.5 | 8,715 | 6.9 | 3.8 | |
| MNES high-English | 17,522 | 7.0 | 3.9 | 21,459 | 8.8 | 4.7 | |
| MNES low-English | 30,602 | 28.9 | 18.7 | 20,306 | 28.0 | 20.7 | |
| Observations (person-pairs) | 174,962 | 174,962 | 164,973 | 152,060 | 152,060 | 148,198 | |
| Individuals | 21,549 | 21,549 | 20,618 | 20,031 | 20,031 | 19,747 | |

*Notes:* This table reports the owner–employee gap for own-account workers (self-employed, or owners without employees, in small workplaces), relative to employees of small workplaces of the same origin, from the regressions of Equation (4.3) in

Table 5.2. Panel A reports the gap, $\theta_O$ for the native-born and $\theta_O + \theta_{Og}$ for migrant group $g$. Panel B reports the difference from the native-born gap, $\theta_{Og}$, and Panel C its minimum detectable difference, $2.8 \times \text{SE}(\theta_{Og})$. The dependent variables are income in dollars, and volatility and large falls in percentage points. Each column represents a separate regression on the unrestricted or restricted sample, with the covariates and fixed effects of Table 5.2. The last column is the number of distinct own-account workers in the restricted income regression. All standard errors reported in parentheses are clustered at the individual and industry-year levels. * p < 0.10, ** p < 0.05, *** p < 0.01.

**Table C.4:** *Income level, volatility and downside risk by ownership status, business size and origin group (full estimation results).*

| | Unrestricted sample: status at t | | | Restricted sample: same status at t–1 and t | | |
|---|---|---|---|---|---|---|
| | Income (dollars) | Volatility (pp) | Large fall (share) | Income (dollars) | Volatility (pp) | Large fall (share) |
| Ownership and origin | | | | | | |
| Small employer | 12,730*** | 27.280*** | 0.151*** | 14,369*** | 32.655*** | 0.173*** |
| | (3,136) | (1.209) | (0.007) | (4,168) | (1.407) | (0.008) |
| Large employer | 55,750*** | 22.344*** | 0.102*** | 57,824*** | 25.710*** | 0.117*** |
| | (18,871) | (3.413) | (0.019) | (21,708) | (3.745) | (0.021) |
| Own-account worker | -9,379*** | 33.742*** | 0.198*** | -13,463*** | 41.012*** | 0.223*** |
| | (1,169) | (1.068) | (0.005) | (1,536) | (1.264) | (0.006) |
| MES | 913 | 1.464*** | 0.000 | 909 | 1.102** | -0.001 |
| | (1,109) | (0.534) | (0.003) | (1,140) | (0.458) | (0.003) |
| MNES high-English | -9,279*** | 2.696*** | 0.005* | -9,415*** | 2.425*** | 0.004 |
| | (1,105) | (0.563) | (0.003) | (1,182) | (0.472) | (0.003) |
| MNES low-English | -16,655*** | 5.656* | -0.008 | -16,970*** | 2.911 | -0.012 |
| | (2,330) | (3.175) | (0.014) | (2,619) | (2.448) | (0.013) |
| MES x Small employer | -3,468 | -1.886 | 0.004 | -2,704 | -3.542 | -0.000 |
| | (6,054) | (3.043) | (0.018) | (7,407) | (3.595) | (0.020) |
| MNES high-English x Small employer | 4,487 | -5.461* | -0.028** | 5,929 | -6.891* | -0.034** |
| | (5,773) | (2.821) | (0.014) | (6,622) | (3.586) | (0.017) |
| MNES low-English x Small employer | 11,810 | 11.805 | 0.044 | 4,800 | 11.629 | 0.034 |
| | (10,666) | (12.299) | (0.071) | (8,305) | (14.537) | (0.085) |
| MES x Large employer | 116,497 | 2.170 | 0.056 | 147,203 | -1.093 | 0.073 |
| | (126,299) | (7.748) | (0.050) | (146,738) | (8.486) | (0.055) |
| MNES high-English x Large employer | 6,779 | 10.934 | 0.078* | 11,681 | 11.332 | 0.093* |
| | (26,389) | (8.252) | (0.046) | (29,620) | (9.360) | (0.051) |
| MES x Own-account worker | -2,908 | -3.650* | 0.002 | -949 | -5.361** | 0.001 |
| | (2,767) | (2.120) | (0.012) | (3,112) | (2.477) | (0.014) |
| MNES high-English x Own-account worker | 5,086 | -2.830 | -0.023 | 6,848 | -3.370 | -0.027 |
| | (6,258) | (2.485) | (0.014) | (7,664) | (3.133) | (0.017) |
| MNES low-English x Own-account worker | 4,535 | -10.717 | 0.015 | -5,444 | -19.144* | -0.112 |
| | (10,929) | (10.331) | (0.067) | (7,252) | (10.000) | (0.074) |
| Demographics | | | | | | |
| Age | 3,524*** | -2.860*** | -0.006*** | 3,471*** | -2.521*** | -0.005*** |
| | (194) | (0.129) | (0.001) | (181) | (0.122) | (0.001) |
| Age squared | -40.408*** | 0.032*** | 0.000*** | -39.427*** | 0.027*** | 0.000*** |
| | (2.361) | (0.001) | (0.000) | (2.202) | (0.001) | (0.000) |

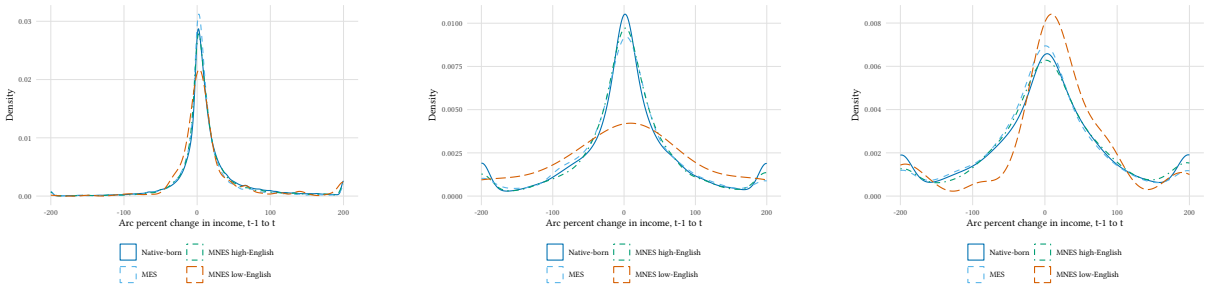
**Table C.4:** *Income level, volatility and downside risk by ownership status, business size and origin group (full estimation results). (continued)*

| | Unrestricted sample: status at t | | | Restricted sample: same status at t–1 and t | | |
|---|---|---|---|---|---|---|
| | Income (dollars) | Volatility (pp) | Large fall (share) | Income (dollars) | Volatility (pp) | Large fall (share) |
| Female | -20,495*** | 3.962*** | 0.019*** | -20,651*** | 3.240*** | 0.014*** |
| | (916) | (0.384) | (0.002) | (958) | (0.358) | (0.002) |
| De facto | -1,384* | -4.247*** | -0.009*** | -1,562** | -3.030*** | -0.008*** |
| | (766) | (0.482) | (0.003) | (789) | (0.415) | (0.002) |
| Separated | -504 | 1.081 | -0.014*** | 531 | 0.376 | -0.013*** |
| | (1,149) | (0.894) | (0.005) | (1,186) | (0.814) | (0.005) |
| Divorced | -1,861* | 0.267 | -0.015*** | -1,328 | 0.335 | -0.013*** |
| | (990) | (0.721) | (0.004) | (948) | (0.625) | (0.004) |
| Widowed | 2,432 | -4.002** | -0.016 | 2,830 | -4.635*** | -0.010 |
| | (3,178) | (1.742) | (0.010) | (3,221) | (1.647) | (0.010) |
| Never married | -8,254*** | 1.431*** | -0.011*** | -8,093*** | 1.712*** | -0.008*** |
| | (780) | (0.545) | (0.003) | (777) | (0.475) | (0.003) |
| Education (reference: year 11 and below) | | | | | | |
| Postgraduate | 37,991*** | -0.973 | -0.002 | 39,845*** | -0.673 | -0.003 |
| | (2,433) | (0.930) | (0.004) | (2,548) | (0.814) | (0.004) |
| Graduate diploma or certificate | 26,435*** | -0.978 | 0.004 | 27,693*** | -0.344 | -0.001 |
| | (1,782) | (0.848) | (0.004) | (1,805) | (0.744) | (0.004) |
| Bachelor or honours | 20,323*** | -2.770*** | -0.001 | 21,829*** | -1.931*** | -0.002 |
| | (1,045) | (0.688) | (0.003) | (1,092) | (0.595) | (0.003) |
| Advanced diploma or diploma | 10,345*** | -3.705*** | -0.001 | 11,152*** | -3.222*** | -0.006* |
| | (1,029) | (0.756) | (0.004) | (1,074) | (0.654) | (0.003) |
| Certificate III or IV | 4,052*** | -4.081*** | -0.002 | 4,463*** | -3.138*** | -0.002 |
| | (656) | (0.585) | (0.003) | (661) | (0.528) | (0.003) |
| Year 12 | 4,055*** | -5.553*** | 0.000 | 4,483*** | -4.034*** | 0.001 |
| | (881) | (0.676) | (0.003) | (857) | (0.595) | (0.003) |
| Location and job characteristics | | | | | | |
| Inner regional | -2,209*** | 3.053*** | 0.009*** | -2,244*** | 2.583*** | 0.009*** |
| | (635) | (0.470) | (0.003) | (656) | (0.438) | (0.002) |
| Outer regional | 409 | 2.257*** | 0.007* | 544 | 2.244*** | 0.006 |
| | (860) | (0.695) | (0.004) | (898) | (0.641) | (0.004) |
| Remote | 5,081 | 5.999*** | 0.036*** | 5,374* | 6.137*** | 0.033*** |
| | (3,500) | (1.574) | (0.009) | (3,115) | (1.586) | (0.009) |
| Very remote | 12,782*** | 2.234 | 0.019 | 12,156*** | 3.944 | 0.022 |
| | (3,561) | (2.980) | (0.018) | (3,510) | (3.102) | (0.019) |
| SEIFA disadvantage decile (1 = most disadvantaged) | 1,874*** | 0.009 | -0.000 | 1,910*** | 0.124** | 0.000 |
| | (121) | (0.065) | (0.000) | (127) | (0.058) | (0.000) |
| Occupational tenure (years) | 533*** | -0.411*** | -0.001*** | 460*** | -0.261*** | -0.001*** |
| | (43.128) | (0.024) | (0.000) | (41.422) | (0.020) | (0.000) |
| Small single-site workplace | -12,343*** | 6.164*** | 0.045*** | -11,743*** | 3.053*** | 0.033*** |
| | (832) | (0.553) | (0.003) | (871) | (0.475) | (0.003) |
| Small site of multi-site employer | -4,466*** | 3.206*** | 0.021*** | -4,223*** | 1.902*** | 0.015*** |
| | (694) | (0.461) | (0.003) | (724) | (0.397) | (0.003) |
| Chain-like workplace | 5,951*** | -1.685*** | -0.010*** | 5,767*** | -1.058*** | -0.005** |
| | (890) | (0.364) | (0.002) | (912) | (0.306) | (0.002) |
| Prior income decile (reference: decile 5) | | | | | | |

**Table C.4:** *Income level, volatility and downside risk by ownership status, business size and origin group (full estimation results). (continued)*

| | Unrestricted sample: status at t | | | Restricted sample: same status at t−1 and t | | |
|---|---|---|---|---|---|---|
| | Income (dollars) | Volatility (pp) | Large fall (share) | Income (dollars) | Volatility (pp) | Large fall (share) |
| Decile 1 (lowest) | | 63.087*** | 0.044*** | | 57.309*** | 0.032** |
| | | (2.023) | (0.014) | | (2.712) | (0.014) |
| Decile 2 | | 67.491*** | 0.016 | | 57.139*** | -0.002 |
| | | (1.840) | (0.012) | | (2.263) | (0.012) |
| Decile 3 | | 35.678*** | 0.019*** | | 30.167*** | 0.015*** |
| | | (1.143) | (0.006) | | (1.110) | (0.006) |
| Decile 4 | | 12.669*** | 0.008* | | 10.995*** | 0.009** |
| | | (0.802) | (0.005) | | (0.765) | (0.005) |
| Decile 6 | | -7.908*** | -0.012*** | | -6.756*** | -0.011*** |
| | | (0.594) | (0.004) | | (0.574) | (0.004) |
| Decile 7 | | -10.979*** | -0.017*** | | -9.230*** | -0.017*** |
| | | (0.647) | (0.004) | | (0.599) | (0.004) |
| Decile 8 | | -12.481*** | -0.022*** | | -10.366*** | -0.020*** |
| | | (0.702) | (0.004) | | (0.649) | (0.004) |
| Decile 9 | | -12.122*** | -0.018*** | | -10.285*** | -0.017*** |
| | | (0.767) | (0.004) | | (0.718) | (0.004) |
| Decile 10 (highest) | | -9.716*** | -0.005 | | -8.384*** | -0.009** |
| | | (0.903) | (0.004) | | (0.824) | (0.004) |
| Fewer than two prior waves | | 10.885*** | 0.004 | | 7.111*** | 0.002 |
| | | (1.223) | (0.004) | | (1.041) | (0.004) |
| Sample | Unrestricted | Unrestricted | Unrestricted | Restricted | Restricted | Restricted |
| Same status at t−1 required | No | No | No | Yes | Yes | Yes |
| Year FEs | Yes | Yes | Yes | Yes | Yes | Yes |
| Industry-year FEs | Yes | Yes | Yes | Yes | Yes | Yes |
| State-year FEs | Yes | Yes | Yes | Yes | Yes | Yes |
| F-test (df) | 0.65 (8, 379) | 1.43 (8, 379) | 1.44 (8, 379) | 0.53 (8, 379) | 1.92 (8, 379) | 1.66 (8, 379) |
| F-test (p-value) | 0.737 | 0.182 | 0.180 | 0.837 | 0.055 | 0.106 |
| Individuals | 21,549 | 21,549 | 20,618 | 20,031 | 20,031 | 19,747 |
| Observations (person-pairs) | 174,962 | 174,962 | 164,973 | 152,060 | 152,060 | 148,198 |
| R-squared | 0.300 | 0.216 | 0.072 | 0.312 | 0.211 | 0.074 |

*Notes:* This table reports OLS estimates of Equation (4.3). These are the full results of Table 5.2 and Table 5.3. The dependent variables are annual labour and business income at $t$ in dollars, income volatility (the absolute arc per cent change of Equation (4.2), 0 to 200) and an indicator for an income fall of 25 per cent or more, in shares (0.145 is 14.5 percentage points). Each column represents a separate regression; the first three use the unrestricted sample, which classifies individuals by their status at $t$, and the last three the restricted sample of individuals with the same status at $t-1$ and $t$. The owner-type rows are $\theta_S$, $\theta_L$ and $\theta_O$, the origin rows are $\gamma_g$ and the interaction rows are $\theta_{Sg}$, $\theta_{Lg}$ and $\theta_{Og}$; the owner–employee gap of group $g$ in Table 5.2 is the owner-type row plus the interaction row. All columns include year, industry-year and state-year fixed effects; the volatility and large-fall columns also include the prior income decile, which ranks average income over waves $t-5$ to $t-2$, with a separate category for individuals with fewer than two waves observed. The reference categories are native-born employees, decile 5 and Year 11 or below. The single low-English large employer is excluded, so that interaction is not estimated. The F-test, with its degrees of freedom in parentheses and its p-value in the row below, tests the null hypothesis that all origin by owner-type interactions are jointly zero, using the clustered covariance matrix. All standard errors reported in parentheses are clustered at the individual and industry-year levels. * p < 0.10, ** p < 0.05, *** p < 0.01.




(a) Employees. (b) Owners with employees. (c) Own-account workers.

**Figure C.2:** *Distribution of year-to-year income changes by ownership status and origin group.*

*Notes:* This figure plots kernel densities of the arc per cent change in annual labour and business income between consecutive waves, which ranges from −200 to 200, by origin group, in the restricted sample of Table 5.2. Values of ±200 arise when income moves to or from zero or changes sign, and the concentration at zero reflects identical incomes reported in consecutive years. Densities for low-English owners rest on few observations.

# D Entry into Business Ownership

This section reports unconditional entry rates, the full estimates behind Table 5.4 and the robustness checks discussed in Section 5.3.

**Table D.1:** *Transitions into business ownership by migrant and language proficiency group.*

| Group | At-risk person-pairs | Entries | Entry rate (%) | 95% CI | p vs Australian-born | Standardised rate (%) |
|---|---|---|---|---|---|---|
| Australian-born | 230,642 | 4,595 | 1.99 | [1.92, 2.07] | | 1.99 |
| Migrant, main English-speaking country | 26,573 | 591 | 2.22 | [1.99, 2.48] | 0.012 | 2.31 |
| Migrant, other country, high English | 30,589 | 653 | 2.13 | [1.93, 2.35] | 0.095 | 2.18 |
| Migrant, other country, low English | 3,332 | 37 | 1.11 | [0.73, 1.59] | < 0.001 | 1.75 |

*Notes:* This table reports unconditional entry into business ownership in the transition sample, by origin group. The entry rate is the percentage of at-risk person-pairs (not a business owner at $t-1$) in which the individual owns a business at $t$. The 95% confidence interval is from a bootstrap that resamples individuals (2,000 replications). *p vs Australian-born* is from a Fisher exact test; a chi-square test of equal rates across all groups gives p < 0.001. *Standardised rate* is the entry rate each group would have with the Australian-born mix of age band and industry of most recent job at $t-1$, computed as each group's average predicted entry rate over the Australian-born person-pairs from a logit of entry on origin group, age band and industry.

**Table D.2:** *Full estimation results for entry into business ownership between t−1 and t.*

| | Dependent variable: entry into business ownership (logit) | | | |
|---|---|---|---|---|
| | Origin only | Labour market only | Origin and labour market | Adding human capital |
| *Origin and English proficiency* | | | | |
| Migrant with English-speaking origin | 0.038<br>(0.087) | | 0.075<br>(0.087) | -0.022<br>(0.088) |
| Migrant from other origin with high English proficiency | 0.053<br>(0.084) | | 0.066<br>(0.083) | -0.059<br>(0.084) |
| Migrant from other origin with low English proficiency | -0.098<br>(0.207) | | -0.230<br>(0.208) | -0.256<br>(0.207) |
| *Labour-market position at t−1* | | | | |

**Table D.2:** *Full estimation results for entry into business ownership between t–1 and t. (continued)*

| | Dependent variable: entry into business ownership (logit) | | | |
|---|---|---|---|---|
| | Origin only | Labour market only | Origin and labour market | Adding human capital |
| Unemployed | | 0.777*** | 0.777*** | 0.700*** |
| | | (0.067) | (0.067) | (0.068) |
| Not in labour force | | 0.588*** | 0.590*** | 0.491*** |
| | | (0.048) | (0.048) | (0.052) |
| Bottom wage quintile | | 0.406*** | 0.408*** | 0.430*** |
| | | (0.047) | (0.047) | (0.048) |
| Liquidity constrained | | 0.151*** | 0.152*** | 0.149*** |
| | | (0.038) | (0.038) | (0.038) |
| Liquidity measure missing | | 0.304*** | 0.310*** | 0.322*** |
| | | (0.047) | (0.047) | (0.047) |
| *Human capital and job attributes at t–1* | | | | |
| Postgraduate | | | | 0.648*** |
| | | | | (0.087) |
| Graduate diploma or certificate | | | | 0.472*** |
| | | | | (0.085) |
| Bachelor or honours | | | | 0.419*** |
| | | | | (0.064) |
| Advanced diploma or diploma | | | | 0.474*** |
| | | | | (0.068) |
| Certificate III or IV | | | | 0.376*** |
| | | | | (0.053) |
| Year 12 | | | | 0.241*** |
| | | | | (0.060) |
| Tenure with current employer (years) | | | | -0.018*** |
| | | | | (0.003) |
| *Demographics* | | | | |
| Age | 0.112*** | 0.131*** | 0.130*** | 0.129*** |
| | (0.007) | (0.008) | (0.008) | (0.008) |
| Age squared | -0.001*** | -0.002*** | -0.002*** | -0.001*** |
| | (0.000) | (0.000) | (0.000) | (0.000) |
| Female | -0.264*** | -0.336*** | -0.335*** | -0.313*** |
| | (0.038) | (0.038) | (0.038) | (0.038) |
| Years since arrival in Australia | 0.002 | 0.003* | 0.001 | 0.003 |
| | (0.002) | (0.001) | (0.002) | (0.002) |
| Years since arrival missing | -0.516 | -0.524 | -0.569 | -0.430 |
| | (0.879) | (0.881) | (0.879) | (0.824) |
| De facto | 0.011 | 0.007 | 0.006 | -0.000 |
| | (0.045) | (0.045) | (0.045) | (0.045) |
| Separated | -0.156* | -0.203** | -0.205** | -0.197** |
| | (0.091) | (0.091) | (0.091) | (0.091) |
| Divorced | -0.200** | -0.234*** | -0.234*** | -0.242*** |
| | (0.079) | (0.079) | (0.079) | (0.079) |
| Widowed | -0.618*** | -0.602*** | -0.604*** | -0.589*** |
| | (0.170) | (0.169) | (0.169) | (0.168) |
| Never married and not de facto | -0.428*** | -0.505*** | -0.504*** | -0.467*** |
| | (0.055) | (0.056) | (0.056) | (0.055) |
| *Household and location at t–1* | | | | |
| Homeowner | -0.052 | 0.025 | 0.027 | 0.028 |
| | (0.035) | (0.036) | (0.036) | (0.036) |
| SEIFA disadvantage decile | 0.086*** | 0.098*** | 0.098*** | 0.086*** |
| | (0.007) | (0.007) | (0.007) | (0.007) |

**Table D.2:** *Full estimation results for entry into business ownership between t−1 and t. (continued)*

| | Dependent variable: entry into business ownership (logit) | | | |
|---|---|---|---|---|
| | Origin only | Labour market only | Origin and labour market | Adding human capital |
| Inner regional Australia | 0.330*** | 0.309*** | 0.309*** | 0.315*** |
| | (0.047) | (0.046) | (0.047) | (0.047) |
| Outer regional Australia | 0.275*** | 0.257*** | 0.257*** | 0.272*** |
| | (0.065) | (0.065) | (0.065) | (0.065) |
| Remote Australia | 0.366*** | 0.345** | 0.346** | 0.355*** |
| | (0.137) | (0.136) | (0.136) | (0.137) |
| Very remote Australia | 1.030*** | 1.004*** | 1.015*** | 1.002*** |
| | (0.202) | (0.206) | (0.204) | (0.206) |
| Observations (person-pairs) | 289,616 | 289,616 | 289,616 | 289,616 |
| AIC | 54352 | 53974 | 53976 | 53786 |
| BIC | 61460 | 61102 | 61136 | 61020 |
| Individuals | 28,626 | 28,626 | 28,626 | 28,626 |
| Year FEs | Yes | Yes | Yes | Yes |
| Industry-year FEs | Yes | Yes | Yes | Yes |
| State-year FEs | Yes | Yes | Yes | Yes |
| Wald $\chi^2$ (p-value) | | 268.05 (< 0.001) | 268.83 (< 0.001) | 197.35 (< 0.001) |

*Notes:* This table reports logit estimates of Equation (4.5). These are the full results of Table 5.4. The dependent variable equals one if an individual who is not a business owner at $t-1$ is an own-account worker or employer at $t$. Each column adds regressors in steps, starting with origin group only, then labour-market position and liquidity at $t-1$ only, then both and finally both with education and employer tenure at $t-1$. All columns include year, industry-year and state-year fixed effects. Coefficients are log-odds. Reference categories are the Australian-born, employees who were not low-paid and not liquidity constrained at $t-1$, Year 11 or below as highest education, male, married, not a homeowner and living in a major city. The Wald $\chi^2$ statistic, with its p-value in parentheses, tests the null hypothesis that the four labour-market and liquidity coefficients are jointly zero. All standard errors reported in parentheses are clustered at the individual level. * p < 0.10, ** p < 0.05, *** p < 0.01.

**Table D.3:** *Timing of labour market status and within-person estimates of entry into business ownership (logit coefficients).*

| | Dependent variable: entry into business ownership (logit) | | | |
|---|---|---|---|---|
| | Status at t−1 | Status at t−2 | Status at t−1 and t−2 | With person means |
| *Push from the labour market, by origin* | | | | |
| Unemployed at t−1 | 0.649*** | | 0.834*** | 0.370*** |
| | (0.083) | | (0.087) | (0.085) |
| Bottom wage quintile at t−1 | 0.356*** | | 0.519*** | 0.087 |
| | (0.056) | | (0.058) | (0.056) |
| MES x Unemployed at t−1 | -0.095 | | -0.242 | 0.178 |
| | (0.252) | | (0.263) | (0.219) |
| MNES high English x Unemployed at t−1 | 0.305 | | 0.364* | 0.332* |
| | (0.197) | | (0.206) | (0.178) |
| MNES low English x Unemployed at t−1 | 1.137* | | 0.546 | 1.377*** |
| | (0.628) | | (0.663) | (0.502) |
| MES x Bottom wage quintile at t−1 | 0.419*** | | 0.382** | 0.476*** |
| | (0.156) | | (0.166) | (0.145) |
| MNES high English x Bottom wage quintile at t−1 | 0.222 | | 0.172 | 0.192 |
| | (0.149) | | (0.164) | (0.138) |
| MNES low English x Bottom wage quintile at t−1 | 0.309 | | 0.389 | 0.570 |
| | (0.515) | | (0.497) | (0.461) |
| Unemployed at t−2 | | -0.027 | -0.321*** | |
| | | (0.086) | (0.091) | |
| Bottom wage quintile at t−2 | | -0.301*** | -0.469*** | |
| | | (0.058) | (0.062) | |

**Table D.3:** *Timing of labour market status and within-person estimates of entry into business ownership (logit coefficients). (continued)*

| | Dependent variable: entry into business ownership (logit) | | | |
|---|---|---|---|---|
| | Status at t−1 | Status at t−2 | Status at t−1 and t−2 | With person means |
| MES x Unemployed at t−2 | | 0.394* | 0.429* | |
| | | (0.236) | (0.250) | |
| MNES high English x Unemployed at t−2 | | -0.166 | -0.303 | |
| | | (0.236) | (0.248) | |
| MNES low English x Unemployed at t−2 | | 1.697*** | 1.492*** | |
| | | (0.488) | (0.511) | |
| MES x Bottom wage quintile at t−2 | | 0.111 | -0.021 | |
| | | (0.198) | (0.213) | |
| MNES high English x Bottom wage quintile at t−2 | | 0.243 | 0.174 | |
| | | (0.160) | (0.177) | |
| MNES low English x Bottom wage quintile at t−2 | | -0.211 | -0.435 | |
| | | (0.654) | (0.677) | |
| *Origin and English proficiency* | | | | |
| Migrant with English-speaking origin | -0.072 | -0.094 | -0.093 | -0.089 |
| | (0.103) | (0.099) | (0.103) | (0.095) |
| Migrant from other origin with high English proficiency | -0.120 | -0.096 | -0.117 | -0.120 |
| | (0.098) | (0.095) | (0.098) | (0.092) |
| Migrant from other origin with low English proficiency | -0.329 | -0.243 | -0.338 | -0.543** |
| | (0.276) | (0.276) | (0.295) | (0.258) |
| *Other labour-market position* | | | | |
| Not in labour force at t−1 | 0.493*** | | 0.954*** | 0.290*** |
| | (0.054) | | (0.059) | (0.060) |
| Not in labour force at t−2 | | -0.563*** | -1.025*** | |
| | | (0.062) | (0.069) | |
| Liquidity constrained at t−1 | 0.115*** | | 0.128*** | 0.042 |
| | (0.042) | | (0.042) | (0.043) |
| Liquidity constrained at t−2 | | 0.113*** | | |
| | | (0.041) | | |
| Liquidity measure missing at t−1 | 0.305*** | | 0.315*** | 0.296*** |
| | (0.051) | | (0.050) | (0.047) |
| Liquidity measure missing at t−2 | | 0.253*** | | |
| | | (0.052) | | |
| *Person means* | | | | |
| Person mean of unemployment | | | | 0.848*** |
| | | | | (0.188) |
| Person mean of bottom wage quintile | | | | 0.964*** |
| | | | | (0.116) |
| Person mean of not in labour force | | | | 0.496*** |
| | | | | (0.093) |
| Person mean of liquidity constraint | | | | 0.264*** |
| | | | | (0.089) |
| Person mean of homeownership | | | | -0.044 |
| | | | | (0.076) |
| Person mean of SEIFA decile | | | | 0.063*** |
| | | | | (0.014) |
| *Human capital and job attributes at t−1* | | | | |
| Postgraduate | 0.627*** | 0.507*** | 0.581*** | 0.699*** |
| | (0.092) | (0.092) | (0.091) | (0.088) |
| Graduate diploma or certificate | 0.447*** | 0.359*** | 0.407*** | 0.521*** |
| | (0.090) | (0.090) | (0.089) | (0.086) |

**Table D.3:** *Timing of labour market status and within-person estimates of entry into business ownership (logit coefficients). (continued)*

| | Dependent variable: entry into business ownership (logit) | | | |
|---|---|---|---|---|
| | Status at t−1 | Status at t−2 | Status at t−1 and t−2 | With person means |
| Bachelor or honours | 0.378*** | 0.279*** | 0.336*** | 0.463*** |
| | (0.069) | (0.069) | (0.068) | (0.066) |
| Advanced diploma or diploma | 0.438*** | 0.381*** | 0.411*** | 0.501*** |
| | (0.073) | (0.073) | (0.072) | (0.068) |
| Certificate III or IV | 0.337*** | 0.279*** | 0.311*** | 0.398*** |
| | (0.057) | (0.057) | (0.056) | (0.053) |
| Year 12 | 0.212*** | 0.188*** | 0.202*** | 0.249*** |
| | (0.065) | (0.066) | (0.065) | (0.061) |
| Tenure with current employer (years) | -0.017*** | -0.039*** | -0.021*** | -0.016*** |
| | (0.004) | (0.004) | (0.004) | (0.003) |
| *Demographics* | | | | |
| Age | 0.120*** | 0.094*** | 0.102*** | 0.135*** |
| | (0.008) | (0.008) | (0.009) | (0.008) |
| Age squared | -0.001*** | -0.001*** | -0.001*** | -0.002*** |
| | (0.000) | (0.000) | (0.000) | (0.000) |
| Female | -0.296*** | -0.200*** | -0.257*** | -0.345*** |
| | (0.041) | (0.041) | (0.040) | (0.039) |
| Years since arrival in Australia | 0.003 | 0.004 | 0.004 | 0.003 |
| | (0.003) | (0.003) | (0.003) | (0.003) |
| Years since arrival missing | -0.171 | -0.169 | -0.173 | -0.478 |
| | (0.792) | (0.796) | (0.806) | (0.823) |
| De facto | -0.023 | -0.051 | -0.037 | -0.021 |
| | (0.049) | (0.048) | (0.048) | (0.045) |
| Separated | -0.216** | -0.201** | -0.205** | -0.212** |
| | (0.098) | (0.098) | (0.097) | (0.092) |
| Divorced | -0.241*** | -0.254*** | -0.241*** | -0.264*** |
| | (0.083) | (0.082) | (0.082) | (0.080) |
| Widowed | -0.637*** | -0.698*** | -0.657*** | -0.596*** |
| | (0.172) | (0.173) | (0.171) | (0.169) |
| Never married and not de facto | -0.472*** | -0.410*** | -0.442*** | -0.534*** |
| | (0.058) | (0.057) | (0.057) | (0.056) |
| *Household and location at t−1* | | | | |
| Homeowner | -0.000 | -0.033 | -0.014 | 0.084 |
| | (0.038) | (0.038) | (0.038) | (0.052) |
| SEIFA disadvantage decile | 0.084*** | 0.074*** | 0.080*** | 0.047*** |
| | (0.008) | (0.007) | (0.007) | (0.011) |
| Inner regional Australia | 0.313*** | 0.337*** | 0.320*** | 0.326*** |
| | (0.050) | (0.050) | (0.049) | (0.047) |
| Outer regional Australia | 0.250*** | 0.258*** | 0.247*** | 0.284*** |
| | (0.070) | (0.070) | (0.069) | (0.065) |
| Remote Australia | 0.393*** | 0.386*** | 0.380*** | 0.398*** |
| | (0.144) | (0.144) | (0.142) | (0.137) |
| Very remote Australia | 1.066*** | 1.031*** | 1.053*** | 1.036*** |
| | (0.216) | (0.212) | (0.211) | (0.207) |
| Observations (person-pairs) | 256,524 | 256,524 | 256,524 | 289,616 |
| Individuals | 25,257 | 25,257 | 25,257 | 28,626 |
| Labour market status measured at | t−1 | t−2 | t−1 and t−2 | t−1, with person means |
| Sample | Observed at t−2 | Observed at t−2 | Observed at t−2 | Full |
| Joint test of interactions (p-value) | 0.027 | 0.004 | 0.002 | 0.001 |

**Table D.3:** *Timing of labour market status and within-person estimates of entry into business ownership (logit coefficients). (continued)*

*Notes:* This table reports logit estimates of Equation (4.6) with labour market status measured at different times. The dependent variable equals one if an individual who is not a business owner at $t-1$ is an own-account worker or employer at $t$. The first three columns use the person-pairs whose labour market status is also observed two years before entry, so that differences between them reflect timing and not sample composition, and measure unemployment and low wages at $t-1$, at $t-2$ and at both. *With person means* uses the full transition sample and adds each individual's mean of the time-varying regressors, so that the $t-1$ coefficients compare an individual with themselves at other times (Mundlak, 1978). All columns include the covariates of Table 5.5 and year, industry-year and state-year fixed effects. Coefficients are log-odds, with reference categories as in Table D.2. The joint test reports the p-value of a Wald test of the null hypothesis that all origin by labour-market interactions in the column are jointly zero. All standard errors reported in parentheses are clustered at the individual level. $^{*}$ p < 0.10, $^{**}$ p < 0.05, $^{***}$ p < 0.01.

**Table D.4:** *Linear probability models of entry into business ownership (percentage points).*

| | **Dependent variable: entry into business ownership × 100 (OLS)** | | | | | |
|---|---|---|---|---|---|---|
| | Origin only | Labour market only | Origin and labour market | Adding human capital | Origin × push | Origin × push, individual FEs |
| *Origin and English proficiency* | | | | | | |
| Migrant with English-speaking origin | 0.247 | | 0.322* | 0.124 | -0.128 | |
| | (0.187) | | (0.187) | (0.188) | (0.195) | |
| Migrant from other origin with high English proficiency | 0.207 | | 0.236 | -0.025 | -0.231 | |
| | (0.175) | | (0.174) | (0.179) | (0.185) | |
| Migrant from other origin with low English proficiency | -0.107 | | -0.317 | -0.410 | -0.689*** | |
| | (0.249) | | (0.252) | (0.253) | (0.237) | |
| *Labour-market position at t–1* | | | | | | |
| Unemployed | | 1.651*** | 1.651*** | 1.548*** | 1.311*** | 0.874*** |
| | | (0.181) | (0.181) | (0.184) | (0.191) | (0.223) |
| Not in labour force | | 0.929*** | 0.936*** | 0.784*** | 0.782*** | 0.577*** |
| | | (0.111) | (0.111) | (0.122) | (0.122) | (0.166) |
| Bottom wage quintile | | 0.727*** | 0.733*** | 0.797*** | 0.620*** | 0.672*** |
| | | (0.112) | (0.112) | (0.113) | (0.117) | (0.127) |
| Liquidity constrained | | 0.351*** | 0.356*** | 0.338*** | 0.338*** | 0.120 |
| | | (0.081) | (0.081) | (0.081) | (0.080) | (0.088) |
| Liquidity measure missing | | 0.687*** | 0.708*** | 0.727*** | 0.726*** | 0.366*** |
| | | (0.106) | (0.107) | (0.107) | (0.107) | (0.113) |
| *Origin × labour-market position at t–1* | | | | | | |
| MES × Unemployed | | | | | 0.996 | 1.279 |
| | | | | | (0.763) | (0.841) |
| MNES high English × Unemployed | | | | | 1.338** | -0.160 |
| | | | | | (0.651) | (0.674) |
| MNES low English × Unemployed | | | | | 3.339 | -1.161 |
| | | | | | (2.200) | (2.194) |
| MES × Bottom wage quintile | | | | | 1.739*** | 1.426** |
| | | | | | (0.554) | (0.584) |
| MNES high English × Bottom wage quintile | | | | | 0.737* | -0.262 |
| | | | | | (0.399) | (0.444) |
| MNES low English × Bottom wage quintile | | | | | 1.487 | 3.112* |
| | | | | | (1.338) | (1.847) |
| *Human capital and job attributes at t–1* | | | | | | |
| Postgraduate | | | | 1.400*** | 1.402*** | 1.433*** |
| | | | | (0.210) | (0.210) | (0.492) |
| Graduate diploma or certificate | | | | 0.929*** | 0.929*** | 0.756 |
| | | | | (0.177) | (0.177) | (0.483) |
| Bachelor or honours | | | | 0.799*** | 0.794*** | 0.667** |
| | | | | (0.122) | (0.122) | (0.290) |

**Table D.4:** *Linear probability models of entry into business ownership (percentage points). (continued)*

| | Dependent variable: entry into business ownership × 100 (OLS) | | | | | |
|---|---|---|---|---|---|---|
| | Origin only | Labour market only | Origin and labour market | Adding human capital | Origin × push | Origin × push, individual FEs |
| Advanced diploma or diploma | | | | 0.895*** (0.137) | 0.890*** (0.137) | 1.260*** (0.360) |
| Certificate III or IV | | | | 0.697*** (0.099) | 0.692*** (0.099) | 0.929*** (0.238) |
| Year 12 | | | | 0.303*** (0.097) | 0.296*** (0.097) | 0.637*** (0.170) |
| Tenure with current employer (years) | | | | -0.032*** (0.006) | -0.032*** (0.006) | 0.014* (0.008) |
| *Demographics* | | | | | | |
| Age | 0.133*** (0.009) | 0.151*** (0.010) | 0.151*** (0.010) | 0.149*** (0.010) | 0.146*** (0.010) | -7.627 (43917.949) |
| Age squared | -0.002*** (0.000) | -0.002*** (0.000) | -0.002*** (0.000) | -0.002*** (0.000) | -0.002*** (0.000) | -0.004*** (0.000) |
| Female | -0.521*** (0.073) | -0.582*** (0.072) | -0.579*** (0.072) | -0.525*** (0.073) | -0.526*** (0.073) | 1.135 (26711.081) |
| Years since arrival in Australia | -0.001 (0.004) | 0.003 (0.002) | -0.003 (0.004) | 0.001 (0.004) | 0.004 (0.004) | 0.013 (0.016) |
| Years since arrival missing | -0.987 (0.874) | -1.047 (0.937) | -1.215 (0.919) | -0.960 (0.831) | -0.975 (0.805) | 0.531 (0.432) |
| De facto | -0.038 (0.113) | -0.047 (0.112) | -0.046 (0.113) | -0.076 (0.112) | -0.076 (0.112) | -0.542*** (0.187) |
| Separated | -0.377** (0.181) | -0.477*** (0.181) | -0.477*** (0.181) | -0.446** (0.181) | -0.444** (0.181) | -0.633** (0.260) |
| Divorced | -0.420*** (0.131) | -0.494*** (0.131) | -0.491*** (0.131) | -0.479*** (0.131) | -0.478*** (0.131) | -0.879*** (0.274) |
| Widowed | -0.046 (0.101) | 0.012 (0.103) | 0.013 (0.103) | 0.058 (0.103) | 0.053 (0.103) | 0.103 (0.212) |
| Never married and not de facto | -1.014*** (0.106) | -1.153*** (0.106) | -1.147*** (0.106) | -1.065*** (0.106) | -1.058*** (0.106) | -1.250*** (0.220) |
| *Household and location at t−1* | | | | | | |
| Homeowner | -0.147** (0.069) | -0.033 (0.070) | -0.023 (0.070) | -0.002 (0.070) | 0.001 (0.070) | 0.003 (0.111) |
| SEIFA disadvantage decile | 0.170*** (0.013) | 0.187*** (0.013) | 0.186*** (0.013) | 0.164*** (0.013) | 0.165*** (0.013) | 0.073*** (0.024) |
| Inner regional Australia | 0.594*** (0.092) | 0.553*** (0.091) | 0.556*** (0.092) | 0.564*** (0.091) | 0.567*** (0.091) | 0.329* (0.189) |
| Outer regional Australia | 0.528*** (0.122) | 0.487*** (0.121) | 0.490*** (0.121) | 0.514*** (0.121) | 0.518*** (0.121) | 0.457* (0.274) |
| Remote Australia | 0.759** (0.328) | 0.718** (0.327) | 0.723** (0.327) | 0.749** (0.326) | 0.754** (0.326) | 0.753 (0.511) |
| Very remote Australia | 2.695*** (0.806) | 2.629*** (0.808) | 2.641*** (0.808) | 2.651*** (0.811) | 2.652*** (0.811) | 2.750** (1.177) |
| Observations (person-pairs) | 289,616 | 289,616 | 289,616 | 289,616 | 289,616 | 286,174 |
| Individuals | 28,626 | 28,626 | 28,626 | 28,626 | 28,626 | 25,184 |
| Mean entry rate (%) | 2.03 | 2.03 | 2.03 | 2.03 | 2.03 | 1.93 |
| Year, industry-year and state-year FEs | Yes | Yes | Yes | Yes | Yes | Yes |
| Individual FEs | No | No | No | No | No | Yes |
| Joint test of interactions (p-value) | | | | | 0.002 | 0.091 |

**Table D.4:** *Linear probability models of entry into business ownership (percentage points). (continued)*

*Notes:* This table reports OLS estimates of linear probability versions of Equation (4.5) (first four columns) and Equation (4.6) (last two columns), on the same person-pairs as the logits in Table 5.4 and Table 5.5. The dependent variable is 100 times the entry indicator, so each coefficient is a change in the annual probability of entry in percentage points. Each column matches the corresponding logit specification; the last column adds individual fixed effects, which absorb the origin groups, and drops individuals observed in only one person-pair. All columns include the covariates of the corresponding logit and year, industry-year and state-year fixed effects, with reference categories as in Table D.2. The joint test reports the p-value of a Wald test of the null hypothesis that the six origin by labour-market interactions are jointly zero. All standard errors reported in parentheses are clustered at the individual level. * p < 0.10, ** p < 0.05, *** p < 0.01.

**Table D.5:** *Labour market position and liquidity at t−1 among people at risk of entering business ownership, by origin group.*

| | Unemployed | Not in labour force | Bottom wage quintile | Liquidity constrained | Person-pairs |
|---|---|---|---|---|---|
| *Panel A: Share of person-pairs at risk of entry (%)* | | | | | |
| Native-born | 4.2 | 34.4 | 12.9 | 22.4 | 230,642 |
| MES | 2.8 | 44.0 | 6.2 | 18.0 | 26,573 |
| MNES high-English | 3.7 | 40.5 | 8.6 | 19.0 | 30,589 |
| MNES low-English | 2.8 | 81.1 | 7.3 | 21.6 | 3,332 |
| *Panel B: Difference from the native-born, adjusted (percentage points)* | | | | | |
| MES | 0.2 | −1.2* | −1.1*** | −1.0 | |
| | (0.2) | (0.7) | (0.3) | (0.6) | |
| MNES high-English | 0.5*** | 2.4*** | −0.8** | −2.2*** | |
| | (0.2) | (0.7) | (0.3) | (0.6) | |
| MNES low-English | 1.2*** | 24.7*** | 2.2** | 3.8* | |
| | (0.5) | (2.1) | (1.1) | (2.1) | |
| *Panel C: Share of entrants (%)* | | | | | |
| Native-born | 5.2 | 26.0 | 14.5 | 26.2 | 4,595 |
| MES | 5.2 | 28.9 | 12.7 | 24.9 | 591 |
| MNES high-English | 7.5 | 28.6 | 13.2 | 24.2 | 653 |
| MNES low-English | 13.5 | 29.7 | 24.3 | 4.0 | 37 |

*Notes:* This table reports labour market position and liquidity at $t-1$ by origin group, for person-pairs in the transition sample of Table 5.4. Each column is an indicator defined in Section 3.4. Panel A reports the percentage of each group in each position. Panel B reports the difference from the native-born from an OLS regression of the indicator (multiplied by 100) on the origin groups, age, age squared, sex and year fixed effects, with standard errors clustered at the individual level in parentheses. Panel C repeats Panel A for the person-pairs in which the individual enters ownership at $t$, and rests on few low-English entries. Liquidity is observed only for respondents who returned the self-completion questionnaire, so its shares and differences use those person-pairs alone. * p < 0.10, ** p < 0.05, *** p < 0.01.

# E Hourly Income of Owners and Employees

The regressions in Table 5.2 compare owners with employees in general, conditional on the controls. As a complementary descriptive measure, we compare the income of employees in small independent workplaces with that of owners with employees in the same industry and year.

$$\text{WorkerOwnerGap}_{kt} = \text{Median}_{kt}\left(\log w^{\text{Employee, SmallSingleSite}}\right) - \text{Median}_{kt}\left(\log y^{\text{OwnerWithEmployees}}\right) \tag{E.1}$$

where the medians are taken over employees and owners within industry $k$ and year $t$. Both terms are measured per hour of work. $w$ is the employee's hourly wage in their main job, and $y$ is the owner's annual labour and business income divided by their annual hours, since owners typically work longer hours than employees. Medians are used so that the measure remains defined when some owners

report business losses.[7] A positive gap indicates that employees earn more per hour than owners in the same industry.

Table E.1 shows that the gap varies considerably across industries. Employees earn more per hour than owners in low-margin industries dominated by small businesses, including accommodation and food services, agriculture, retail trade and other services. In accommodation and food services, employees earn around 30 per cent more per hour than the owners of small businesses in the same industry. In professional industries such as health care, finance and professional services, owners earn substantially more per hour than employees. The ordering does not follow the share of migrant owners. The two industries with the largest migrant shares of owners are health care and social assistance (37 per cent) and accommodation and food services (35 per cent), with gaps of −0.66 and 0.27 log points.

Across all 273 industry-years, owners with employees earn more than employees of small independent workplaces, by about 9 per cent per hour (a weighted mean gap of −0.085 log points) and about 60 per cent per year (−0.472 log points). The much larger annual difference reflects the longer hours worked by owners. Employees nonetheless earn more per hour than owners in 37 per cent of industry-years, concentrated in the low-margin industries at the top of Table E.1.

**Table E.1:** *Median hourly income gap between employees and owners with employees, by industry.*

| Industry | Industry-years | Mean gap (log points) | Years with positive gap | Migrant share of owners |
|---|---|---|---|---|
| Accommodation and food services | 21 | 0.267 (0.057) | 76% | 35% |
| Agriculture, forestry and fishing | 21 | 0.216 (0.066) | 76% | 8% |
| Other services | 21 | 0.090 (0.035) | 67% | 18% |
| Retail trade | 21 | 0.083 (0.028) | 81% | 25% |
| Administrative and support services | 21 | −0.017 (0.044) | 48% | 29% |
| Transport, postal and warehousing | 21 | −0.040 (0.045) | 48% | 25% |
| Manufacturing | 21 | −0.049 (0.026) | 38% | 23% |
| Education and training | 9 | −0.078 (0.071) | 22% | 26% |
| Construction | 21 | −0.156 (0.021) | 14% | 23% |
| Wholesale trade | 21 | −0.178 (0.058) | 19% | 35% |
| Rental, hiring and real estate services | 10 | −0.248 (0.056) | 0% | 25% |
| Information media and telecommunications | 3 | −0.271 (0.079) | 0% | 31% |
| Professional, scientific and technical services | 21 | −0.297 (0.023) | 0% | 19% |

[7] The gap is computed only for industry-years with at least 10 owners with employees, to avoid imprecise estimates from small cells.

**Table E.1:** *Median hourly income gap between employees and owners with employees, by industry. (continued)*

| Industry | Industry-years | Mean gap (log points) | Years with positive gap | Migrant share of owners |
|---|---|---|---|---|
| Financial and insurance services | 19 | −0.631 (0.055) | 0% | 31% |
| Health care and social assistance | 21 | −0.663 (0.033) | 0% | 37% |
| All industry-years | 273 | −0.085 | 37% | 23% |

*Notes:* This table reports the median hourly income gap of Equation (E.1) by industry, in HILDA waves 1 to 21. The gap is the median log hourly wage of employees in small single-site workplaces minus the median log hourly income of owners with employees in small workplaces, within each industry and year; a positive gap means that employees earn more per hour than owners. The mean gap for each industry is the average over its industry-years with at least 10 owners, weighted by the number of owners, and industries with fewer than three such years are not listed. The standard error in parentheses treats industry-years as independent and is therefore indicative. The last row pools all industry-years.

# F Assimilation Decay Horizons: Supporting Literature

Section 3.5 searches over the decay parameter $\lambda$ in the range $[0, 0.30]$ for a single pooled rate and $[0, 0.50]$ for origin-specific rates. This range is informed by a broader literature on the horizon over which migrants' earnings, occupational status and linguistic proficiency converge toward native-born benchmarks.

Chiswick (1978) and Schuetze & Antecol (2007) established that immigrant earnings typically converge to native benchmarks within 10 to 15 years, consistent with Akee et al. (2026), who find, using administrative panel data in the United States, that recent cohorts achieve parity within 10 years, though earlier cohorts exhibit strong growth without complete convergence. Examining job search mechanics, Bowlus et al. (2016) estimates that immigrants take an average of 13 years to match native job offer arrival rates. Longer horizons are also evident. Arellano-Bover & San (2026) show that the overall wage gap closes only after 27 to 29 years, with convergence and long-term prosperity driven primarily by firm-to-firm mobility and climbing the employer ladder. In the Australian context, Chiswick et al. (2005) document a U-shaped occupational trajectory, where an initial drop in status due to imperfect skill transferability is followed by gradual earnings recovery. Schuetze & Antecol (2007) found that self-employment rates for specific immigrant cohorts steadily increase with their time spent in the host country (main countries of the study include the United States, Canada and Australia), typically catching up to native-born rates within 10 to 20 years of arrival. Although self-employed immigrants tend to outperform their native counterparts, Schuetze & Antecol (2007) attributes this to the fact that the countries used their analysis, as well as many other developed nations, use special visas and entrepreneurial entry requirements as tools to attract foreign talent, but notes that broader market structures and local economic conditions remain the strongest overall drivers of success. Breunig et al. (2013) examines the wage gap on entry and wage assimilation of immigrants to Australia by comparing results from standard cross-sectional models (Heckman selection model) with panel data estimators (Fixed Effects, Random Effects and Hausman-Taylor (HT) estimator) to account for unobserved individual heterogeneity. They found that there is a wage gap for immigrant men from English-speaking backgrounds, in contrast with previous research which has found no wage gap and wage assimilation occurs slowly for all immigrant groups, but is much faster for MES men than for MNES group.

Evidence on identity and linguistic assimilation provides further guidance on empirical decay horizons. Abramitzky et al. (2020) show that naming convergence between immigrant and native parents occurs over roughly 20 years. Language acquisition follows a similar pattern. Kulkarni & Hu (2014) find that English proficiency rises from 45 to 55% at arrival to 60 to 63% after 5 years, and reaches 66 to 67% after 10 years, reflecting rapid initial gains followed by slow gains thereafter (Espinosa & Massey, 1997; Guven & Islam, 2015; Kossoudji, 1988; Stevens, 1999). Wage convergence, however, proceeds more slowly for non-English-speaking migrants once individual heterogeneity is controlled for, with older arrival cohorts experiencing persistent structural gaps (Breunig et al., 2013).

Furthermore, Chiswick et al. (2005) suggests that immigrants moving between similar institutional contexts, such as an English-speaking Canadian entering the United States, face minimal initial downward mobility. For these main English-speaking migrants, divergence from native-born workers depends primarily on arrival timing, not skill acquisition. Indeed, the literature on Australia and the United States documents "negative" assimilation among immigrants from English-speaking developed countries, where initial wage premiums decay over time (Chiswick & Miller, 2011; 2012). In Australia, migrants from English-speaking developed countries experience a relative earnings decline over their duration of residence, with earnings 5.5% lower for the 1988 cohort than the 1998 cohort over ten years (Chiswick & Miller, 2011).

Broader socio-cultural integration reflects a similar pattern. While shared values facilitate smoother initial settlement (Nakhaie, 2020), full identity assimilation can remain non-converged even after decades abroad (Scott, 2004). Anglophone immigrants in Australia, and UK-born immigrants specifically, nonetheless exhibit employment outcomes and qualification returns nearly identical to those of native-born Australians (Colic-Peisker, 2011). Identity and cultural assimilation operate on much longer horizons than economic, occupational and linguistic assimilation, so our empirical focus in the main text centres strictly on the latter.

Table F.1 reports the five best-fitting values of $\lambda$ within each of the three decay specifications described in Section 3.5, alongside the mean squared error achieved via cross-validation.

**Table F.1:** *Best-fitting $\lambda$ estimates for three specifications.*

| **Type** | $\lambda$ | **Half-life** | $\lambda_{\text{MES}}$ | $\lambda_{\text{MNES}}$ | **Half-life (MES)** | **Half-life (MNES)** | **MSE** | **Difference (in $10^5$ scale)** |
|---|---|---|---|---|---|---|---|---|
| Type 0 | 0.00 | – | | | | | 0.15244606 | +0.00 |
| Type 1 | **0.02** | **34.7** | | | | | **0.15243183** | **−1.42** |
| | 0.03 | 23.1 | | | | | 0.15243205 | −1.40 |
| | 0.04 | 17.3 | | | | | 0.15243557 | −1.05 |
| | 0.01 | 69.3 | | | | | 0.15243610 | −1.00 |
| | 0.05 | 13.9 | | | | | 0.15244133 | −0.47 |
| Type 2 | | | **0.04** | **0.00** | **17.3** | **∞** | **0.15242431** | **−2.17** |
| | | | 0.04 | 0.01 | 17.3 | 69.3 | 0.15242439 | −2.17 |
| | | | 0.05 | 0.00 | 13.9 | ∞ | 0.15242443 | −2.16 |
| | | | 0.05 | 0.01 | 13.9 | 69.3 | 0.15242473 | −2.13 |
| | | | 0.06 | 0.00 | 11.6 | ∞ | 0.15242585 | −2.02 |

**Table F.1:** *Best-fitting $\lambda$ estimates for three specifications. (continued)*

*Notes:* This table reports the five best-fitting decay rates for each specification of Section 3.5, ranked by the five-fold cross-validated mean squared error (MSE) of log hourly wages. Type 0 applies no decay ($\lambda = 0$), Type 1 a single pooled rate to all migrant owners and Type 2 separate rates to MES and MNES owners. Half-lives are $\ln 2/\lambda$ in years. *Difference* is the change in MSE from the Type 0 benchmark, multiplied by $10^5$. The selected values are in bold.

The Type 2 specification, which allows the decay rate to vary by English-speaking background, achieves the lowest mean squared error of the three, but its gain over no decay is small relative to the error itself. We therefore use the plain owner shares (Type 0) for the main employee wage estimates and report Types 1 and 2 as robustness checks.